

Customer Churn Project-Final Business Report

List of Contents :

• Contents	02
• List of Figures	03
• List of Tables	04
• Problem Statement	06
• Overview of the Dataset	08
• Data Cleaning and Preprocessing	10
• Exploratory Data Analysis	12
• Outlier Detection and treatment.....	28
• Feature Engineering	28
• Model Building	29
• Logistic Regression Model Performance Improvement	30
• Logistic Regression Model Performance Comparison	34
• Advanced Model Building.....	35
• Hyperparameter Tuning.....	40
• Model Performance Comparison and Final Model Selection.....	43
• Actionable Insights and Recommendations.....	46
• APPENDIX	49

List of Figures :

Figures	Page NO.
Fig 1: Countplot of the target variable(Churn_New)	13
Fig 2: Observation on tenure	14
Fig 3: Observation on MonthlyCharges	15
Fig 4: Observation on TotalCharges	16
Fig 5: Observation on PaymentMethod_New	17
Fig 6: Observation on Customer Churn and InternetService	18
Fig 7: Observation on Customer Churn and Contract	19
Fig 8: Observation on Customer Churn and PaymentMethod_New	20
Fig 9: Correlation matrix of Numerical variables	21
Fig 10: Observation on Customer Churn and tenure	22
Fig 11: Observation on Customer Churn and TotalCharges	23
Fig 12: Observation on Customer Churn and MonthlyCharges	24
Fig 13: Observation on Churn Rate by Contract	25
Fig 14: Pairplot for Numerical features and Customer Churn	26
Fig 15: Heatmap of Churn Rate for Contract and Payment Method	27
Fig 16: Outliers in the dataset	28
Fig 17: Confusion matrix for Logistic Regression Training data	32
Fig 18: Confusion matrix for Logistic Regression validation data	33
Fig 19: Feature Importance in Model Building	45

List of Tables :

Tables	Page NO.
Table 1: First five rows of the dataset	09
Table 2: Last five rows of the dataset	09
Table 3: Information about the dataset	09
Table 4: Statistical Summary of the dataset	12
Table 5: Logistic Regression Baseline Model	30
Table 6: Logistic Regression Final Model	31
Table 7: Performance Metrics for Logistic Regression Training data	32
Table 8: Performance Metrics for Logistic Regression Test data	33
Table 9: Training Performance Comparison	34
Table 10: Test Performance Comparison	34
Table 11: Decision Tree Classifier Training Performance	35
Table 12: Decision Tree Classifier Validation Performance	36
Table 13: Bagging Classifier Training Performance	36
Table 14: Bagging Classifier Validation Performance	37
Table 15: Random Forest Classifier Training Performance	37
Table 16: Random Forest Classifier Validation Performance	37
Table 17: AdaBoost Classifier Training Performance	38
Table 18: AdaBoost Classifier Validation Performance	38
Table 19: Gradient Boosting Classifier Training Performance	39

Tables	Page NO.
Table 20: Gradient Boosting Classifier Validation Performance	39
Table 21: Tuned Random Forest Classifier Training Performance	41
Table 22: Tuned Random Forest Classifier Validation Performance	41
Table 23: Tuned AdaBoost Classifier Training Performance	42
Table 24: Tuned AdaBoost Classifier Validation Performance	42
Table 25: Tuned Gradient Boosting Classifier Training Performance	42
Table 26: Tuned Gradient Boosting Classifier Validation Performance	43
Table 27: Model Performance Comparison on Training Data	43
Table 28: Model Performance Comparison on Validation Data	43
Table 29: Tuned Gradient Boosting Classifier - Performance on the Test Data	45

Problem Statement

Business Context:

AlphaCom, a leading telecommunications provider, has recently experienced a concerning rise in customer churn despite offering competitive services and a wide product portfolio. This increase is directly impacting revenue and undermining brand reputation in an intensely competitive market. Traditional retention strategies have proven inadequate because customer churn is influenced by a complex mix of factors, including service usage, billing preferences, contract types, and demographics. Without clear insights into these patterns, the company is left reacting to churn instead of preventing it.

Business Significance:

Industry research consistently shows that the cost of acquiring a new customer is 5–6 times higher than retaining an existing one. Therefore, even a marginal improvement in customer retention can lead to substantial cost savings. In addition, the probability of successfully selling new products or services to existing customers is significantly higher (60–70%) compared to new customers (5–20%). This makes customer retention not only a defensive strategy, but also a strong revenue growth opportunity.

From a financial perspective, increasing customer retention by just 5% can boost profits by 25% to 95%, depending on the industry. At a macro level, U.S. companies are estimated to lose approximately \$136.8 billion annually due to avoidable customer churn, highlighting the scale of the problem. These benchmarks emphasize why understanding, predicting, and preventing churn is a high-priority business objective.

Customer churn is not merely the loss of customers; it directly affects revenue, customer lifetime value, operational efficiency, and brand reputation. High churn rates often indicate underlying issues such as poor service quality, pricing dissatisfaction, or weak customer engagement. Hence, a data-driven churn analysis enables organizations to move from reactive churn management to proactive retention strategies.

Objective:

As a data scientist at AlphaCom, you are tasked with developing a predictive model to identify customers at high risk of churn and uncover the key factors driving their decisions. Solving this problem will enable the company to proactively design targeted retention strategies, reduce churn-related losses, and improve customer lifetime value, ultimately safeguarding revenue and strengthening AlphaCom's competitive position.

Data Description:

The data contains different attributes related to churn. The detailed data dictionary is given below:

- **Gender:** The customer's gender (e.g., Male or Female). This demographic feature may correlate with customer behavior.
- **Age Range:** Indicates the customer's age bracket (e.g., 18–25, 26–35, etc.), offering demographic insights that can impact churn analysis.
- **SeniorCitizen:** A binary indicator (if included) that identifies whether the customer is a senior citizen (commonly 1 for senior, 0 for non-senior). Senior status can influence service preferences and retention strategies.
- **Partner:** Indicates whether the customer has a partner. This factor can affect customer loyalty and service usage patterns.
- **Dependents:** Specifies whether the customer has dependents. This information can provide context on the customer's household and influence their service needs.
- **Tenure:** The number of months the customer has been with the company. Longer tenure may indicate higher loyalty, while shorter tenure could be a churn risk indicator.
- **PhoneService:** Denotes whether the customer subscribes to telephone services. This binary feature (Yes/No) helps understand service adoption.
- **MultipleLines:** Indicates if the customer has multiple phone lines. This feature can provide insight into customer behavior and service complexity.
- **InternetService:** Describes the type of internet service the customer uses (e.g., DSL, Fiber optic, or None). The type of internet service can be a critical factor in churn analysis.
- **OnlineSecurity:** Shows whether the customer subscribes to online security services. This value (Yes/No) may influence customer satisfaction and retention.
- **OnlineBackup:** Indicates if the customer has an online backup service. Similar to online security, this can be a part of the overall service bundle affecting churn.
- **DeviceProtection:** Specifies whether the customer is enrolled in a device protection plan, providing an added layer of service value.
- **TechSupport:** Denotes if the customer subscribes to technical support services. Access to tech support can improve customer experience and reduce churn.
- **StreamingTV:** Indicates whether the customer subscribes to a streaming TV service. Media consumption patterns can be a differentiator in customer preferences.
- **StreamingMovies:** Specifies if the customer subscribes to a streaming movies service. This, combined with other services, can highlight trends in customer behavior.

- **Contract:** Describes the type of contract the customer holds (e.g., month-to-month, one-year, or two-year). Contract type is a strong indicator of churn risk—shorter contracts are often associated with higher churn.
- **PaperlessBilling:** Indicates whether the customer is enrolled in paperless billing. This operational feature can sometimes correlate with customer engagement levels.
- **PaymentMethod:** Details the payment method used by the customer (e.g., electronic check, mailed check, bank transfer, or credit card). Payment methods can affect both churn and overall customer satisfaction.
- **MonthlyCharges:** The monthly amount in USD charged to the customer. Higher charges might increase the likelihood of churn if customers perceive the cost as too high for the value provided.
- **TotalCharges:** The cumulative amount in \$ USD charged over the customer's tenure. This helps in understanding the long-term value of each customer and can be a predictor of churn.
- **Churn:** The target variable indicating whether the customer has left (typically denoted as "Yes" or "No"). This is the primary outcome you aim to predict with your machine learning model.

Loading the Dataset

The dataset provided was loaded into a pandas dataframe for analysis.

Data Overview

Getting the first 5 and last 5 rows

- The dataset contains 12055 rows and 20 columns.
- First 5 rows of the dataset:

	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV
0	Female	0	Yes	No	1.0	No	No phone service	DSL	No	Yes	No	No	No
1	Male	0	No	No	34.0	Yes	No	DSL	Yes	No	Yes	No	No
2	Male	0	No	No	2.0	Yes	No	DSL	Yes	Yes	No	No	No
3	Male	0	No	No	45.0	No	No phone service	DSL	Yes	No	Yes	Yes	No
4	Female	0	No	No	2.0	Yes	No	Fiber optic	No	No	No	No	No

Table 1: First five rows of the dataset

- **Last 5 rows of the dataset:**

	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV
12050	Female	0	No	No	6.0	Yes	No	Fiber optic	No	No	No	No	Yes
12051	Female	0	Yes	Yes	1.0	Yes	No	Fiber optic	No	Yes	No	No	No
12052	Male	0	No	No	72.0	Yes	Yes	Fiber optic	Yes	Yes	Yes	Yes	Yes
12053	Female	0	Yes	No	57.0	Yes	Yes	Fiber optic	Yes	No	No	No	Yes
12054	Female	0	Yes	No	NaN	Yes	No	Fiber optic	No	Yes	No	Yes	Yes

Table 2: Last five rows of the dataset

Information about the dataset :

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12055 entries, 0 to 12054
Data columns (total 20 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   gender                                12055 non-null  object
1   SeniorCitizen                        12055 non-null  int64
2   Partner                              12055 non-null  object
3   Dependents                           12055 non-null  object
4   tenure                               11451 non-null  float64
5   PhoneService                         12055 non-null  object
6   MultipleLines                        12055 non-null  object
7   InternetService                      12055 non-null  object
8   OnlineSecurity                       12055 non-null  object
9   OnlineBackup                         12055 non-null  object
10  DeviceProtection                     12055 non-null  object
11  TechSupport                          12055 non-null  object
12  StreamingTV                          12055 non-null  object
13  StreamingMovies                      12055 non-null  object
14  Contract                             12055 non-null  object
15  PaperlessBilling                     12055 non-null  object
16  PaymentMethod                       12055 non-null  object
17  MonthlyCharges                       12055 non-null  object
18  TotalCharges                         12055 non-null  object
19  Churn                               12055 non-null  object
dtypes: float64(1), int64(1), object(18)
```

Table 3: Information about the dataset

❖ Observations :

- ❖ Most of the columns(18) are **categorical** including demographics like **gender**, **Partner**, and **Dependents**, as well as service and billing information such as **InternetService**, **Contract**, and **PaymentMethod**.
- ❖ **Churn** column is the **target** variable of this dataset.
- ❖ **SeniorCitizen**, **tenure** are numerical features. **MonthlyCharges**, **TotalCharges** are currently an object type but contain numerical data.
- ❖ There are some missing values in numerical columns, which typically requires data cleaning before analysis.

Data cleaning and preprocessing

Processing Columns

We process **tenure**, **MonthlyCharges** and **TotalCharges** to extract numerical values.

● Anomalous value check and treatment

Treat invalid and missing tenure

In **tenure** column, some of the data contained with negative or unusual values such as 0.0, -2.0, and -3.0. We have treated these invalids for further analysis.

Remove \$ and Formatting

We have formatting and treated all those special characters(\$,£) and commas(",") and nan's from the numerical columns(MonthlyCharges and TotalCharges).

Methodology : For **tenure**, any negative values were corrected to zero, and all unspecified entries were filled with zero, after which the column was converted to whole numbers. For **MonthlyCharges** and **TotalCharges**, currency symbols were removed, text indicating missing data was replaced with a standard placeholder for missing values, and the columns were then converted to a numerical format.

- **Missing value check and treatment**

All missing values in numerical columns(tenure, MonthlyCharges, TotalCharges) have been approximately identified and treated.

Methodology : Missing values in the numerical column were filled using the median value of their respective columns.

- **Standardizing Categorical data**

We cleaned and standardized the PaymentMethod and Churn columns. This involved removing leading/trailing spaces, converting to lowercase, removing extra spaces, and then applying title case(or capitalizing for 'Churn') to ensure consistent categorical values. The original PaymentMethod and Churn columns were then dropped in favour of new, cleaned PaymentMethod_New and Churn_New columns.

Methodology : categorical columns such as 'PaymentMethod' and 'Churn' were cleaned by removing extra spaces, standardizing the capitalization of entries, and creating new, consistent columns ('PaymentMethod_New', 'Churn_New'), with the original columns being removed.

The **SeniorCitizen** column, which was numeric (0 or 1), was mapped to '**No**' and '**Yes**' and converted to a categorical type for better interpretability.

- **Review Data cleaning and Duplicates**

All duplicate values have been removed and initial data cleaning(handling special characters, consistent categorical values, and basic type conversions) has been performed.

Methodology : Duplicate rows were identified and removed from the dataset.

Note: Detailed Data Preprocessing is in the **APPENDIX** Section.

Exploratory Data Analysis(EDA)

Purpose of EDA

The purpose of Exploratory Data Analysis (EDA) in this project is to identify variables that show clear separation between churned and non-churned customers. This includes analyzing the separation of mean and median for continuous variables, understanding distribution skewness, and evaluating how these statistical characteristics influence customer churn. Univariate, bivariate, and multivariate analyses are used to extract actionable business insights rather than only descriptive statistics.

Note: The EDA section has been covered multiple times in the previous case studies. In this case study, we will mainly focus on the model building aspects. We will only be looking at the key observations from EDA. The detailed EDA can be found in the **APPENDIX** Section.

Statistical Summary of the Dataset

Let's check statistical summary of the numerical columns

	count	mean	std	min	25%	50%	75%	max
tenure	12028.0	29.722897	25.301622	0.00	4.000	25.00	53.0000	74.00
MonthlyCharges	12028.0	64.559845	29.963549	15.29	34.500	71.35	89.0825	121.67
TotalCharges	12028.0	2197.616541	2178.132686	0.00	492.075	1328.75	3601.5375	9039.92

Table 4: Statistical Summary of the dataset

❖ Observations :

tenure

- Customers stay for an average of approximately **30 months**(mean:29.72), but there is a large variation in **loyalty**, as indicated by a high standard deviation of about 25.30 months.
- The **median is 25 months**, which is lower than the mean, suggesting the data is likely right-skewed, meaning a larger portion of customers have **shorter tenures**.

MonthlyCharges

- The average monthly charge is approximately **\$64.55**(mean), with a moderate spread(standard deviation of 29.96).
- The mean and **median(\$71.35)** charges are relatively close, but the median is higher, suggesting a slight left-skewed or a binomial distribution might exist.

TotalCharges

- The **mean** total charge is approximately **\$2197.61**, but the large standard deviation of over 2178.13 indicates a very wide variation in accumulated charges.
- The **median** total charge is **\$1328.75**, which is significantly lower than the mean. This wide difference confirms the data is heavily right-skewed, indicating many customers have low total charges(likely due to short tenures), while a few have very high charges.

Univariate Analysis

In this section, we will analyze the distribution of independent variables. It will help us identify the pattern among the variables and the effects they have on the target variable.

Let's first see how the target variable(**Churn_New**) is distributed.

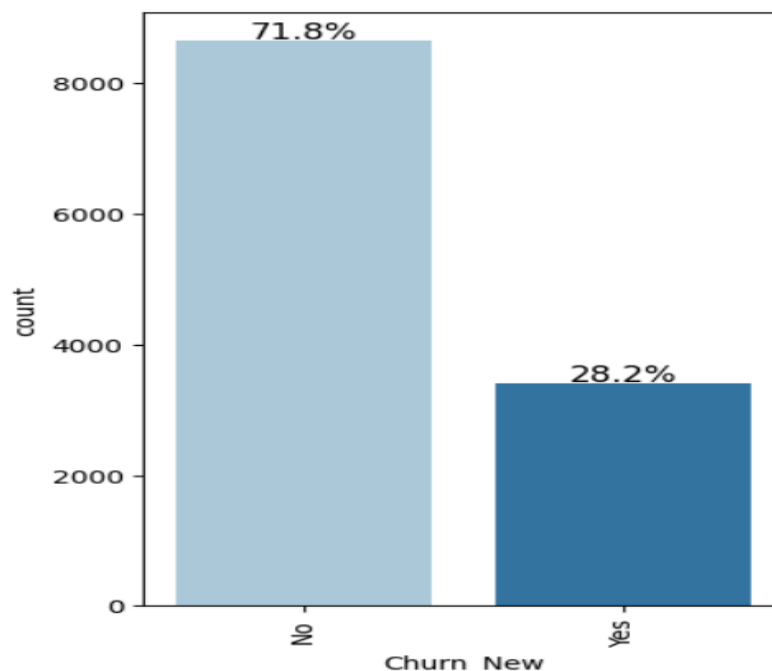


Fig 1: Countplot of the target variable(Churn_New)

- **Significant Churn Rate:** Approximately **28.2%** of AlphaCom's customers have churned. This figure is quite substantial and directly reflects the business problem outlined in the problem statement, indicating a notable loss of customers.

- **Business Urgency:** A nearly 30% churn rate highlights an urgent need for intervention. As previously noted in the business context, retaining existing customers is significantly more cost-effective than acquiring new ones. Reducing this churn rate, even marginally, could lead to substantial financial benefits for AlphaCom.

Now let's understand the distribution of other variables.

Observations on tenure :

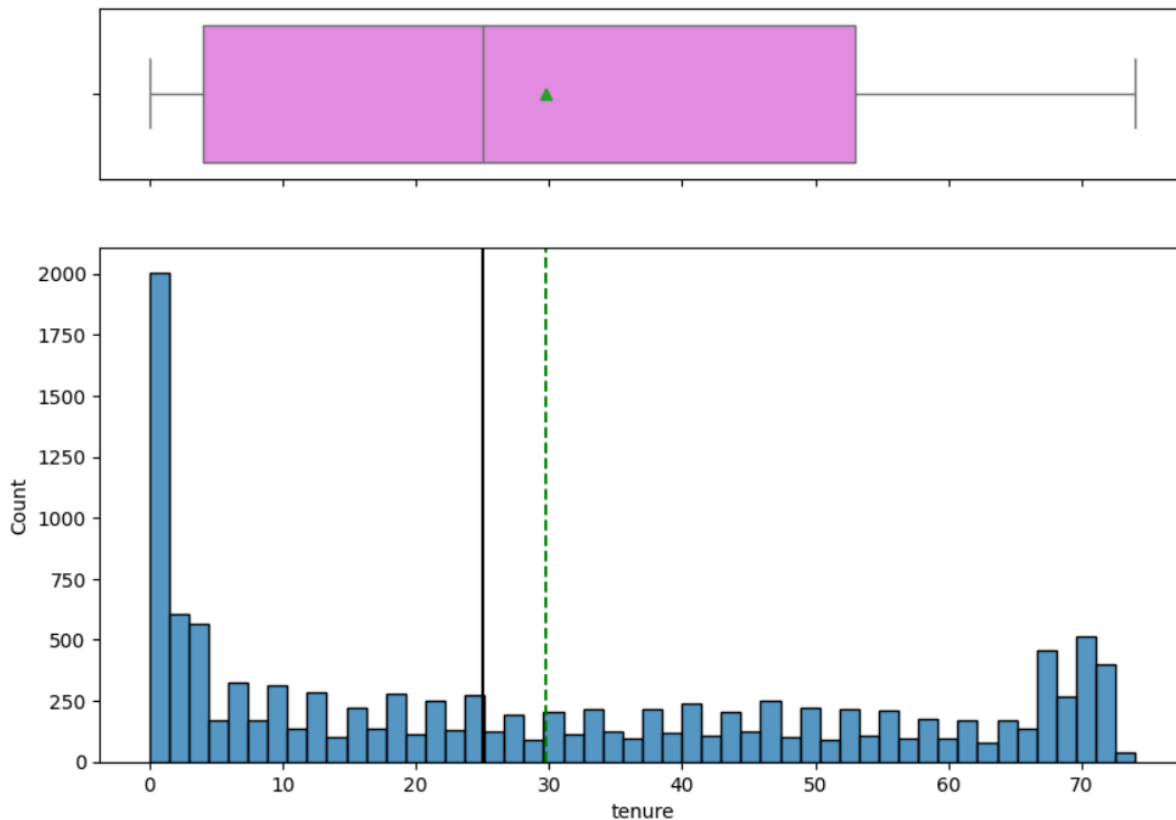


Fig 2: Observation on tenure

- ❖ The histogram displays a **bimodal distribution**, which clearly shows two major concentrations of the datapoints: one very high peak at the beginning of the graph and the second, lower peak towards the end of the scale.
- ❖ There are a very large number of individuals who have extremely short tenure(0-5 months). On the other end a smaller, but significant group has a very long tenure(around 70 months).
- ❖ This pattern indicates a **retention crisis for new customers**, while those who stay past the initial phase are likely very loyal or satisfied.

Business Insight: Customers are most likely to churn during the early stages of their relationship with the company. This highlights the importance of onboarding programs and early engagement strategies to improve retention.

Observation on Monthly Charges :

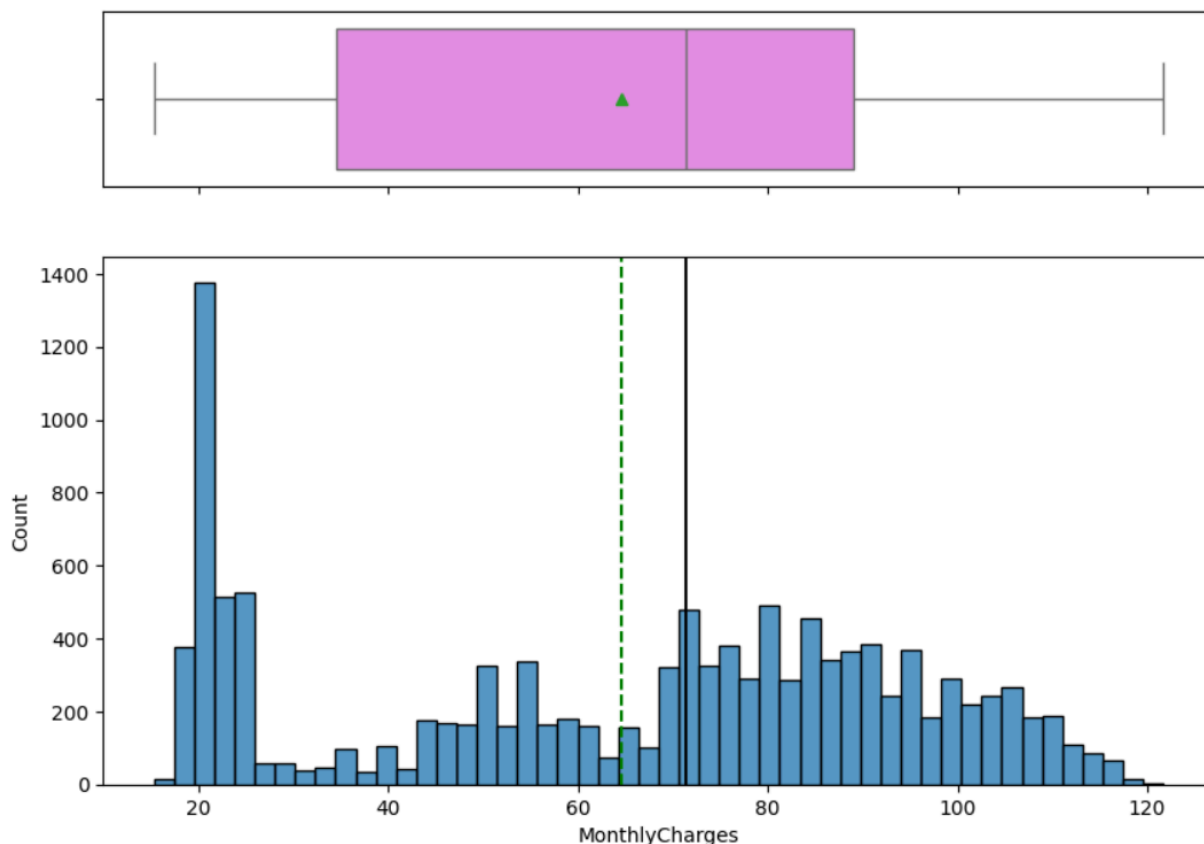


Fig 3: Observation on MonthlyCharges

- ❖ The data for **Monthly Charges** is heavily **right-skewed**. This means a large number of customers have lower monthly charges, while fewer customers have very high charges.
- ❖ The **bulk of customers** have monthly charges between approximately \$60 and \$100. And there is a significant peak in customer count for charges in the lowest range, specifically around the \$20 mark.
- ❖ The wide spread of data indicates high **variability** in customer spending.

Business Insight: Higher recurring charges increase churn risk when customers do not perceive sufficient value, suggesting a need for better pricing plans or bundled offerings.

Observation on Total Charges :

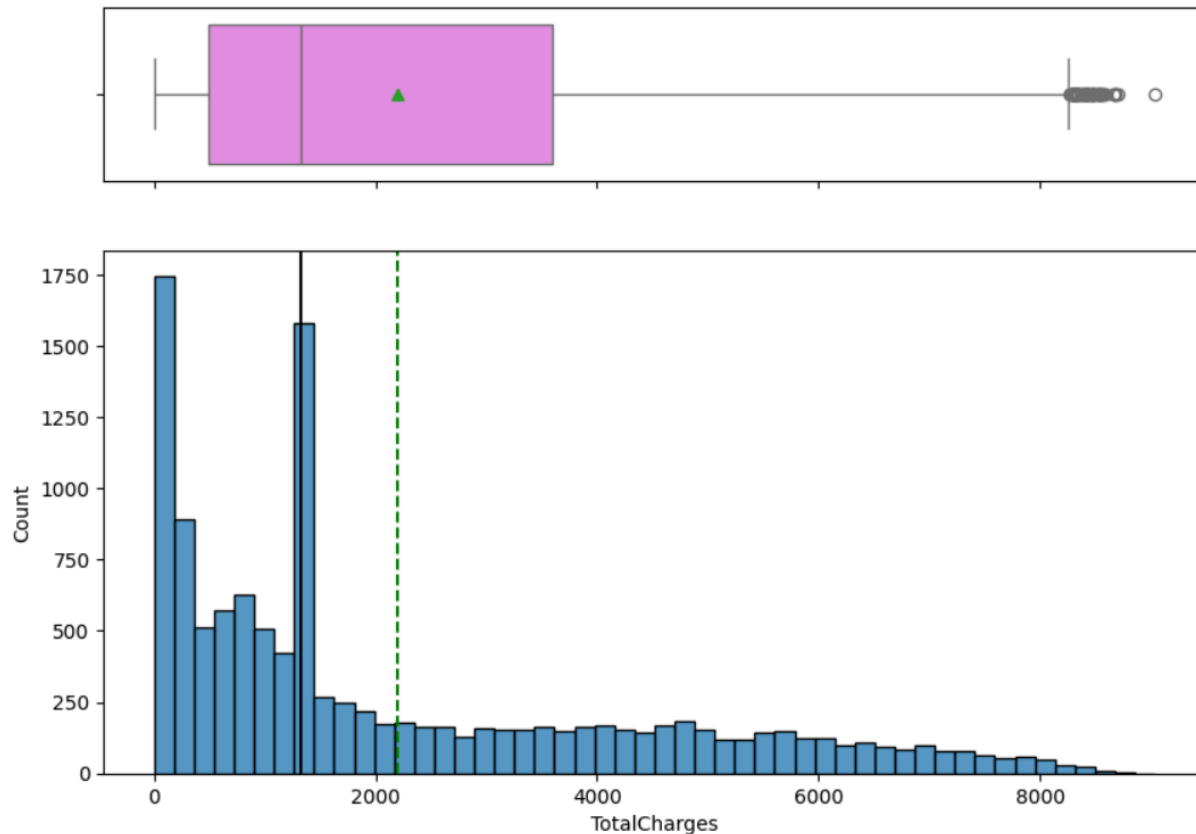


Fig 4: Observation on TotalCharges

- ❖ The data is **right-skewed**, meaning most customers have lower total charges, with fewer customers having very high charges.
- ❖ A very large number of customers have total charges in the low ranges(around \$0 to \$2000). This suggests a significant portion of the customer base might consist of relatively new customers or those on lower-cost plans.
- ❖ There are some outliers in the right end of the boxplot, but they appear to be part of the natural skew of the data rather than extreme. We have to treat these outliers for further analysis.

Business Insight: Early churn prevents the business from realizing long-term customer value, reinforcing the importance of retention in the initial customer lifecycle.

Observation on Payment Method :

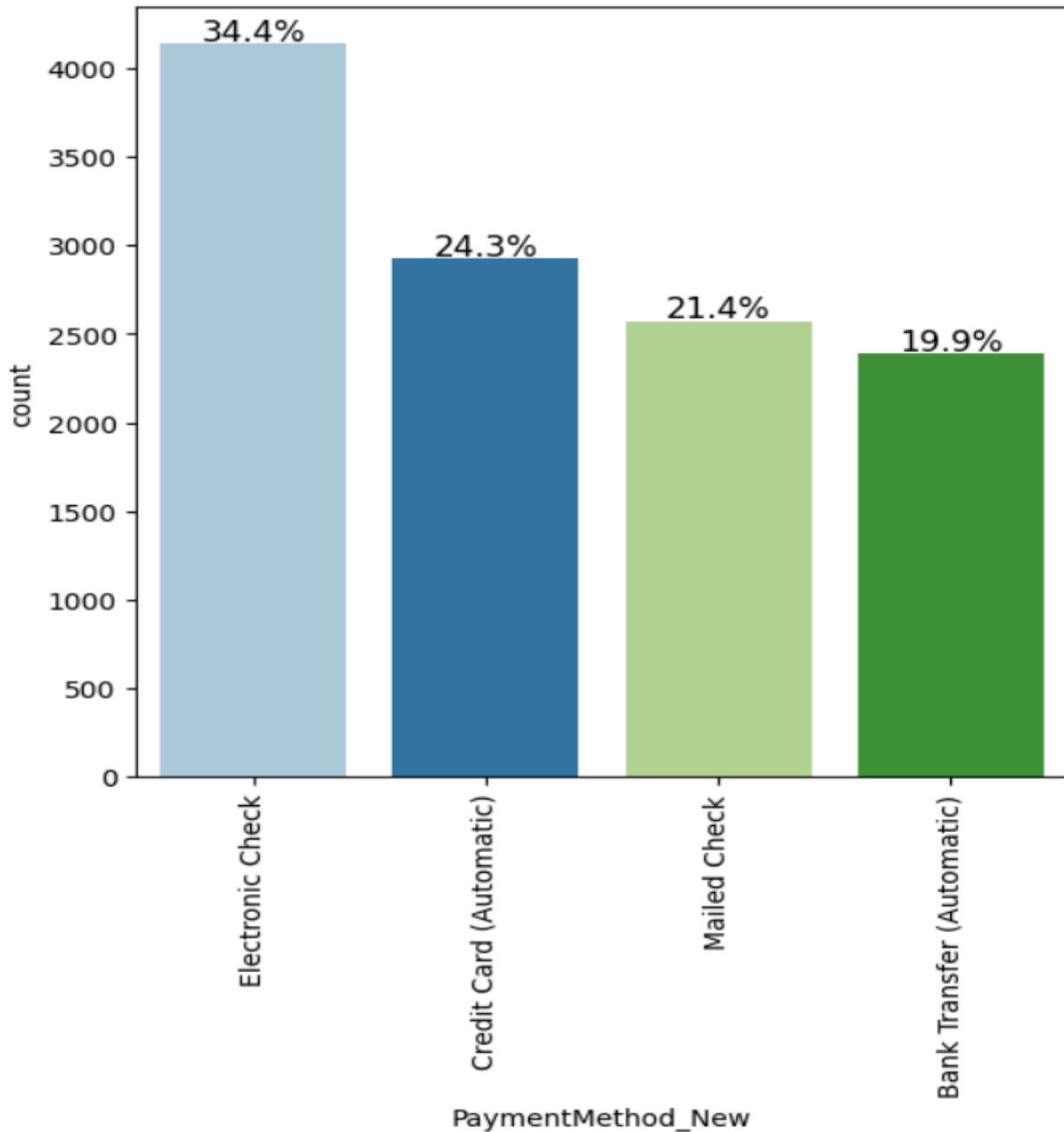


Fig 5: Observation on PaymentMethod_New

- ❖ **Electronic Check(34.4%)** payment method is the most frequently used by the customers. **Credit Card(Automatic)(24.3%)** And **Mailed Check(21.4%)** are good payment options for customers.
- ❖ Very less number of customers go for **Bank Transfer(Automatic)(~20%)** payment option.

Business insights:

- While Electronic Check is the most popular payment method (34.4% of customers), it's strongly associated with the highest churn rate. This suggests potential friction points in the customer journey related to this payment method, or it's preferred by customers with shorter commitment periods.
- Customers using automatic payment methods (Credit Card Automatic: 24.3%, Bank Transfer Automatic: 20%) show significantly lower churn rates. These methods indicate higher customer commitment and convenience, contributing positively to retention.

Bivariate Analysis

For bivariate analysis, we can analyze the contribution of variables in determining the **Churn_New** which is the target variable. This tells us whether the customers are churn or not.

Observations on Customer Churn and Internet Service :

Churn_New	No	Yes	All
InternetService			
All	8634	3394	12028
Fiber optic	2682	2184	4866
DSL	3400	945	4345
No	2552	265	2817

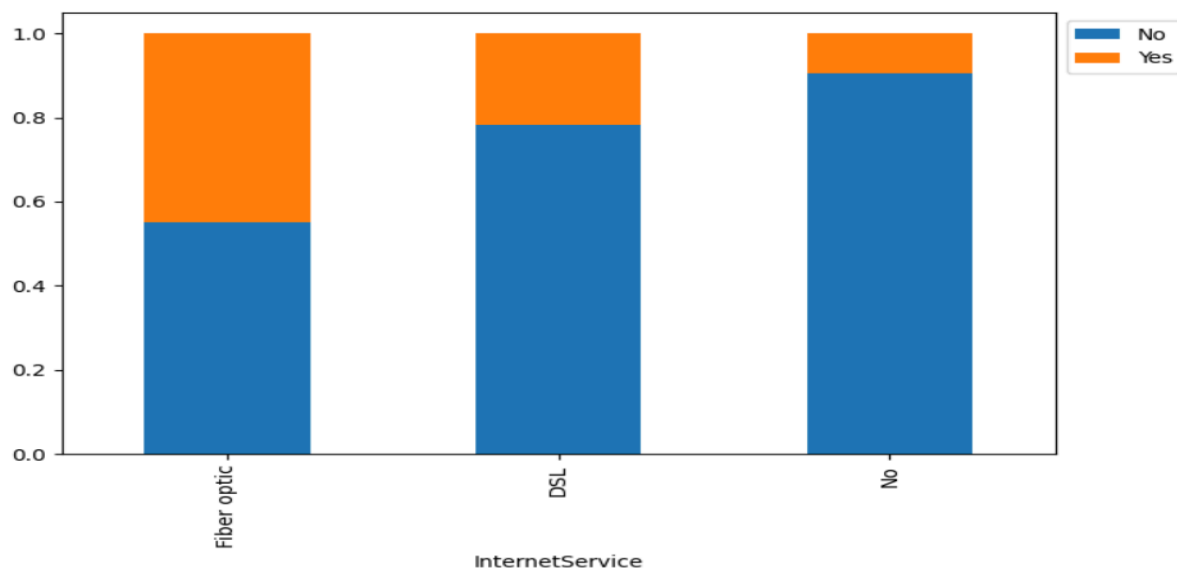


Fig 6: Observation on Customer Churn and InternetService

- ❖ In Internet service, **Fibre optic** customers have the **highest churn rate**(approximately **41-42%**), which is notably higher than customers with **DSL** service(approximately **19%**).

Business Insights :

- Fiber optic customers exhibit the highest churn rate, significantly higher than those with DSL service or no internet service. This is a critical pain point, especially since Fiber Optic is the most prevalent internet service provided by AlphaCom.
- Customers using DSL service have a much lower churn rate compared to Fiber optic users. This indicates a more satisfied or less price-sensitive customer base within this segment.

Observations on Customer Churn and Contract :

Churn_New Contract	No	Yes	All
All	8634	3394	12028
Month-to-month	3527	3006	6533
One year	2265	290	2555
Two year	2842	98	2940

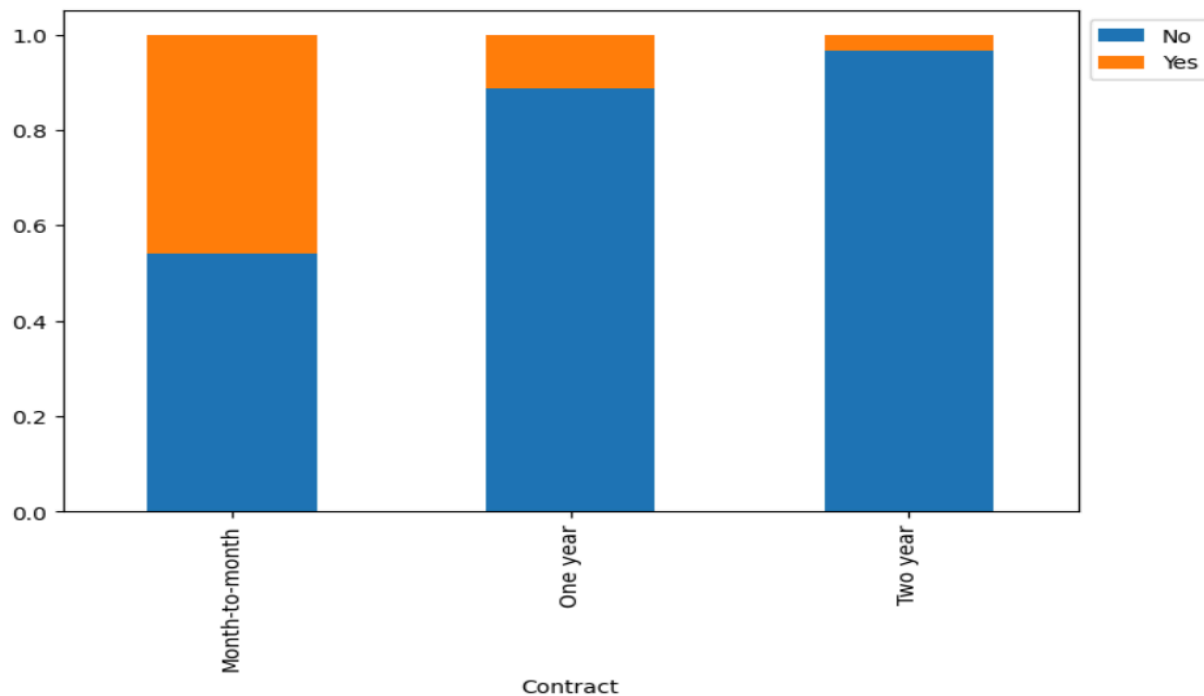


Fig 7: Observation on Customer Churn and Contract

- ❖ Most of the customers who have a Contract on a month-to-month basis are **more likely to churn**(approximately **46%**). But customers with **yearly contracts** are **less likely to churn**(one year ~**11%** & Two year ~**3.3%**).

Business Insights :

- The vast majority of customers on month-to-month contracts are highly prone to churn, with nearly half of them leaving the service. This contract type offers customers maximum flexibility but significantly increases their churn propensity.
- Customers on one-year contracts show a significantly lower churn rate compared to month-to-month customers. This commitment period provides a good balance between customer flexibility and company retention.
- Customers on two-year contracts have the lowest churn rate, indicating very high loyalty and satisfaction. This is AlphaCom's most stable customer segment.

Observations on Customer Churn and Payment Method :

Churn_New PaymentMethod_New	No	Yes	All
All	8634	3394	12028
Electronic Check	2106	2030	4136
Mailed Check	2002	570	2572
Credit Card (Automatic)	2516	412	2928
Bank Transfer (Automatic)	2010	382	2392

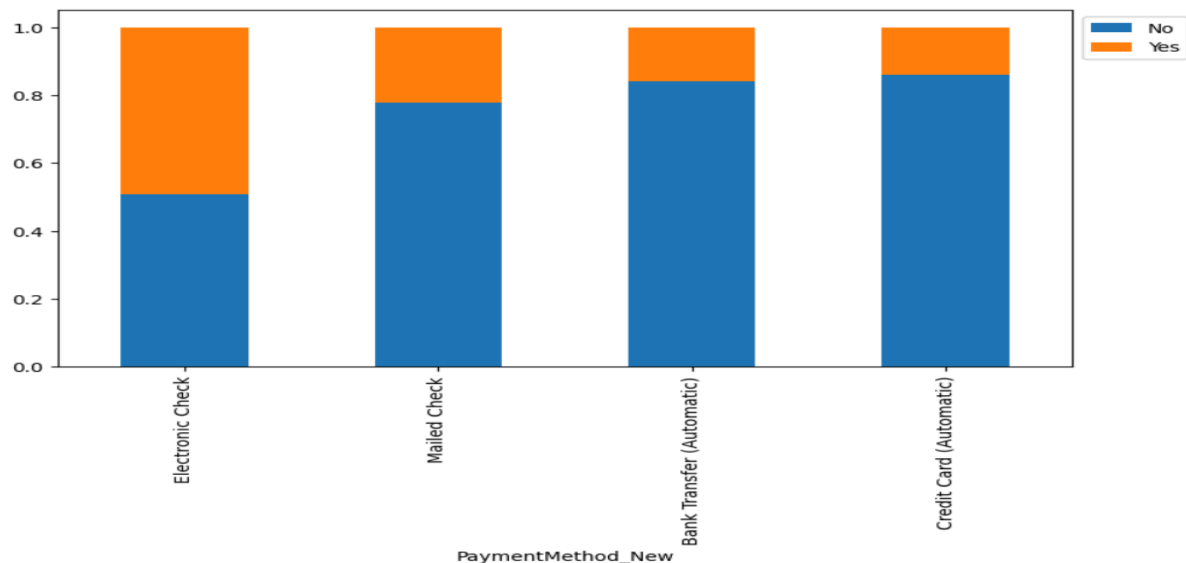


Fig 8: Observation on Customer Churn and PaymentMethod_New

- ❖ A significantly higher proportion of customers using **Electronic Checks** are **churning**(~49%) compared to other payment methods.

Business Insight: Payment friction or lack of automation may contribute to dissatisfaction and increased churn.

Observations on Customer Churn and Numerical Variables :

Correlation matrix

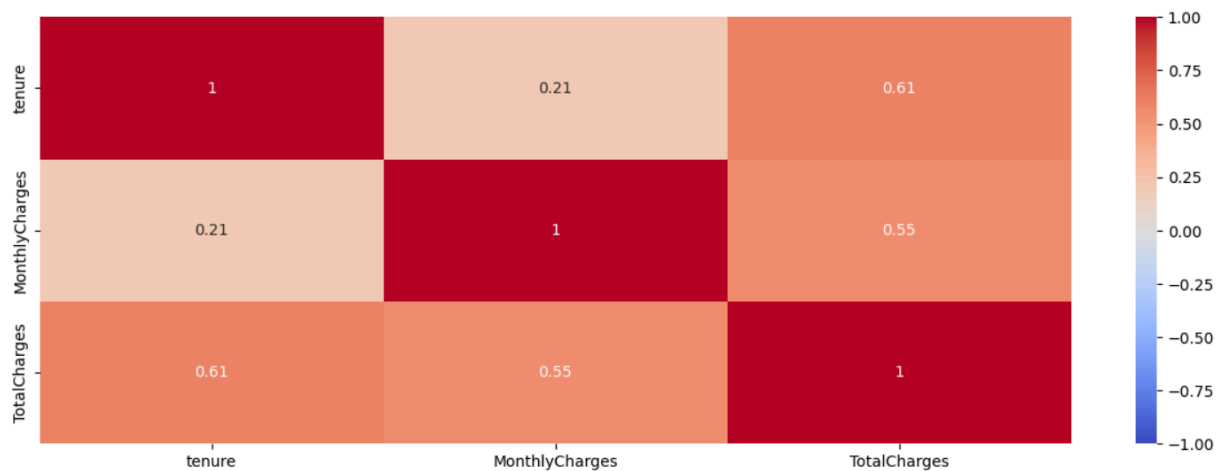


Fig 9: Correlation matrix of Numerical variables

- ❖ **Moderate positive** correlation between **Total charges** and **tenure**, indicating that as a customer's tenure increases, their total accumulated charges tend to increase significantly.
- ❖ **Moderate positive** correlation between **Total charges** and **Monthly charges**, meaning that customers with higher monthly bills also tend to have higher total charges, as expected.
- ❖ **Monthly charges** and **tenure** have no significant correlation between them.

Business Insight:

- Customers who churn early contribute significantly less to lifetime revenue. Retention efforts in the early tenure period can substantially increase customer lifetime value.
- High-priced plans may drive early dissatisfaction if customers do not perceive adequate value, leading to early churn.
- Long-term customers are not necessarily paying higher monthly fees, suggesting price-sensitive customers tend to stay longer.

Customer Churn vs tenure

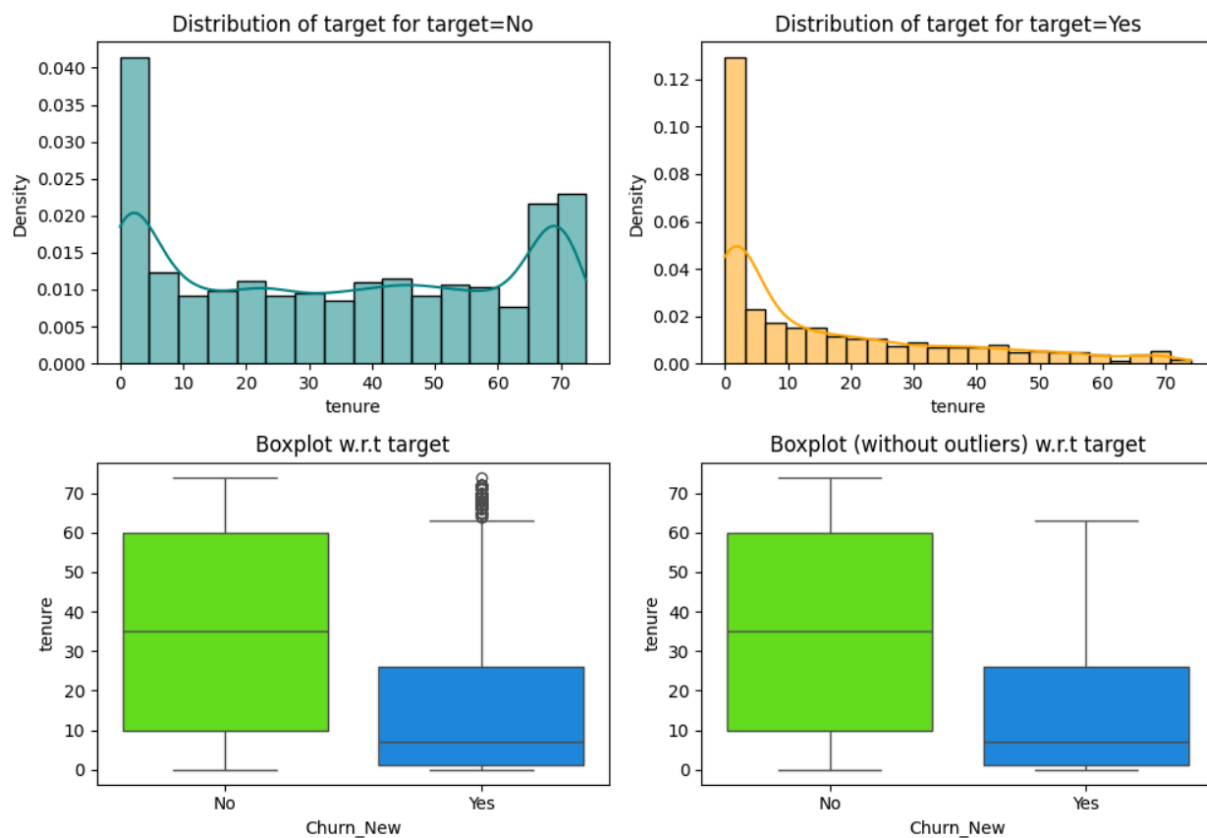


Fig 10: Observation on Customer Churn and tenure

- ❖ A significant portion of churn occurs among newer customers with very short tenures (typically within the first few months)
- ❖ The boxplot for churned customers typically shows a concentration at lower tenure values. However, there are outliers extending to **higher tenure values**.

Customer Churn vs Total Charges

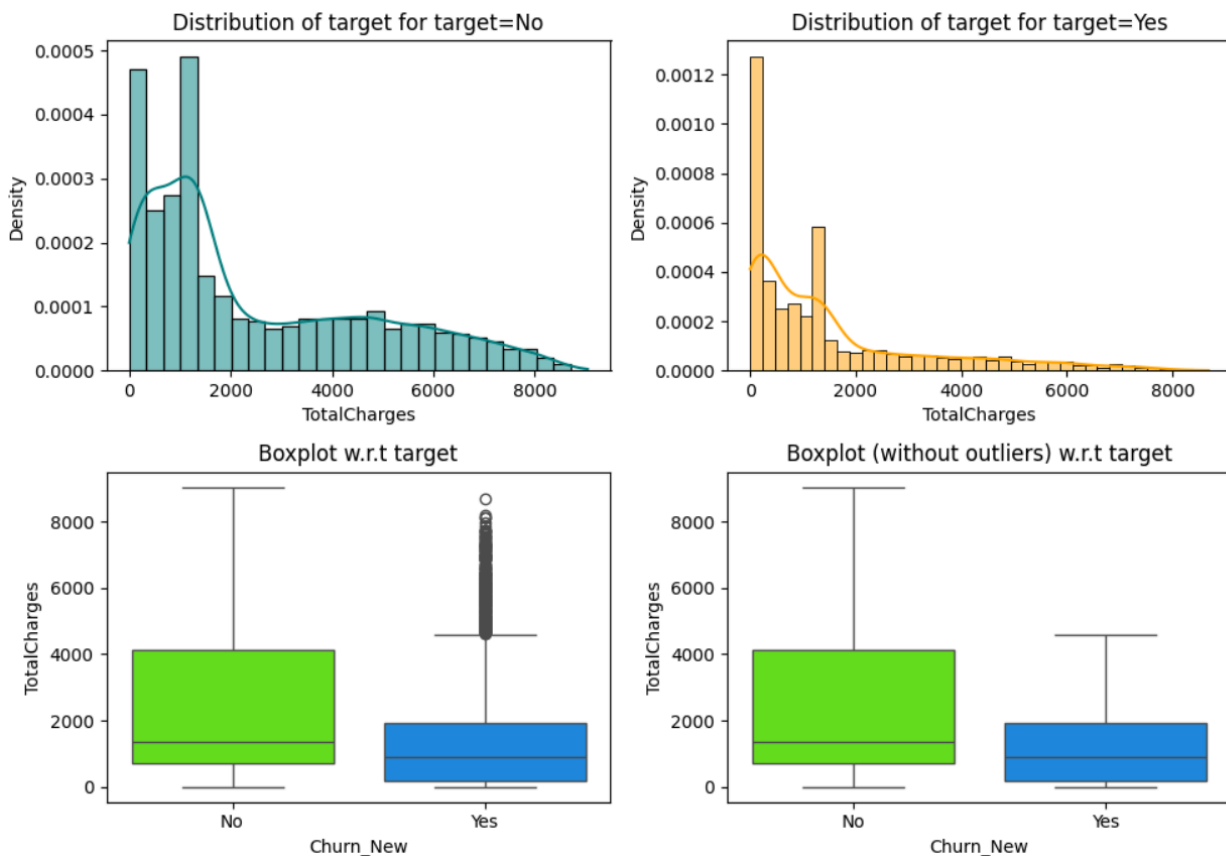


Fig 11: Observation on Customer Churn and TotalCharges

- ❖ The distribution for churn customers is heavily skewed towards the lower end of total charges. The majority of the customers have total charges below approximately 2000.
- ❖ This shows non-churned customers are valuable subscribers who are less likely to leave.
- ❖ There are outliers in boxplots of churned customers (Yes).

Customer Churn vs Monthly Charges

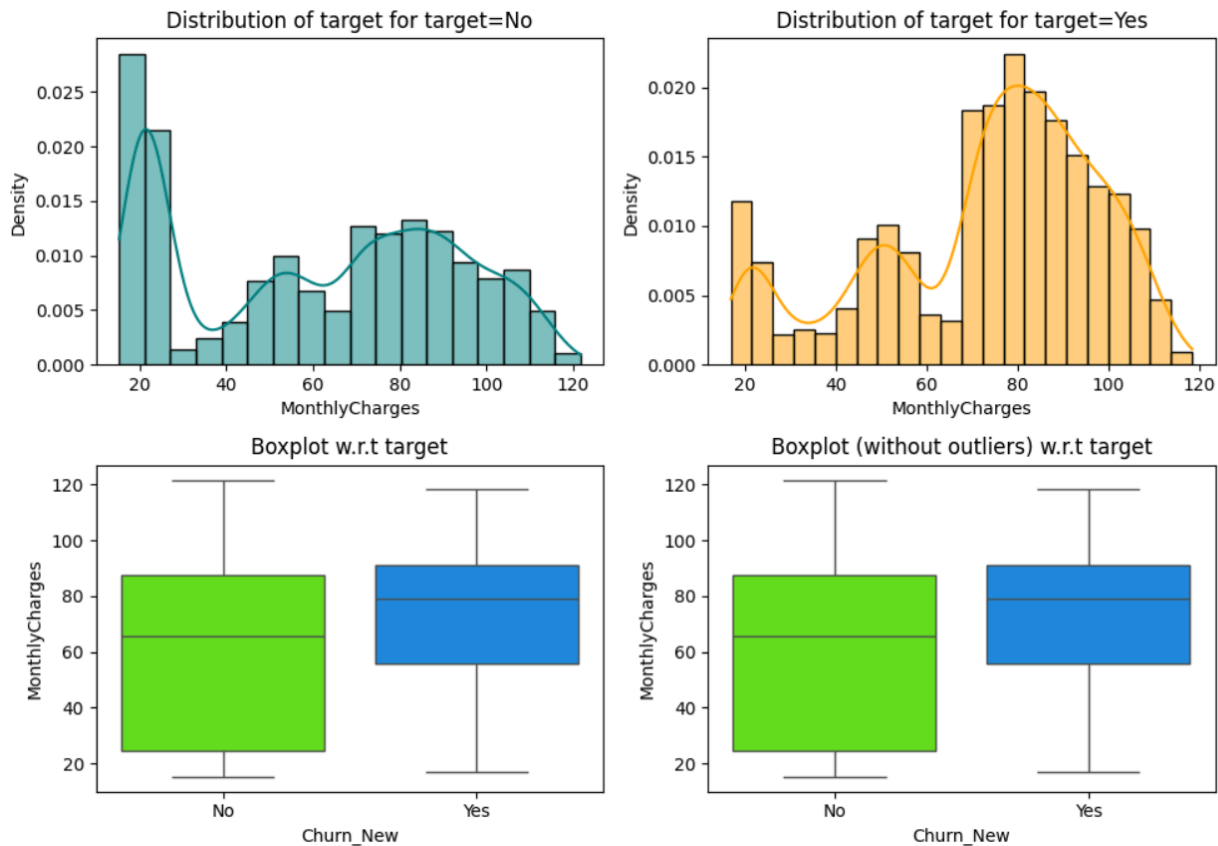


Fig 12: Observation on Customer Churn and MonthlyCharges

- ❖ The median and average monthly charges for churned customers are notably higher than for non-churned customers.
- ❖ Customers paying monthly charges between approximately \$70 and the maximum of around \$120 are more likely to churn.
- ❖ There are several outliers in boxplots of both class distributions.

Let's see churn rates by contract

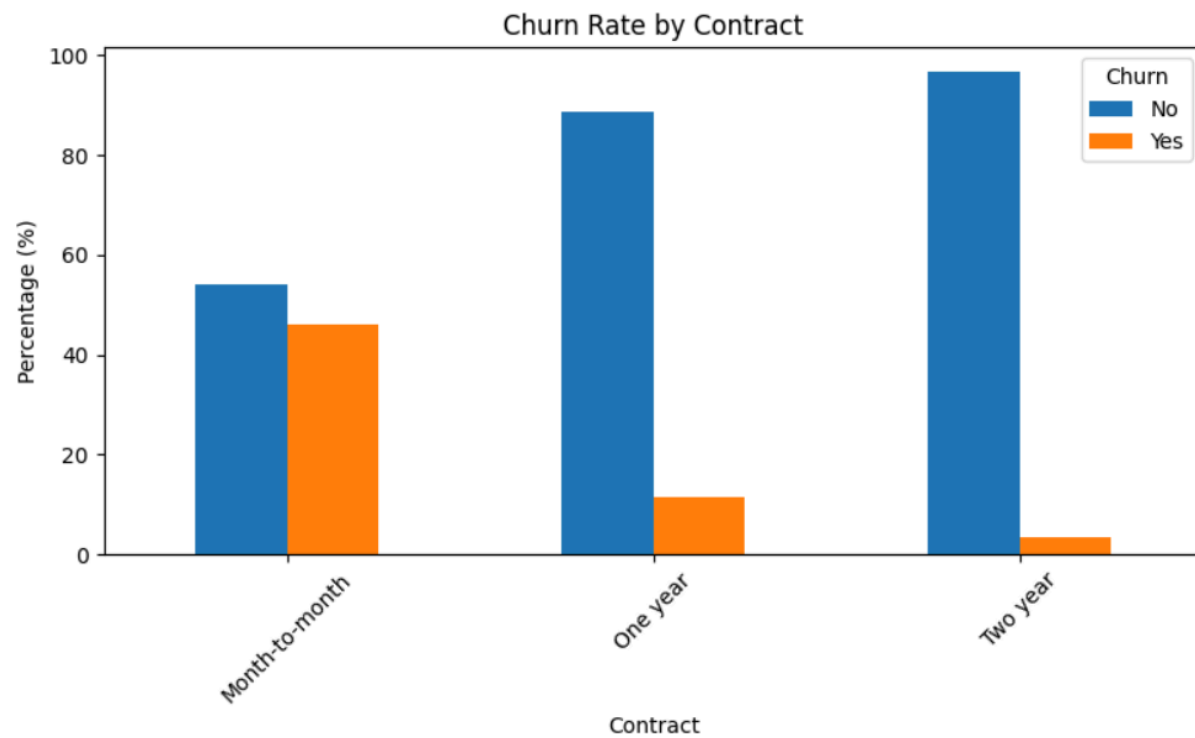


Fig 13: Observation on Churn Rate by Contract

- ❖ Longer contract durations effectively increase customer retention but in month-to-month contracts customers are more likely to leave due to dissatisfaction.

Multivariate Analysis

Pairplot for Numerical features and Customer Churn

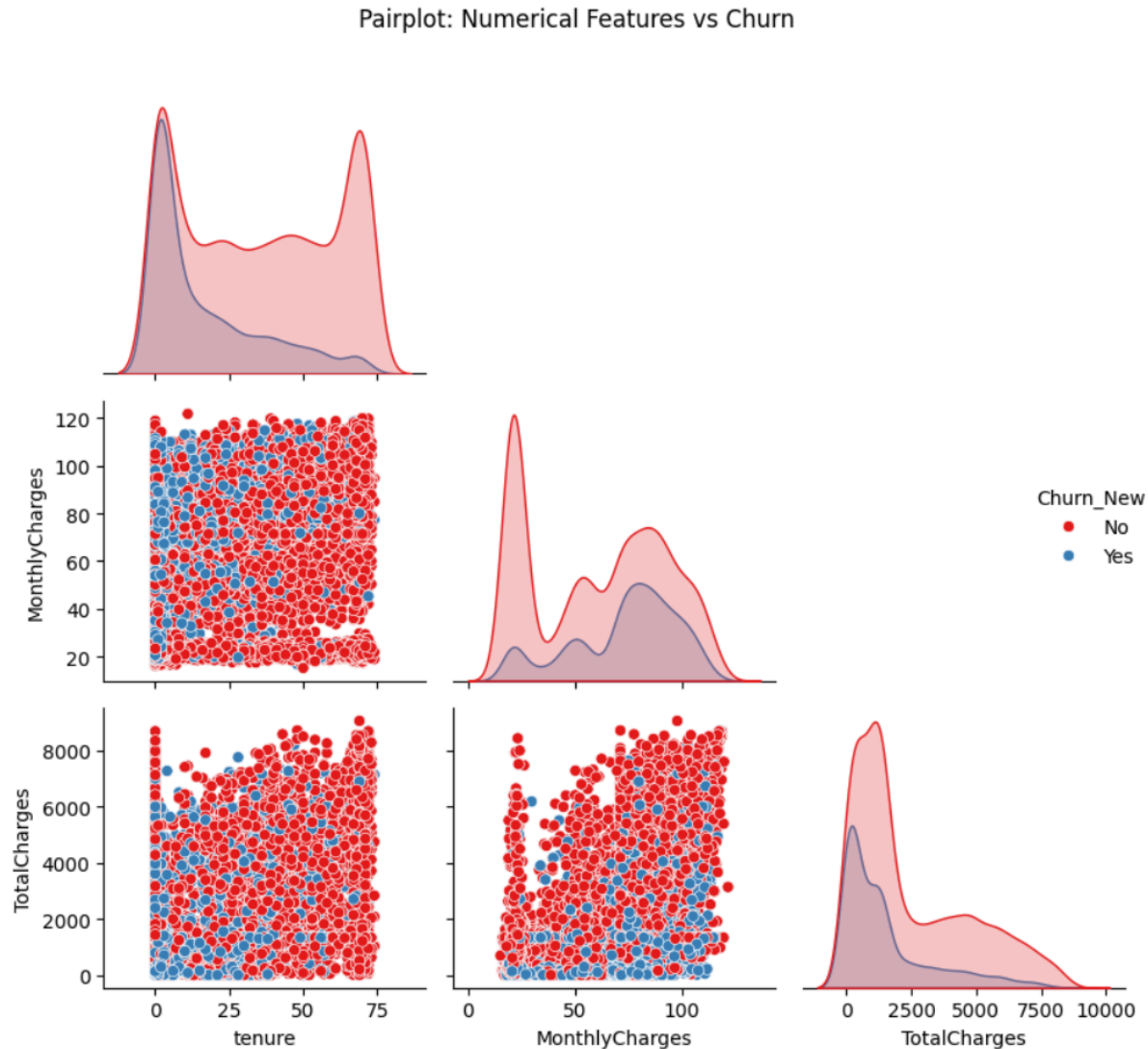


Fig 14: Pairplot for Numerical features and Customer Churn

- ❖ Churned customers are mostly visible at **low tenure across all monthly charge levels**.
- ❖ Churned customers are more densely distributed in the **higher MonthlyCharges range**, suggesting that **pricing sensitivity** plays a major role in customer attrition.

- ❖ Churn is highly concentrated among customers with **low TotalCharges**, which aligns with the fact that many churn early before accumulating large lifetime value.

Business Insight: High pricing without long-term engagement increases churn risk, emphasizing the need for value justification and loyalty incentives.

Let's visualize various combinations to find highest churn percentages

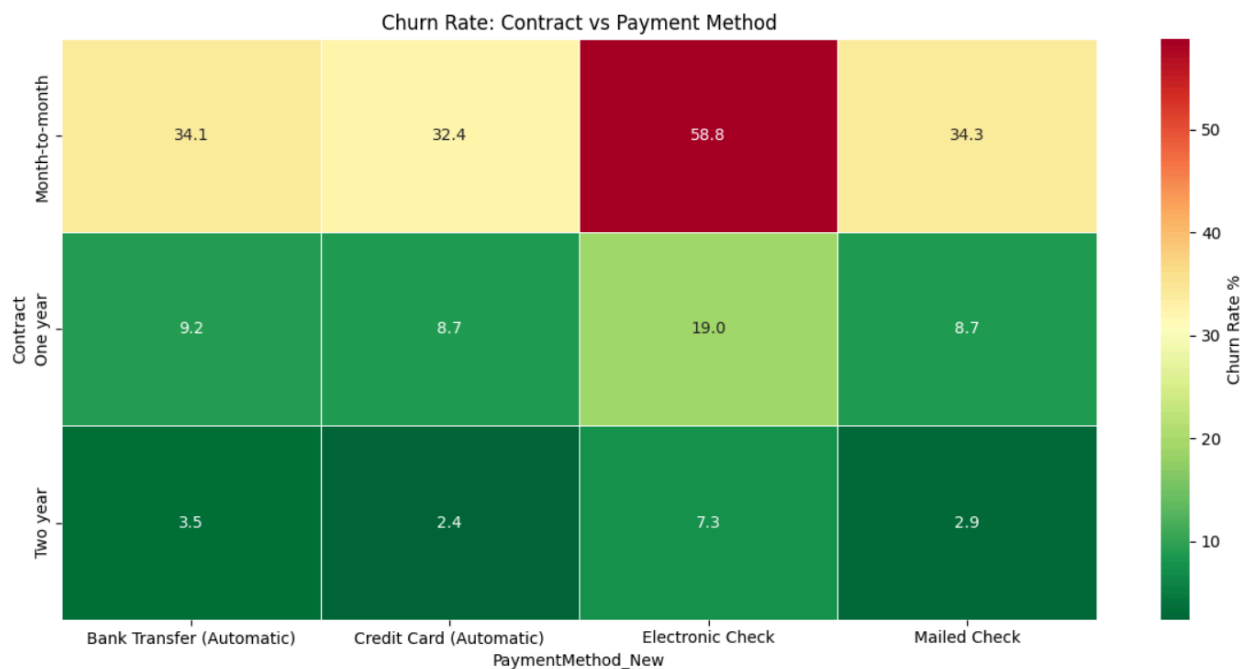


Fig 15: Heatmap of Churn Rate for Contract and Payment Method

- ❖ The **highest churn rate** is observed for **month-to-month contracts** using **Electronic Check payments(58.8%)**.
- ❖ The **lowest churn rate** is observed for **Two-year contracts** using **Credit Card(Automatic payments)(2.4%)**.

Using multiple feature combinations to find highest churn risk :

- ❖ The analysis identifies that **Contract : month-to-month, InternetService : Fibre optic, TechSupport : No, OnlineSecurity : No, OnlineBackup : No** are the **highest risk customer segments** with a significant **churn rate of 68.10%**.

Outlier Detection and Treatment :

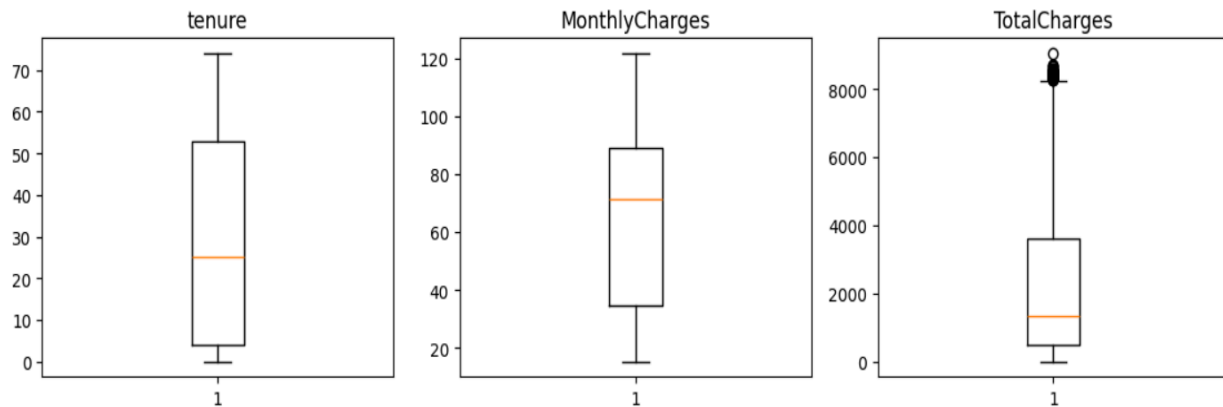


Fig 16: Outliers in the dataset

- ❖ Although outliers exist so we will keep them as they may have a valuable input.

Methodology : Numerical features will be thoroughly examined for outliers using visual tools like boxplots and statistical measures. A decision will be made on whether to modify these extreme values, transform the data, or retain them, based on their severity and potential impact on business insights and model accuracy.

Feature Engineering

- The `tenure_group` feature has been successfully created. Here we store continuous variables of tenure into four groups and they are 'New (0-12)', 'Medium (13-36)', 'Long (37-60)', 'Very Long (60+)'.

Methodology : The 'tenure' column, representing customer loyalty in months, was categorized into several distinct groups: 'New' (0-12 months), 'Medium' (13-36 months), 'Long' (37-60 months), and 'Very Long' (over 60 months). This new categorical feature, `tenure_group`, was derived from these defined ranges.

Data preparation for Modeling :

- We want to predict which customers will churn.

- Before we proceed to build a model, we will have to encode categorical features.
- We will split the data into three parts: Train, Test and Validation to be able to evaluate the model that we build on the Training data.

Methodology : The final set of features will be prepared, including the one-hot encoding for all categorical variables, converting them into a numerical format suitable for machine learning models. The data will then be divided into training, testing and validation subsets, ensuring that all subsequent data transformations are applied consistently across these subsets to avoid any inadvertent data leakage.

Model Building

Note : We have focused on important features of model building aspects so the detailed Model Building can be found in the **APPENDIX Section**.

Model evaluation criterion

A Customer Churn prediction model can make incorrect predictions in the following ways:

- **False Positive** : The model predicts non-churner as a churner. This may lead to unnecessary retention efforts or offers being extended to customers who would not have churned.
- **False Negative** : The model predicts a churner as a non-churner. In this case a customer who is likely to leave is not identified in the right time that could result in a potential revenue loss.

Which Loss is greater?

- **False negatives** are more costly for the company because they represent customers who actually churn but were not flagged by the model.

How to reduce this loss i.e need to reduce False Negatives?

- The company wants **recall** to be maximized i.e. we need to reduce the number of false negatives.

Logistic Regression :

A constant term is added to the independent variable matrix to account for the intercept in the linear regression model.

The logistic regression(logit function) model helps predict whether the customer will be churned or not because it can model the non-linear relationship between predictors and the probability of conversion.

We use the statsmodel library for building the logistic regression model. The statsmodel library for logistic regression offers detailed statistical inferences and diagnostics, aiding in hypothesis testing and interpretation, while sklearn focuses more on predictive modeling without the same level of statistical analysis.

Here is what the model summary looks like after the model is built.

Logit Regression Results						
Dep. Variable:	Churn_New	No. Observations:	8419			
Model:	Logit	Df Residuals:	8386			
Method:	MLE	Df Model:	32			
Date:	Thu, 29 Jan 2026	Pseudo R-squ.:	0.2922			
Time:	08:06:00	Log-Likelihood:	-3546.0			
converged:	True	LL-Null:	-5009.6			
Covariance Type:	nonrobust	LLR p-value:	0.000			
	coef	std err	z	P> z	[0.025	0.975]
const	-0.5628	0.241	-2.339	0.019	-1.034	-0.091
MonthlyCharges	0.0068	0.003	2.594	0.009	0.002	0.012
TotalCharges	-0.0001	2.33e-05	-5.835	0.000	-0.000	-9.03e-05
gender_Male	0.0710	0.060	1.191	0.234	-0.046	0.188
SeniorCitizen_Yes	0.1997	0.088	2.269	0.023	0.027	0.372
Partner_Yes	-0.0947	0.070	-1.358	0.175	-0.231	0.042
Dependents_Yes	-0.2060	0.081	-2.553	0.011	-0.364	-0.048
PhoneService_Yes	-0.0921	0.184	-0.501	0.617	-0.453	0.269
MultipleLines_No phone service	0.0046	0.187	0.025	0.980	-0.362	0.371
MultipleLines_Yes	0.2881	0.072	3.990	0.000	0.147	0.430
InternetService_Fiber optic	0.6802	0.094	7.201	0.000	0.495	0.865
InternetService_No	0.7650	0.585	1.308	0.191	-0.381	1.911
OnlineSecurity_No internet service	0.3288	0.608	0.541	0.589	-0.863	1.521
OnlineSecurity_Yes	-0.4714	0.082	-5.751	0.000	-0.632	-0.311
OnlineBackup_No internet service	-0.5468	0.563	-0.972	0.331	-1.649	0.556
OnlineBackup_Yes	-0.3262	0.073	-4.440	0.000	-0.470	-0.182
DeviceProtection_No internet service	0.3982	0.555	0.717	0.473	-0.690	1.487
DeviceProtection_Yes	-0.0917	0.069	-1.337	0.181	-0.226	0.043
TechSupport_No internet service	0.1183	0.583	0.203	0.839	-1.025	1.261
TechSupport_Yes	-0.5367	0.086	-6.252	0.000	-0.705	-0.368
StreamingTV_No internet service	-0.7608	0.675	-1.127	0.260	-2.083	0.562
StreamingTV_Yes	0.0796	0.078	1.018	0.309	-0.074	0.233
StreamingMovies_No internet service	-1.2389	0.681	-1.818	0.069	-2.575	0.097
StreamingMovies_Yes	-0.0153	0.077	-0.198	0.843	-0.167	0.136

Table 5: Logistic Regression Baseline Model

We have created a baseline model. And now we checked for high multicollinearity and p-value(>0.05).

Logistic Regression Model Performance Improvement

Variance Inflation Factor :

We have checked VIF for multicollinearity and found most variables had relatively low VIF values while some variables showed higher VIF. Despite the high VIF for these specific features, we decided to **retain them**. The reasoning was that these are crucial 'business

drivers' for customer churn. Removing them, even if statistically correlated, would lead to a loss of valuable business insights and model interpretability, which was prioritized in this analytical context.

P-value Based Feature Selection : After the initial VIF assessment (and the decision to keep business-critical, high-VIF features), we did a backward elimination strategy based on p-values from the logistic regression model. And this process resulted in a refined set of **selected_features** that are statistically significant predictors of churn at the 5% significance level, after accounting for their relationships with each other.

Final Model Selection :

Logit Regression Results						
Dep. Variable:	Churn_New	No. Observations:	8419			
Model:	Logit	Df Residuals:	8401			
Method:	MLE	Df Model:	17			
Date:	Thu, 29 Jan 2026	Pseudo R-squ.:	0.2907			
Time:	08:06:03	Log-Likelihood:	-3553.2			
converged:	True	LL-Null:	-5009.6			
Covariance Type:	nonrobust	LLR p-value:	0.000			
	coef	std err	z	P> z	[0.025	0.975]
const	-0.5365	0.133	-4.040	0.000	-0.797	-0.276
MonthlyCharges	0.0063	0.002	2.808	0.005	0.002	0.011
TotalCharges	-0.0001	2.26e-05	-6.197	0.000	-0.000	-9.59e-05
SeniorCitizen_Yes	0.1911	0.087	2.190	0.029	0.020	0.362
Dependents_Yes	-0.2454	0.076	-3.241	0.001	-0.394	-0.097
MultipleLines_Yes	0.2738	0.071	3.870	0.000	0.135	0.412
InternetService_Fiber optic	0.6583	0.091	7.203	0.000	0.479	0.837
OnlineSecurity_Yes	-0.4793	0.082	-5.873	0.000	-0.639	-0.319
OnlineBackup_Yes	-0.3242	0.073	-4.433	0.000	-0.468	-0.181
TechSupport_Yes	-0.5403	0.085	-6.335	0.000	-0.708	-0.373
StreamingMovies_No internet service	-0.9418	0.127	-7.426	0.000	-1.190	-0.693
Contract_One year	-0.8269	0.092	-8.980	0.000	-1.007	-0.646
Contract_Two year	-1.6062	0.146	-10.982	0.000	-1.893	-1.320
PaperlessBilling_Yes	0.3056	0.064	4.792	0.000	0.181	0.431
PaymentMethod_New_Electronic Check	0.4369	0.065	6.748	0.000	0.310	0.564
tenure_group_Medium (13-36)	-0.7806	0.075	-10.350	0.000	-0.928	-0.633
tenure_group_Long (37-60)	-0.9157	0.099	-9.285	0.000	-1.109	-0.722
tenure_group_Very Long (60+)	-1.1322	0.152	-7.471	0.000	-1.429	-0.835

Table 6: Logistic Regression Final Model

In this model no feature has p-value > 0.05, so we will consider this as our final model.

Checking Model Performance on the training set :

Confusion Matrix

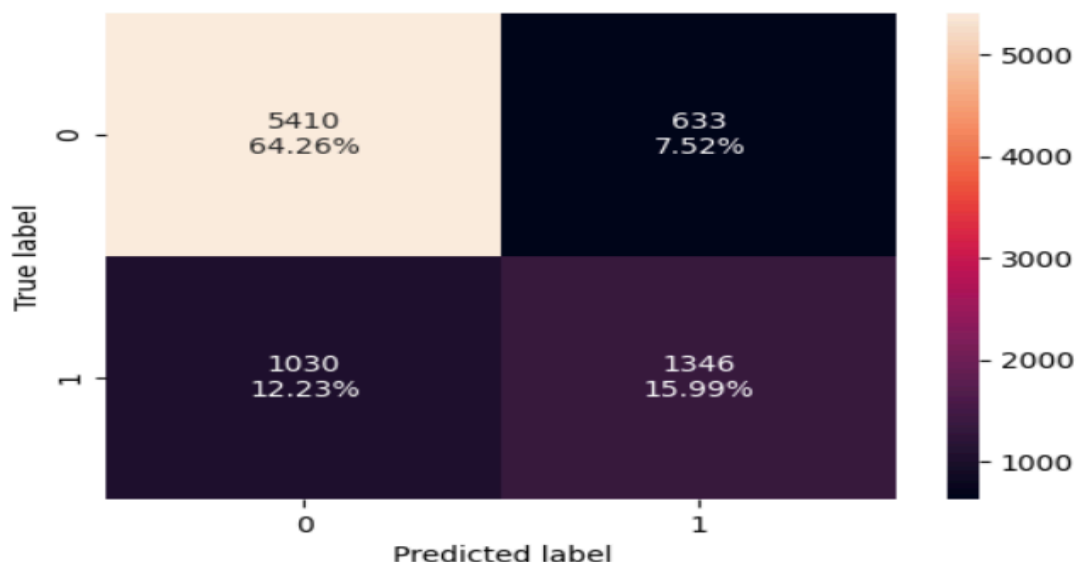


Fig 17: Confusion matrix for Logistic Regression Training data

Training Performance

	Accuracy	Recall	Precision	F1
0	0.802	0.566	0.680	0.618

Table 7: Performance Metrics for Logistic Regression Training data

After training, the model shows a good overall accuracy of 80%. The model is less effective at identifying all actual positive instances (Recall), capturing only 56% of them. The precision score is 0.68, indicating a moderate rate of false positives. The F1 score is 0.61, suggesting that the model has reasonable overall accuracy.

- **AUC-ROC Curve** : We plotted the Receiver Operating Characteristic (ROC) curve and identified an optimal threshold (around **0.34**) where the difference between True Positive Rate (TPR) and False Positive Rate (FPR) is maximized.
- **Precision-Recall Curve** : We also plotted the Precision-Recall curve, noting that a threshold around **0.40** would provide a good balance or higher recall depending on the specific trade-off desired.

Checking Model Performance on Validation set :

Confusion Matrix

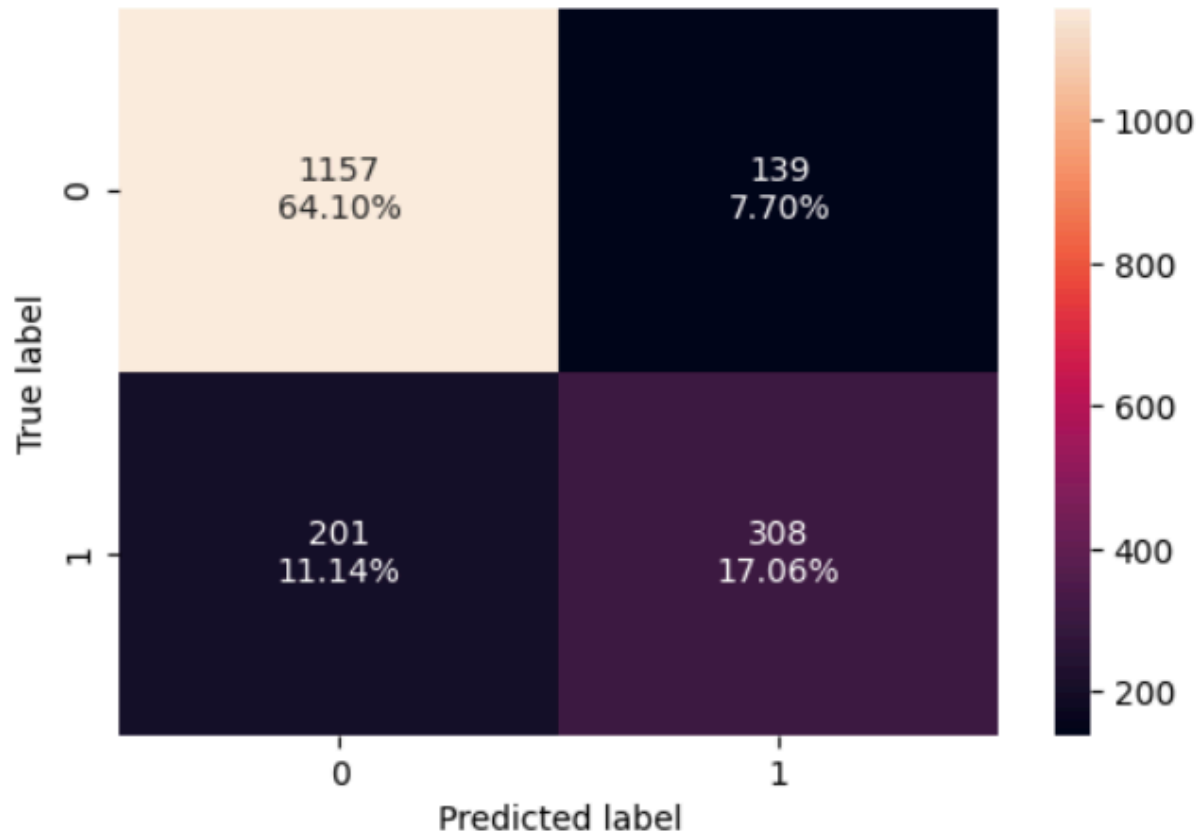


Fig 18: Confusion matrix for Logistic Regression validation data

Validation performance:

	Accuracy	Recall	Precision	F1
0	0.812	0.605	0.689	0.644

Table 8: Performance Metrics for Logistic Regression validation data

The model achieved an accuracy of approximately 81%. The recall for the churn class is about 60.5%. This means the model correctly identified nearly 60% of all actual churners. The precision for the churn class is approximately **68.9%**. This indicates that about 67% of the customers predicted as churners by the model actually did churn. The F1-score, which is the harmonic mean of precision and recall, is approximately **64%**. Overall, the model shows reasonable performance, but there's a trade-off between identifying churners (recall) and the certainty of those predictions (precision).

Model Performance Comparison :

We systematically compared the model's performance on both the training and validation sets using:

- The **default threshold of 0.5**.
- The **0.34 threshold** derived from the AUC-ROC curve.
- The **0.40 threshold** derived from the Precision-Recall curve.

Training performance comparison:

	Logistic Regression statsmodel	Logistic Regression-0.34 Threshold	Logistic Regression-0.40 Threshold
Accuracy	0.802	0.784	0.795
Recall	0.566	0.746	0.702
Precision	0.680	0.593	0.620
F1	0.618	0.661	0.659

Table 9: Training Performance Comparison

Validation set performance comparison:

	Logistic Regression statsmodel	Logistic Regression-0.34 Threshold	Logistic Regression-0.40 Threshold
Accuracy	0.812	0.780	0.797
Recall	0.605	0.764	0.715
Precision	0.689	0.584	0.622
F1	0.644	0.662	0.665

Table 10: Validation Performance Comparison

Observations: Based on the performance results, especially the recall values, the Logistic Regression model (with 0.34 Threshold) emerged as the most suitable choice. While it slightly reduced accuracy and precision compared to the default, it significantly increased the model's ability to correctly identify actual churners.

Limitations of Baseline model : Though Logistic Regression model is a good model, particularly in the context of the business objective (maximizing recall to reduce false negatives). Discuss where the current model might fall short, such as in capturing complex relationships or handling data imbalances effectively without explicit techniques.

We have to build advanced models (like Decision Trees, Random Forests, Gradient Boosting, AdaBoost) which can potentially offer better performance metrics, especially higher recall, by learning more complex patterns, reducing bias, and handling non-linear interactions in the data that simpler models might miss.

Advanced Model Building :

In the advanced model building process, we will build the Ensemble models listed below with the original data.

- Decision Tree Classifier
- Bagging Classifier
- Random Forest Classifier
- AdaBoost Classifier
- Gradient Boosting Classifier

After these models are built, we will assess their performance by analyzing their recall scores on the original data as we want the recall score to be maximized.

Methodology: After establishing a baseline with Logistic Regression, we proceeded to explore more complex, tree-based ensemble models to capture non-linear relationships and potentially improve churn prediction, particularly recall.

Decision Tree Classifier:

A Decision Tree Classifier is a supervised machine learning algorithm used for classification problems. It works by splitting the dataset into smaller subsets based on feature values, forming a tree-like structure of decisions. Each internal node represents a condition on a feature, each branch represents the outcome of that condition, and each leaf node represents a final class label.

One of the main advantages of Decision Trees is their interpretability. The decision-making process is easy to understand and visualize, making the model highly suitable for business applications where explainability is important. Additionally, Decision Trees can handle both numerical and categorical data and do not require feature scaling.

The performance of the Decision Tree model is evaluated below using multiple metrics, including the metric of choice (recall).

Training Performance :

	Accuracy	Recall	Precision	F1
0	0.999	0.997	1.000	0.999

Table 11: Decision Tree Classifier Training Performance

Validation Performance :

	Accuracy	Recall	Precision	F1
0	0.731	0.550	0.521	0.535

Table 12: Decision Tree Classifier Validation Performance

Observations : The Decision Tree model showed excellent performance on the training data but a noticeable drop on the validation data. This indicates overfitting, where the model learns the training data too closely and fails to generalize. Although recall improved compared to the baseline model, its unstable validation performance limits its suitability for deployment.

Methodology : The Decision Tree model was implemented as a baseline approach. It creates rule-based splits in the data that are easy to interpret and explain. While the model performs well on training data, it has a higher tendency to overfit, making it less reliable when applied to new data.

Bagging Classifier :

A Bagging Classifier (Bootstrap Aggregating) is an ensemble learning technique used to improve the performance and stability of machine learning models. It works by creating multiple subsets of the training data using bootstrap sampling (random sampling with replacement) and training a separate base model on each subset.

Bagging is particularly effective when the base learners are unstable, meaning small changes in the training data can lead to significantly different models. Since each model is trained independently, bagging also supports parallel processing, making it computationally efficient.

The performance of the Bagging model is evaluated below using multiple metrics, including the metric of choice (recall).

Training Performance:

	Accuracy	Recall	Precision	F1
0	0.982	0.944	0.992	0.967

Table 13: Bagging Classifier Training Performance

Validation Performance:

	Accuracy	Recall	Precision	F1
0	0.796	0.527	0.677	0.592

Table 14: Bagging Classifier Validation Performance

Observations : The Bagging model reduced the overfitting observed in the Decision Tree by aggregating multiple trees. Validation performance improved, particularly in recall and F1-score. However, while more stable, the model still lagged behind advanced ensemble methods.

Methodology : The Bagging approach was applied to reduce the instability of a single Decision Tree. Multiple models were trained on different subsets of the data, and their predictions were combined. This process improves consistency and reduces the risk of overfitting.

Random Forest Classifier:

Random Forest is an ensemble learning algorithm based on the bagging (Bootstrap Aggregating) technique. It builds multiple decision trees using different bootstrapped samples of the training data and combines their predictions using majority voting for classification problems.

Random Forest is robust to noise, handles both numerical and categorical variables effectively, and performs well even when the dataset contains non-linear relationships. It also provides feature importance scores, which help in understanding which variables have the most influence on predictions—an important advantage for business decision-making.

The performance of the Bagging model is evaluated below using multiple metrics, including the metric of choice (recall).

Training Performance :

	Accuracy	Recall	Precision	F1
0	0.999	0.999	0.998	0.999

Table 15: Random Forest Classifier Training Performance

Validation Performance :

	Accuracy	Recall	Precision	F1
0	0.803	0.566	0.681	0.618

Table 16: Random Forest Classifier Validation Performance

Observations : The Random Forest model delivered strong and consistent performance across all evaluation metrics. It significantly improved recall and F1-score

while maintaining good accuracy. Overfitting was minimal, making it a reliable and robust model for business use cases.

Methodology : Random Forest builds on the Bagging approach by introducing additional randomness in feature selection. This reduces correlation between models and improves generalization. The method delivers strong and stable performance across multiple evaluation metrics.

AdaBoost Classifier :

AdaBoost (Adaptive Boosting) is an ensemble learning algorithm that combines multiple weak learners to create a strong predictive model. It works by training models sequentially, where each new model focuses more on the observations that were misclassified by previous models.

AdaBoost is particularly effective for problems where simple models struggle to capture complex patterns. It often achieves higher accuracy than bagging-based methods but is more sensitive to outliers and noisy data, as misclassified points receive increasing importance.

The performance of the Bagging model is evaluated below using multiple metrics, including the metric of choice (recall).

Training Performance :

	Accuracy	Recall	Precision	F1
0	0.798	0.551	0.674	0.606

Table 17: AdaBoost Classifier Training Performance

Validation Performance :

	Accuracy	Recall	Precision	F1
0	0.811	0.576	0.701	0.632

Table 18: AdaBoost Classifier Validation Performance

Observations : AdaBoost demonstrated improved recall by focusing on difficult-to-classify observations. While precision remained strong, performance was sensitive to noisy data. Overall, it performed better than basic tree models but slightly below Random Forest and Gradient Boosting in terms of stability.

Methodology : AdaBoost was implemented to improve model performance by focusing on observations that were difficult to classify. Each iteration gives higher importance to

previously misclassified cases, allowing the model to improve recall and capture subtle patterns.

Gradient Boosting Classifier :

Gradient Boosting Classifier is an ensemble learning algorithm that builds models in a sequential and additive manner, where each new model corrects the errors made by the previous ones. Unlike bagging methods, gradient boosting focuses on reducing both bias and variance by optimizing a loss function using gradient descent.

Gradient Boosting is highly effective for structured and tabular datasets and often delivers **state-of-the-art performance** in classification tasks. However, it is computationally intensive, sensitive to hyperparameters, and requires careful tuning to avoid overfitting.

The performance of the Bagging model is evaluated below using multiple metrics, including the metric of choice (recall).

Training Performance :

	Accuracy	Recall	Precision	F1
0	0.815	0.591	0.707	0.644

Table 19: Gradient Boosting Classifier Training Performance

Validation Performance :

	Accuracy	Recall	Precision	F1
0	0.818	0.605	0.706	0.652

Table 20: Gradient Boosting Classifier Validation Performance

Observations : Gradient Boosting showed a strong balance between bias and variance. It achieved high recall and F1-score on the validation set, indicating strong generalization capability. The model effectively captured complex patterns in the data and performed consistently across metrics.

Methodology: Gradient Boosting follows a sequential learning approach where each new model corrects the errors of the previous one. This method is effective in capturing complex relationships and improving predictive accuracy while maintaining reasonable model stability.

Model Performance Improvement :

Hyperparameter Tuning:

Hyperparameter tuning is the process of selecting the optimal values for model hyperparameters to achieve the best possible performance. Hyperparameters are configuration settings that are defined before training the model, such as tree depth, number of estimators, or learning rate, and they directly influence how the model learns from data.

Tuning is necessary because default hyperparameter values are generic and may not be suitable for a specific dataset. Improper settings can lead to **underfitting**, where the model fails to capture patterns, or **overfitting**, where the model performs well on training data but poorly on unseen data. Hyperparameter tuning helps strike the right balance between **bias and variance**.

By optimizing hyperparameters using techniques such as **Grid Search** or **Random Search** along with cross-validation, models become more **robust, accurate, and generalizable**. This results in improved predictive performance and more reliable outputs for business decision-making.

We selected 3 models for hyperparameter tuning because of their strong potential to improve predictive performance, especially 'recall' which is crucial for identifying churners.

Methodology : For the Random Forest, AdaBoost, and Gradient Boosting Classifier models, hyperparameter tuning was performed to address overfitting and optimize performance, particularly focusing on Recall as the primary business metric (minimizing False Negatives). The strategy involved using RandomizedSearchCV to efficiently explore the hyperparameter space.

Random Forest Classifier :

Untuned Performance Observation: Analyze its high training accuracy/recall but a noticeable drop in performance on the validation set, indicating potential overfitting.

Reason for Tuning: Elaborate on Random Forest's nature as a powerful ensemble (bagging) method that reduces variance. After tuning we find the optimal balance between model complexity and generalization, ensuring it learns true patterns rather than noise.

Training Performance :

	Accuracy	Recall	Precision	F1
0	0.753	0.810	0.542	0.649

Table 21: Tuned Random Forest Classifier Training Performance

Validation Performance :

	Accuracy	Recall	Precision	F1
0	0.753	0.817	0.541	0.651

Table 22: Tuned Random Forest Classifier Validation Performance

Results : After tuning, it reduced overfitting, improved generalization, and a higher, more stable recall on unseen data, which is crucial for identifying churners.

AdaBoost Classifier :

Untuned Performance Observation: its solid initial validation recall, often outperforming simpler models, showcasing its potential.

Reason for Tuning : AdaBoost is a boosting algorithm that sequentially builds a strong classifier by focusing on misclassified instances. Tuning is vital to control its learning rate and the number of weak learners to prevent overfitting and enhance its error-correcting capabilities.

Training Performance :

	Accuracy	Recall	Precision	F1
0	0.706	0.858	0.488	0.622

Table 23: Tuned AdaBoost Classifier Training Performance

Validation Performance :

	Accuracy	Recall	Precision	F1
0	0.717	0.878	0.499	0.637

Table 24: Tuned AdaBoost Classifier Validation Performance

Results : After tuning, the model performance improved and generalizes well on both training and validation data. Now it is more effective at identifying at-risk customers.

Gradient Boosting Classifier :

Untuned Performance Observation : The untuned Gradient Boosting model's initial performance on both training and validation sets. Highlight its inherent strength as a powerful ensemble method that often achieves good baseline performance, but there are some disparities between training and validation metrics that might indicate potential for improvement or initial signs of overfitting.

Reason for Tuning : Explain that Gradient Boosting, while highly effective, is sequential and prone to overfitting if parameters are not carefully controlled. Tuning is essential to find the optimal balance among parameters like `n_estimators`, `learning_rate`, `max_depth`, `min_samples_leaf`, and `min_samples_split`. This process aims to prevent the model from memorizing the training data, improve its generalization capability, and significantly enhance its ability to identify the minority class (churners) by leveraging `class_weight='balanced'` during fitting.

Training Performance :

	Accuracy	Recall	Precision	F1
0	0.781	0.830	0.578	0.682

Table 25: Tuned Gradient Boosting Classifier Training Performance

Validation Performance :

	Accuracy	Recall	Precision	F1
0	0.768	0.817	0.561	0.666

Table 26: Tuned Gradient Boosting Classifier Validation Performance

Results : After tuning, the model performance improved, showing better generalization on both training and validation data. Emphasize that the tuned model is now more

effective at identifying at-risk customers, especially due to improvements in recall, and discuss the overall balance achieved across various metrics.

Model Comparison and Final Model Selection :

Based on the evaluation results of the improved models for customer churn prediction, it is evident that the models demonstrate enhancements in performance metrics compared to the default counterparts. We can analyze this with the help of the performance comparison data below:

Training Data :

	Logistic Regression (0.34 Threshold)	Decision Tree	Bagging Classifier	Random Forest	AdaBoost Classifier	Gradient Boosting Classifier	Random Forest Tuned	AdaBoost Tuned	Gradient Boosting Tuned
Accuracy	0.784	0.999	0.982	0.999	0.798	0.815	0.753	0.706	0.781
Recall	0.746	0.997	0.944	0.999	0.551	0.591	0.810	0.858	0.830
Precision	0.593	1.000	0.992	0.998	0.674	0.707	0.542	0.488	0.578
F1	0.661	0.999	0.967	0.999	0.606	0.644	0.649	0.622	0.682

Table 27: Model Performance Comparison on Training Data

Validation Data :

	Logistic Regression (0.34 Threshold)	Decision Tree	Bagging Classifier	Random Forest	AdaBoost Classifier	Gradient Boosting Classifier	Random Forest Tuned	AdaBoost Tuned	Gradient Boosting Tuned
Accuracy	0.780	0.731	0.796	0.803	0.811	0.818	0.753	0.717	0.768
Recall	0.764	0.550	0.527	0.566	0.576	0.605	0.817	0.878	0.817
Precision	0.584	0.521	0.677	0.681	0.701	0.706	0.541	0.499	0.561
F1	0.662	0.535	0.592	0.618	0.632	0.652	0.651	0.637	0.666

Table 28: Model Performance Comparison on Validation Data

Methodology : All models were compared using a consistent evaluation framework. Validation-set metrics were analyzed to identify trade-offs between recall, precision, and overall accuracy. Overfitting was assessed by comparing training and validation performance. The final model was selected based on recall performance, stability, and suitability for deployment.

Final Model Selection :

After conducting hyperparameter tuning across multiple models, the **Tuned Gradient Boosting model** was selected as the final model due to its **superior and well-balanced performance on the validation dataset**. This model demonstrates a strong trade-off

between recall and precision, making it particularly suitable for the business objective of accurately identifying positive cases while minimizing misclassification risks.

1. Strong Overall Performance (Highest F1 Score)

The Tuned Gradient Boosting model achieved an **F1-score of 0.666**, which is the **highest among all evaluated models**. Since the F1-score balances both precision and recall, this result indicates that the model provides the most reliable overall classification performance rather than excelling in only one metric.

2. Balanced Recall and Precision

With a **recall of 0.817** and a **precision of 0.561**, the model maintains a healthy balance between identifying actual positive cases and controlling false positives. While some models achieve higher recall, they do so at the cost of significantly lower precision. The Tuned Gradient Boosting model avoids this extreme trade-off, making its predictions more actionable from a business perspective.

3. Improvement Through Hyperparameter Tuning

Compared to the untuned Gradient Boosting model (F1-score of 0.652), hyperparameter optimization led to a **measurable improvement in F1-score**, demonstrating better control over model complexity and improved generalization on unseen data. This confirms that tuning played a critical role in enhancing the model's effectiveness.

4. Superior to Alternative Tuned Models

Although the Tuned Random Forest and Tuned AdaBoost models show higher recall values, they suffer from **lower precision and reduced accuracy**, resulting in inferior F1-scores. In contrast, the Tuned Gradient Boosting model offers a more consistent and robust performance across all evaluation metrics.

Due to the above factors, the Tuned Gradient Boosting model is a suitable choice for our needs and effectively applicable for Customer Churn prediction.

Now that we have our final model, let's analyze the performance of the model on the Test set to see how it performs on unseen data. Here are the results.

Test Performance :

	Accuracy	Recall	Precision	F1
0	0.773	0.790	0.571	0.663

Table 29: Tuned Gradient Boosting Classifier - Performance on the Test Data

For tuned Gradient Boosting model, the most important features utilized in identifying the target variable, i.e., **Churn_New**, are :

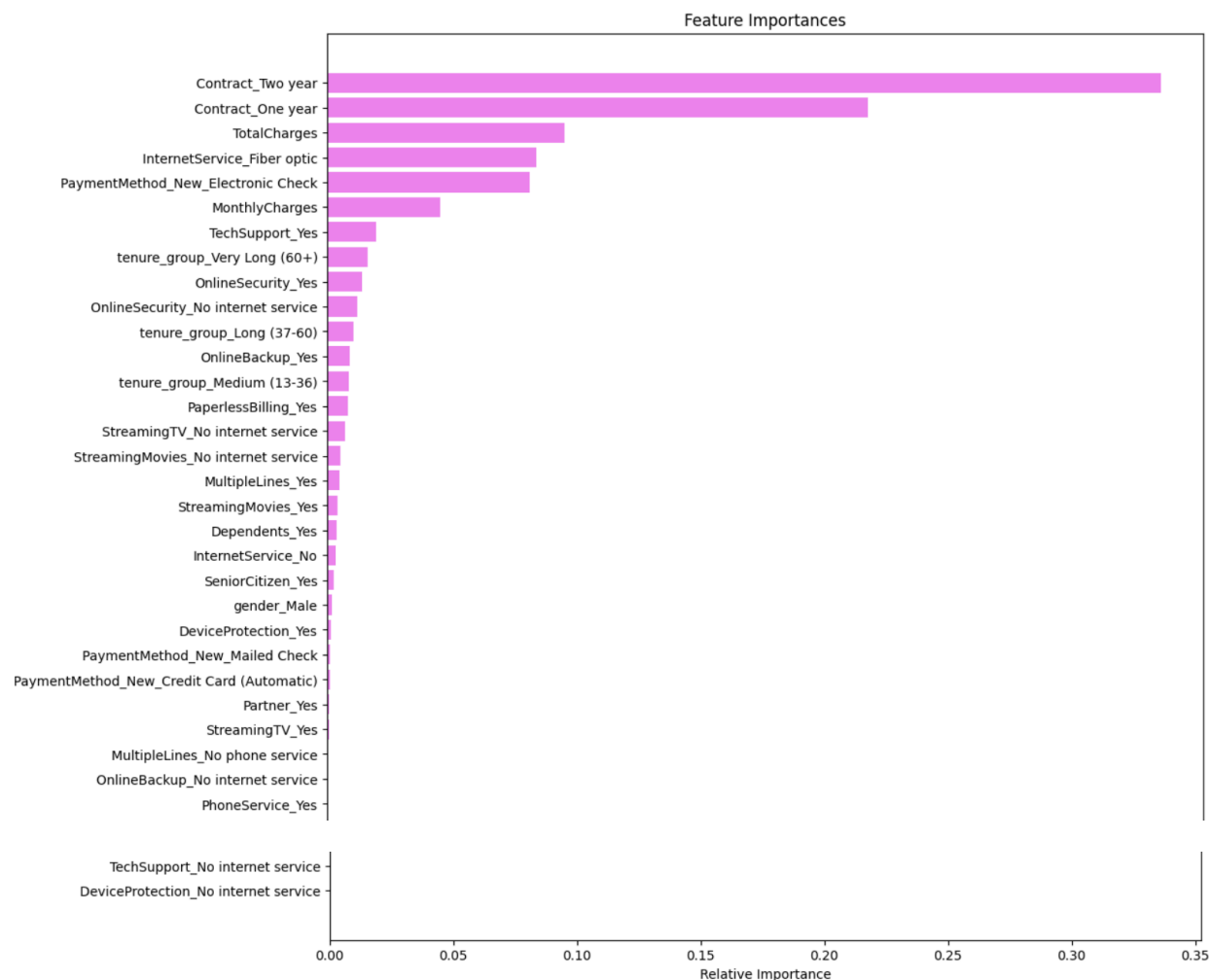


Fig 19: Feature Importance in Model Building

The top 5 attributes affecting the model's prediction are :

1. **Contract_Two Year**
2. **Contract_One Year**
3. **TotalCharges**

4. `InternetService_Fibre Optic`

5. `PaymentMethod_New_Electronic_Check`

Although other attributes may influence the model's performance, the above-mentioned attributes are more likely to be the primary deciding factors in determining whether a customer will churn or remain with the company.

Actionable Insights and Recommendations

Insights:

The analysis identifies several critical factors that drive customer churn at AlphaCom:

- **Contract Type Influence:** Contract type is a primary indicator of churn risk. Customers on month-to-month contracts exhibit significantly higher churn rates compared to those on one-year or two-year contracts.
- **Service Bundle Impact:** Lack of specialized services like Online Security, Tech Support, and Online Backup is strongly correlated with increased churn. Conversely, customers who subscribe to these services are more likely to stay.
- **Internet Service Quality:** Customers using Fiber Optic internet services show a higher propensity for churn than those using DSL. This often suggests potential issues with service quality, pricing, or perceived value for fiber customers.
- **Tenure and Loyalty:** Customer tenure is a significant predictor; the risk of churn is highest during the initial months of service. As tenure increases, customers generally become more loyal.
- **Financial Factors:** High Monthly Charges are a major driver of churn, particularly if customers do not perceive high value for the price.
- **Payment Methods:** Customers using Electronic Checks as their payment method tend to churn at a higher rate than those using automated payment methods like bank transfers or credit cards.

Actionable Business Recommendations:

To mitigate churn and improve retention, AlphaCom should implement the following strategies:

- **Incentivize Long-Term Contracts:**
 1. Offer discounts or loyalty rewards to encourage month-to-month customers to migrate to one-year or two-year contracts.

2. Introduce "early-bird" renewal benefits for customers approaching the end of their long-term contracts.
- **Promote Security and Support Bundles:**
 1. Create service bundles that include Online Security and Tech Support at a discounted rate for new customers.
 2. Target high-risk fiber optic customers with free trials of premium tech support or device protection to increase the "stickiness" of their plans.
- **Targeted Outreach for Fiber Optic Users:**
 1. Conduct targeted surveys specifically for Fiber Optic customers to identify pain points regarding service stability or pricing.
 2. Implement a "proactive outreach" program where technical teams verify line quality for fiber users in high-churn areas.
- **Focus on Early-Stage Retention:**
 1. Implement an onboarding program for the first 3–6 months that includes regular check-ins and tailored tutorials on how to maximize their service value.
 2. Offer a "loyalty bonus" (e.g., a one-time bill credit) after the first 6 months of continuous service.
- **Optimize Payment and Billing:**
 1. Encourage the adoption of Automated Payment Methods (Credit Card/Bank Transfer) by offering a small monthly discount (e.g., \$2–\$5 off)
 2. Streamline the paperless billing interface to ensure customers can easily understand their monthly charges, reducing frustration that leads to churn.

Conclusion

The analysis of AlphaCom's customer data demonstrates that churn is not random but driven by specific service gaps and contractual structures. While high monthly charges and short-term contracts are the biggest risks, the data shows that **bundling technical support and security services** significantly increases customer retention. By shifting from a reactive approach to a proactive, data-driven strategy—focusing on early-tenure engagement and incentivizing long-term commitments—AlphaCom can safeguard its revenue and significantly reduce the high cost of customer acquisition.

APPENDIX

This section details out Data Preprocessing, EDA and Model Performance of each model built for the exercise.

Data Preprocessing :

Processing Columns

Let's process tenure, MonthlyCharges and TotalCharges to extract numerical values.

```
tenure
[ 1. 34.  2. 45.  8. 22. 10. 28. 62. 13. 16. 58. 49. 25. 69. 52. 71. 21.
 nan 12. 30. 72. 17. 27.  5. 46. 11. 70. 63. 43. 15. 60. 18. 66. 47.  9.
  3. 31. 50. 64. 56.  7. 42. 35. 48. 29. 65. 38. 68. 32. 55. 37. 36. 41.
  6.  4. 33. 67. 23. 57. 61. 14. 20. 53. 40. 59. 24. 44. 19. 54. 51. 26.
  0. 39. -1. 73. -2. 74. -3.]
```

```
-----
MonthlyCharges
['$29.85' '$56.95' '$53.85' ... '$111.69' '$76.26' '$79.59']
```

```
-----
TotalCharges
['$29.85' '$1889.5' '$108.15' ... '$4849.07' '$1397.33' '$1426.71']
```

- In the tenure column, the most frequent values are 1.0(993 occurrences), 2.0(607 occurrences), and 72.0(401 occurrences). The data also contains some potentially problematic or unusual values, such as 0.0, -2.0, and -3.0. We have to treat it for further analysis.
- The TotalCharges and MonthlyCharges column contains missing values and different scales so we have to treat these before analysis.

Treat invalid and missing tenure

```
array([ 1, 34,  2, 45,  8, 22, 10, 28, 62, 13, 16, 58, 49, 25, 69, 52, 71,
        21,  0, 12, 30, 72, 17, 27,  5, 46, 11, 70, 63, 43, 15, 60, 18, 66,
        47,  9,  3, 31, 50, 64, 56,  7, 42, 35, 48, 29, 65, 38, 68, 32, 55,
        37, 36, 41,  6,  4, 33, 67, 23, 57, 61, 14, 20, 53, 40, 59, 24, 44,
        19, 54, 51, 26, 39, 73, 74])
```

- All the invalid and missing tenure have been treated.

Remove \$ and Formatting :

We first replaced common string representations of missing values like 'nan', '\$nan' and '£nan' with np.nan. This prepares the columns for numerical conversion. Then removes the '\$' and '£' symbols from the strings. And also removes any comma ' , ' characters, which are often used as thousands separators in numerical strings, to ensure proper conversion to numeric types.

Convert TotalCharges and MonthlyCharges to numeric :

We have converted all variables into numeric from the TotalCharges and MonthlyCharges column.

Handling null values in Numerical column :

We have filled null values with the median value of their respective columns.

Checking Unique values in Categorical column :

The PaymentMethod and Churn column is not structured format so we create new columns(PaymentMethod_New and Churn_New) and store them in this new column after cleaning.

Here are the result after cleaning

	count
PaymentMethod_New	
Electronic Check	4145
Credit Card (Automatic)	2930
Mailed Check	2585
Bank Transfer (Automatic)	2395

Unique values of PaymentMethod_New column

Churn_New		count
No		8650
Yes		3405

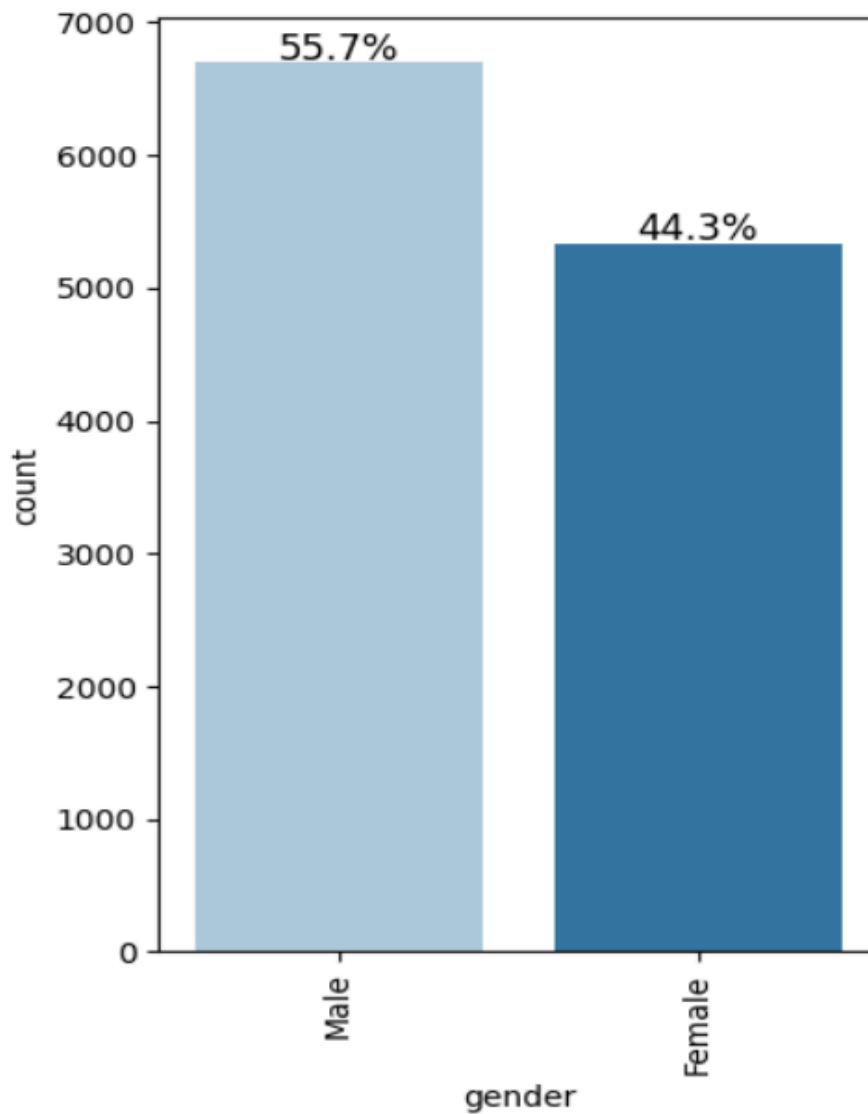
Unique values of Churn_New column

Then we map the SeniorCitizen column as 0(No) and 1(Yes).

Detailed Exploratory Data Analysis (EDA):

Univariate Analysis:

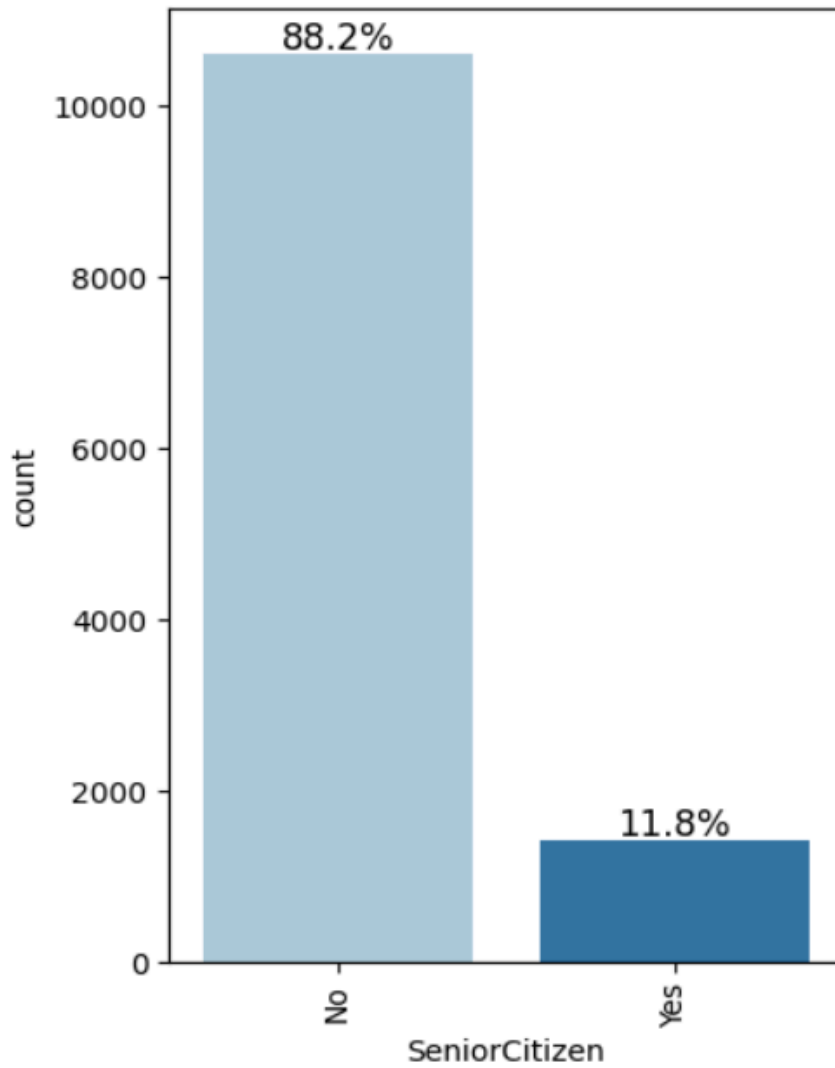
Observation on Gender :



Observation on gender

- ❖ **Male** customers constitute **55.7%** of the total count, where **Female** customers constitute **44.3%** of the total count.

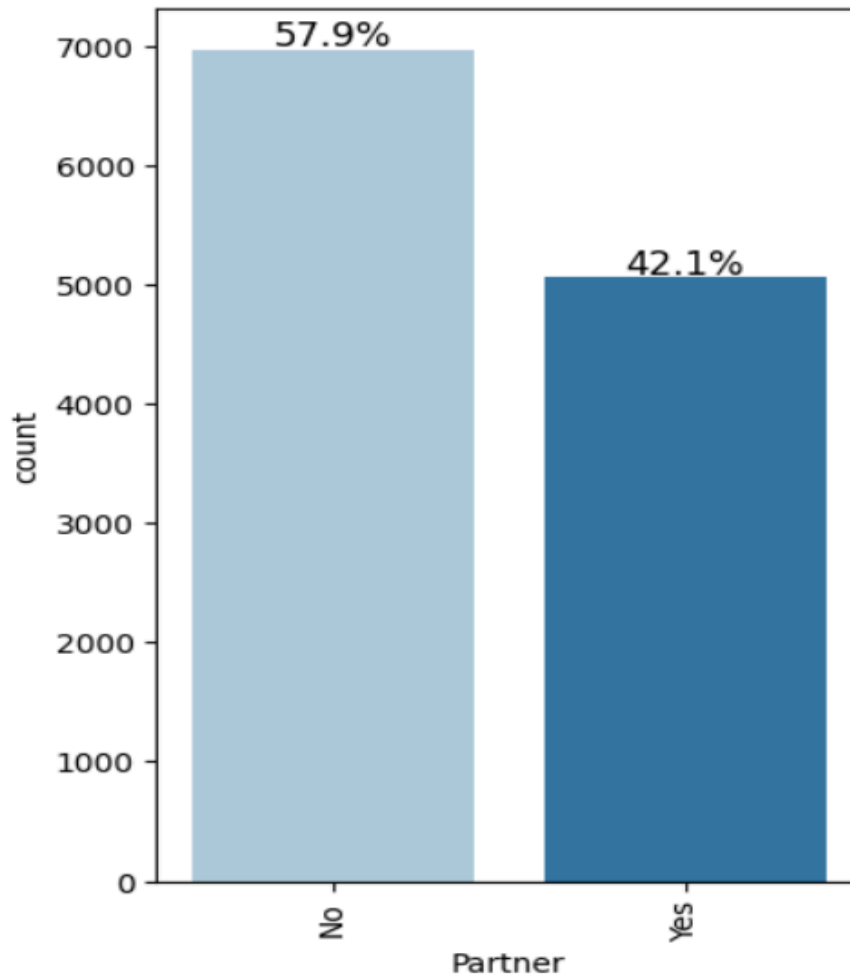
Observation on Senior Citizen :



Observation on SeniorCitizen

- ❖ The majority of the customers(88.2%) are not in the Senior citizen category and remaining customers(11.8%) are classified as Senior citizens.

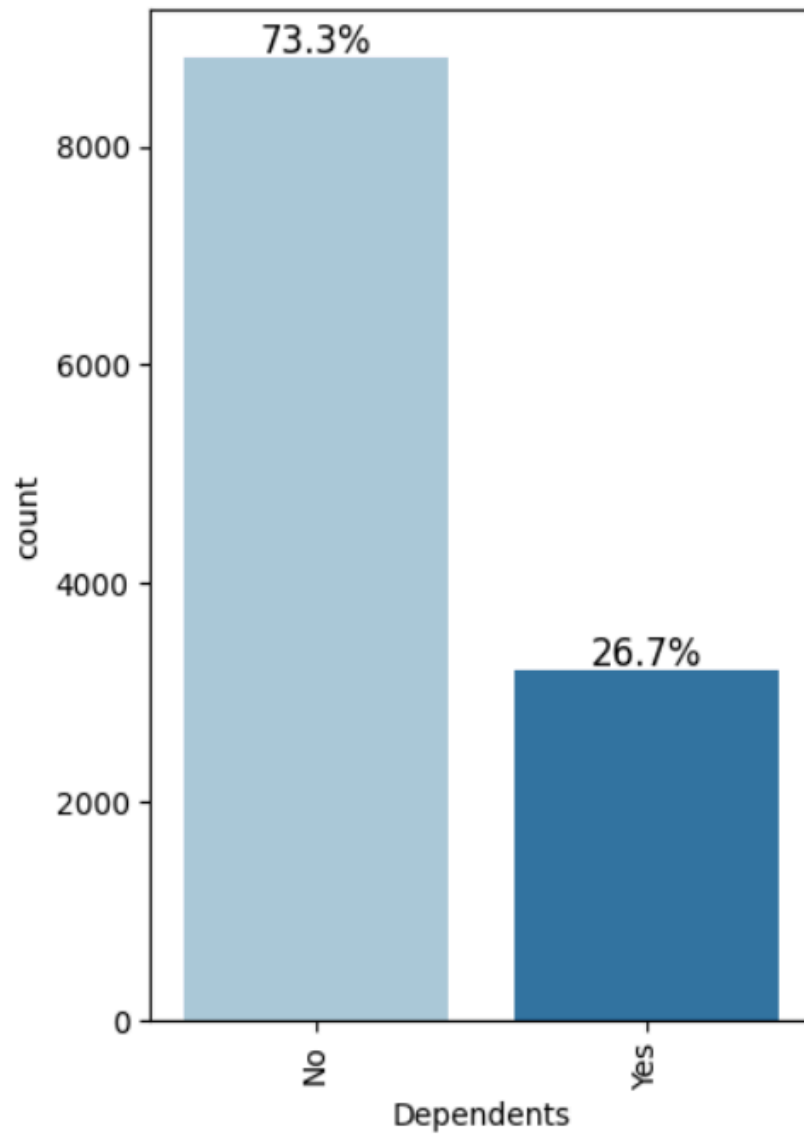
Observation on Partner :



Observation on Partner

- ❖ The majority of the customers have no partner(**57.9%**) while the remaining customers have partners(**42.1%**).

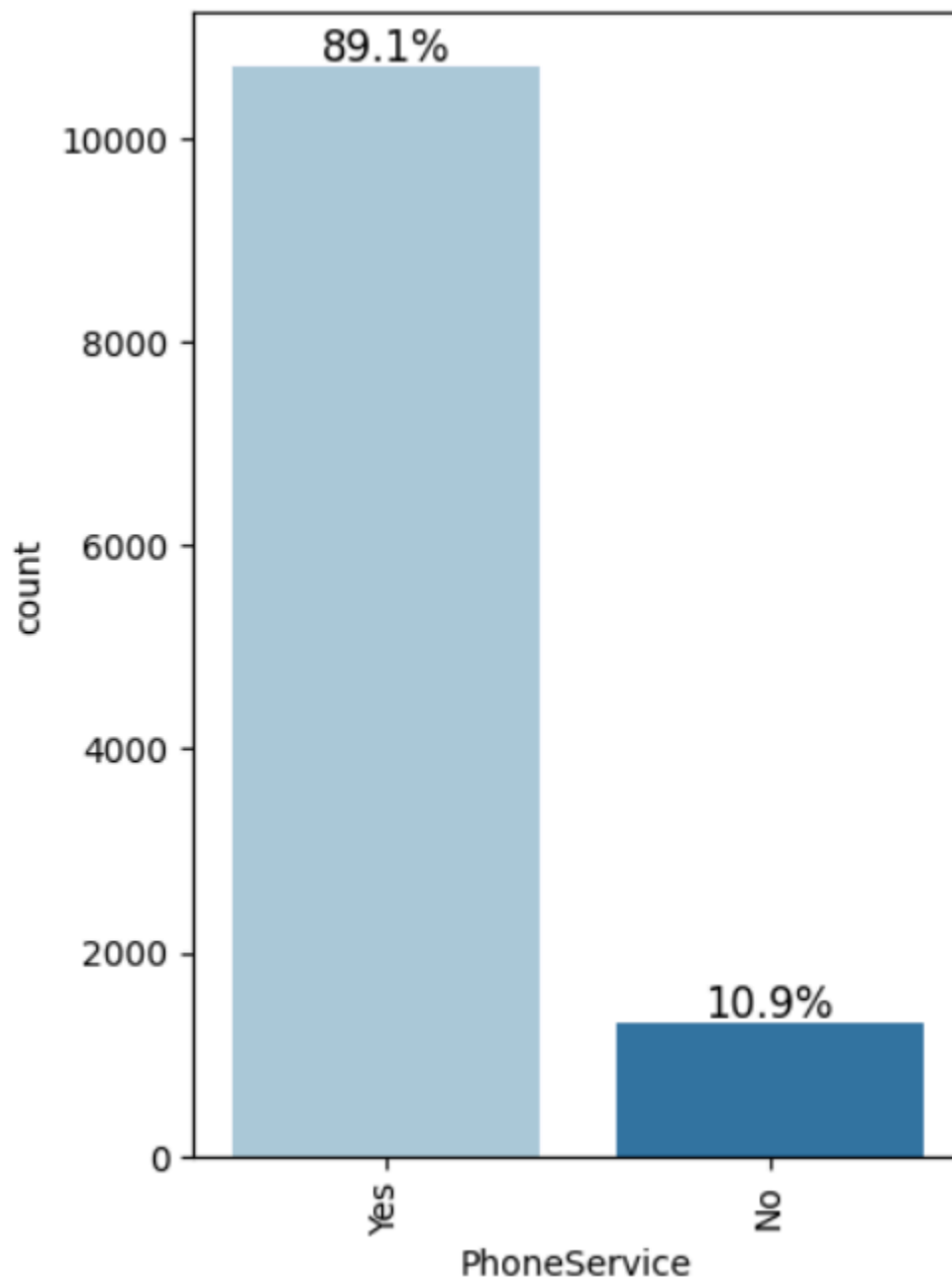
Observation on Dependents :



Observation on Dependents

- ❖ The maximum number of customers do not have dependents(**73.3%**) and remaining customers have dependents(**26.7%**).

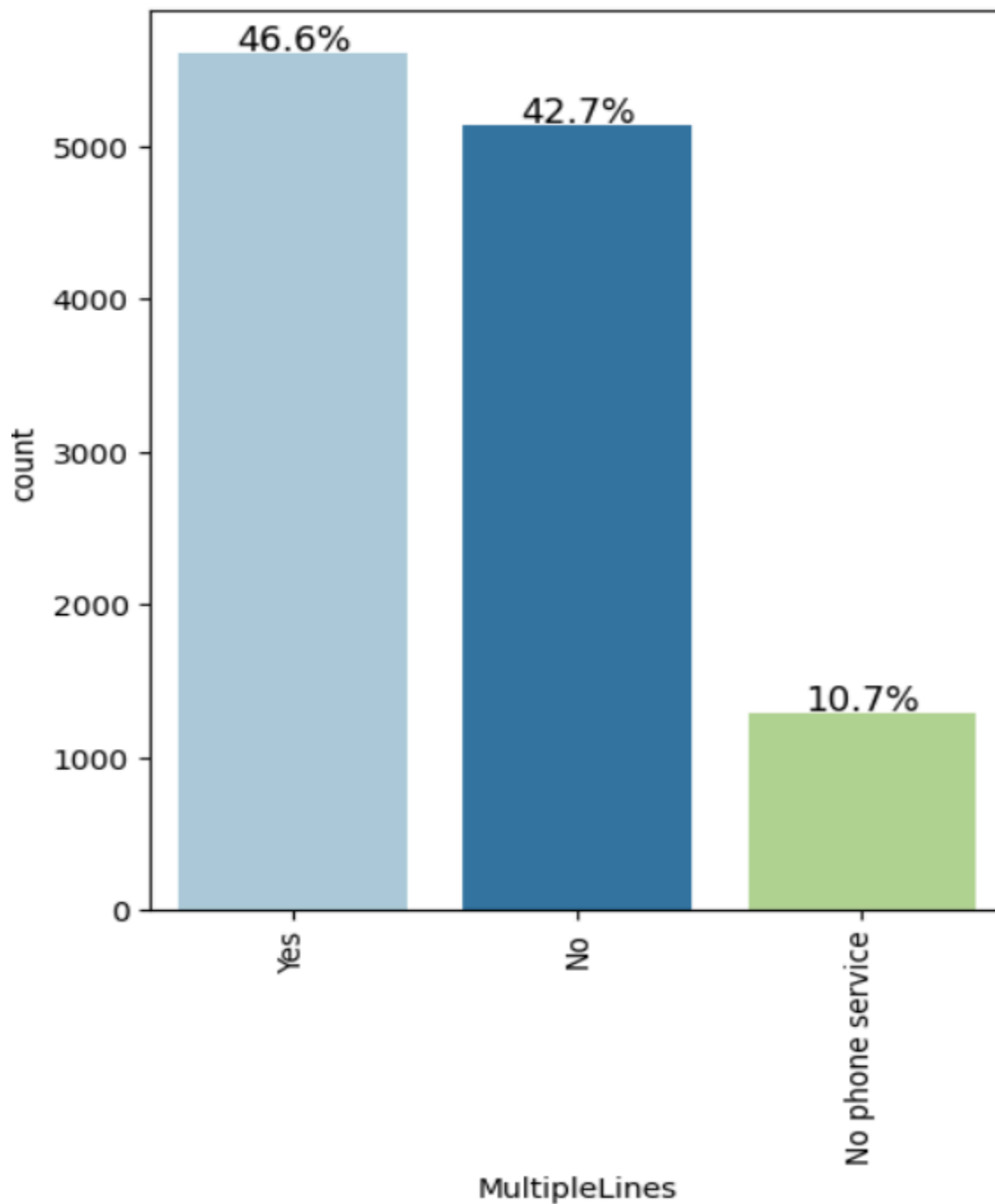
Observation on PhoneService :



Observation on PhoneService

- ❖ A huge number of customers subscribed for Phone services(**89.1%**) while less number of customers have not subscribed for Phone services(**10.9%**).

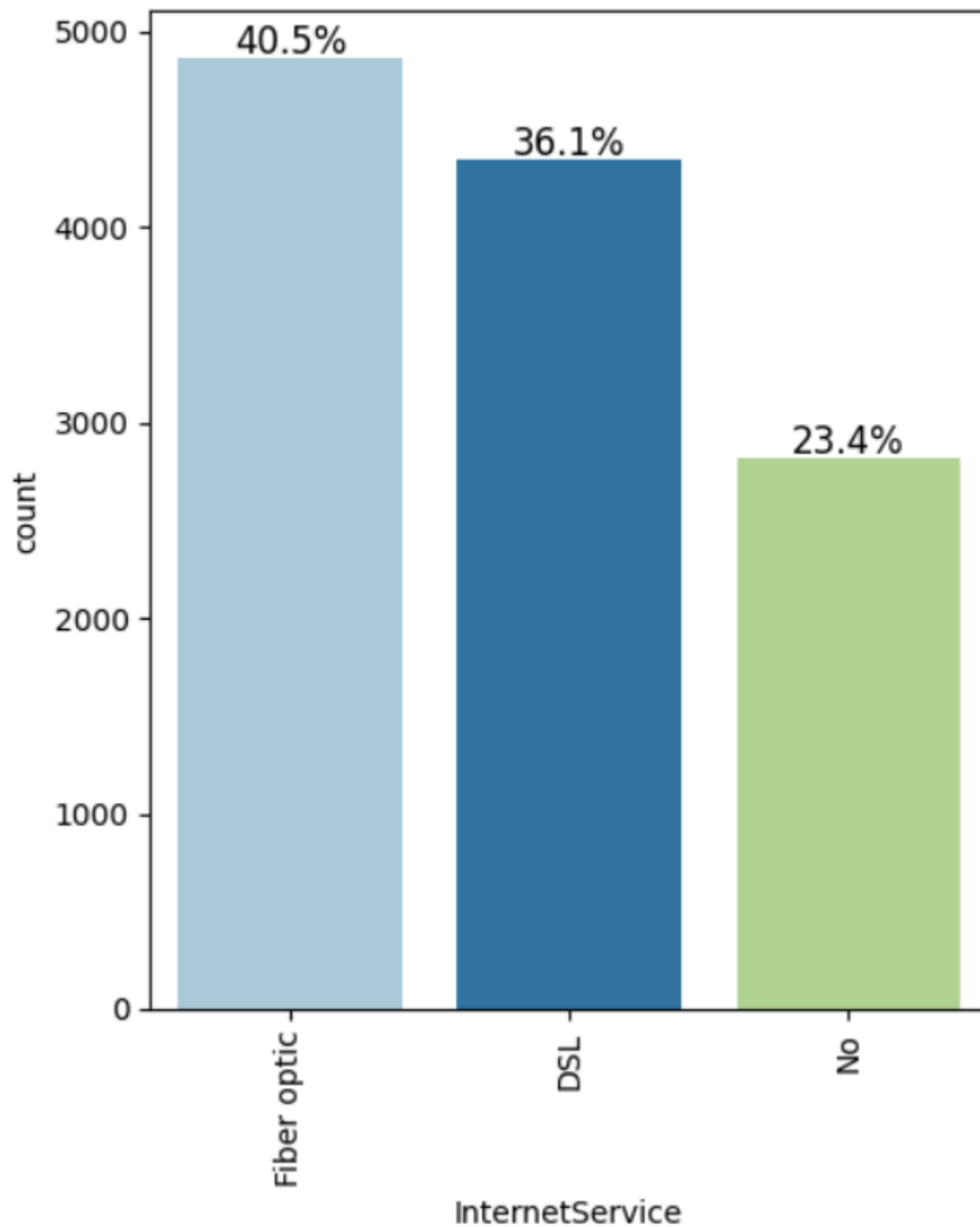
Observation on Multiple Lines :



Observation on MultipleLines

- ❖ **46.6% of customers** are using multiple phone lines and **42.7%** customers are using one phone line. And the remaining **10.7%** customers have no phone services.

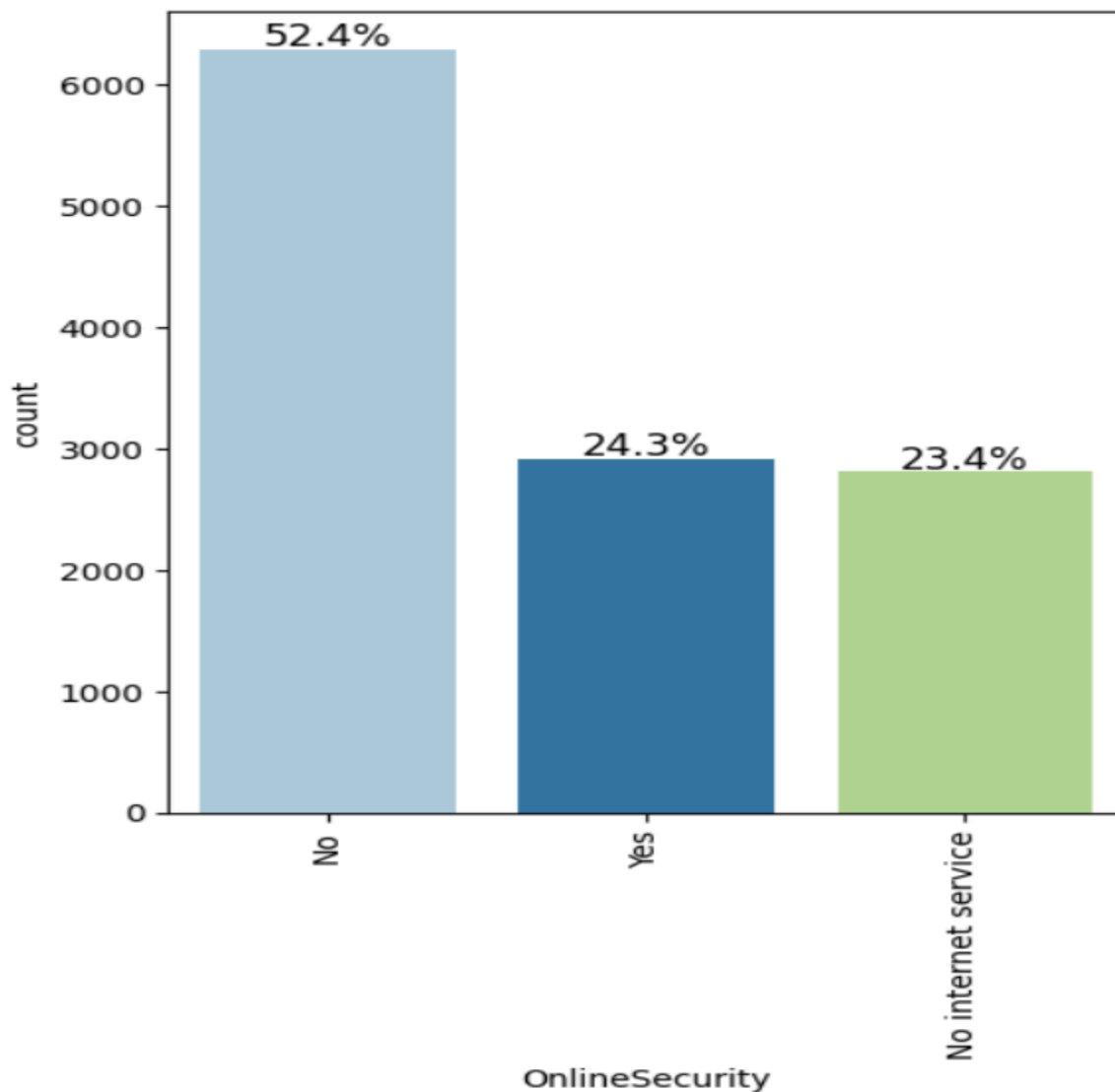
Observations on Internet Service :



Observation on InternetService

- ❖ Most of the customers get internet services from Fibre optic(**40.5%**) followed by DSL(**36.1%**) and the remaining customers have no internet services(**23.4%**).

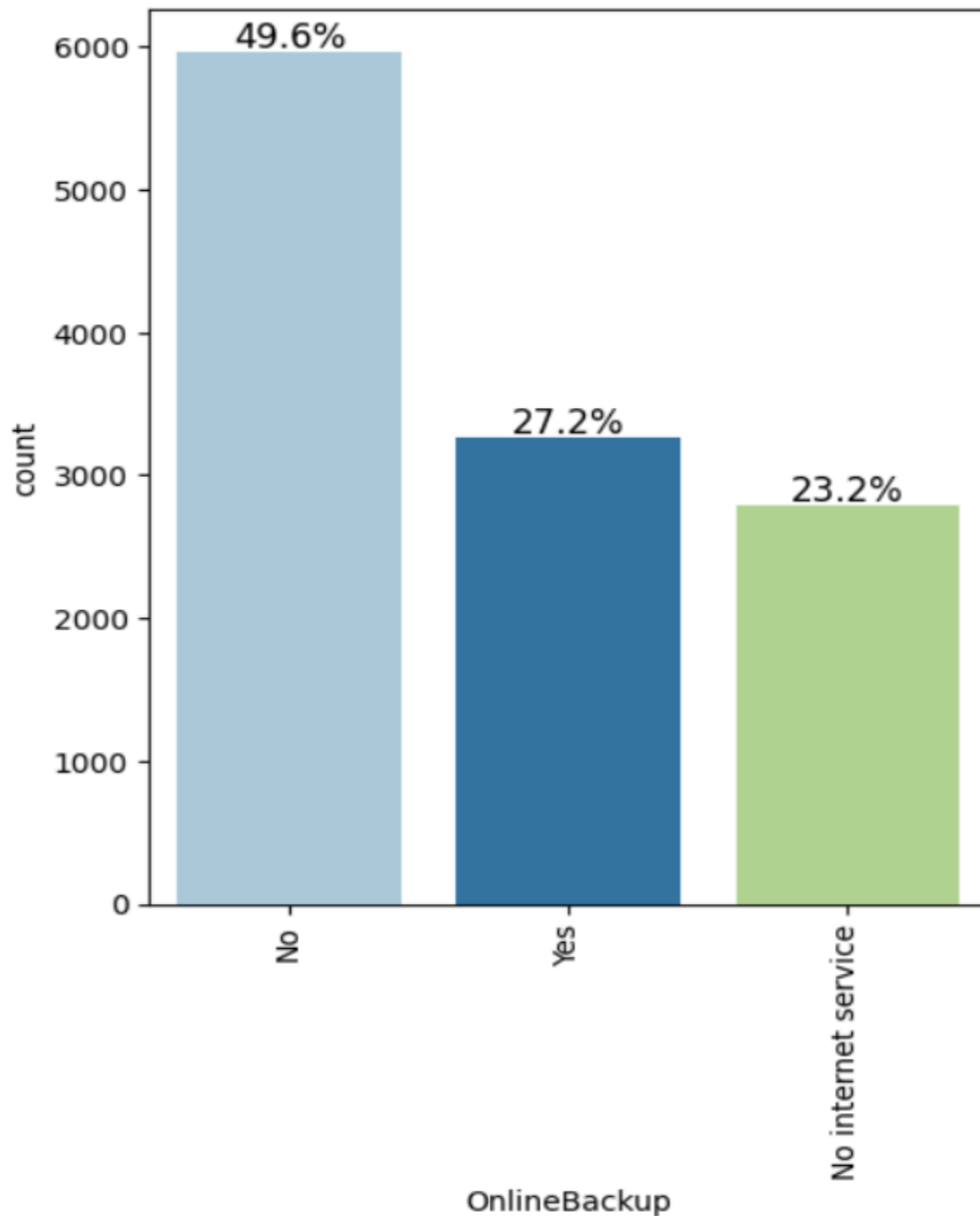
Observation on Online Security :



Observation on OnlineSecurity

- ❖ Very few customers subscribed for online security services(**23.4%**) while a huge number of customers did not subscribe to online security services(**52.4%**)and the remaining customers have no internet services to subscribe to online security(**23.4%**).

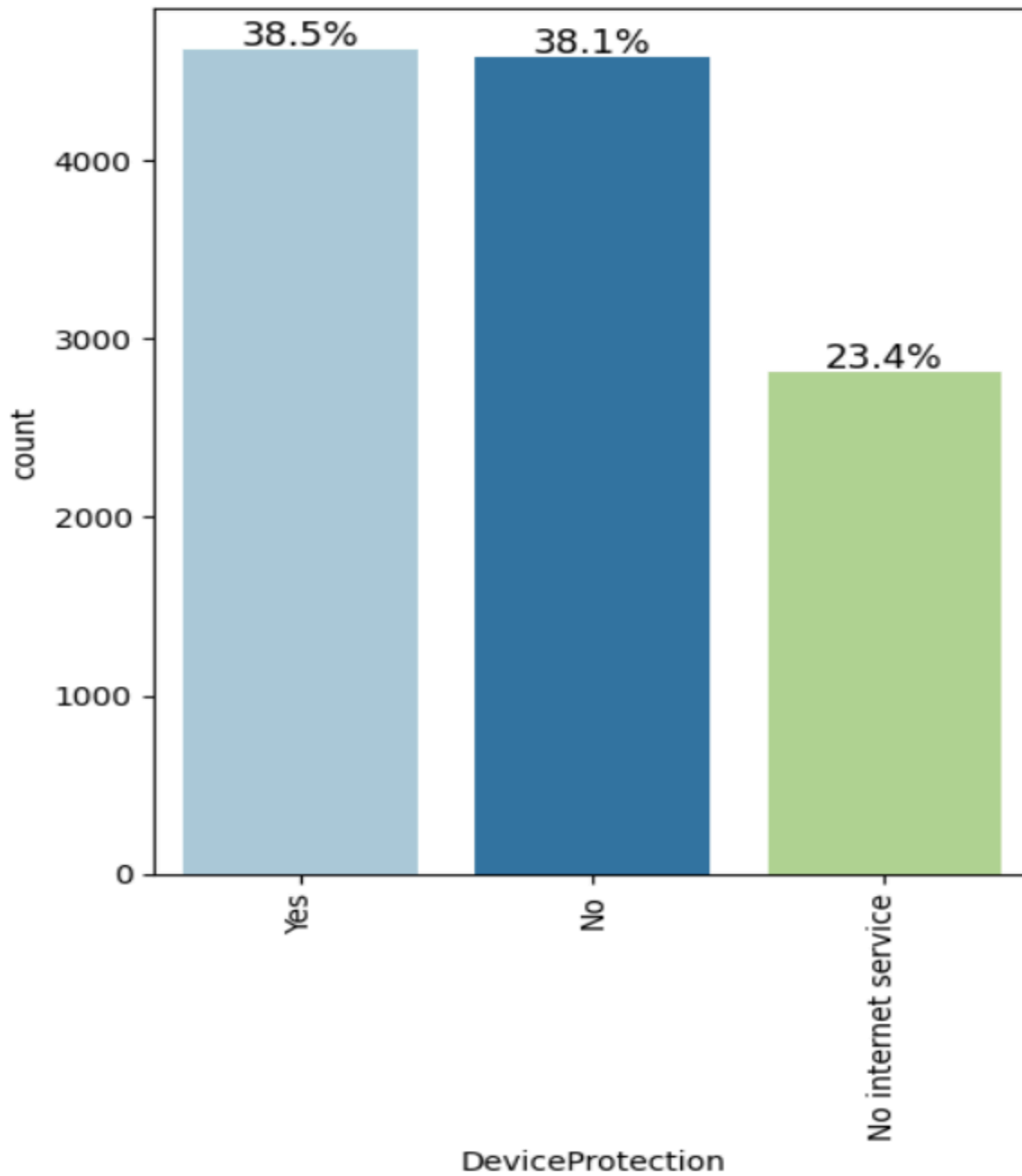
Observation on Online Backup :



Observation on OnlineBackup

- ❖ A large number of customers have no online backup services(**49.6%**) and some of the customers have online backup services(**27.2%**)and the remaining customers have no internet services to get online backup(**23.2%**).

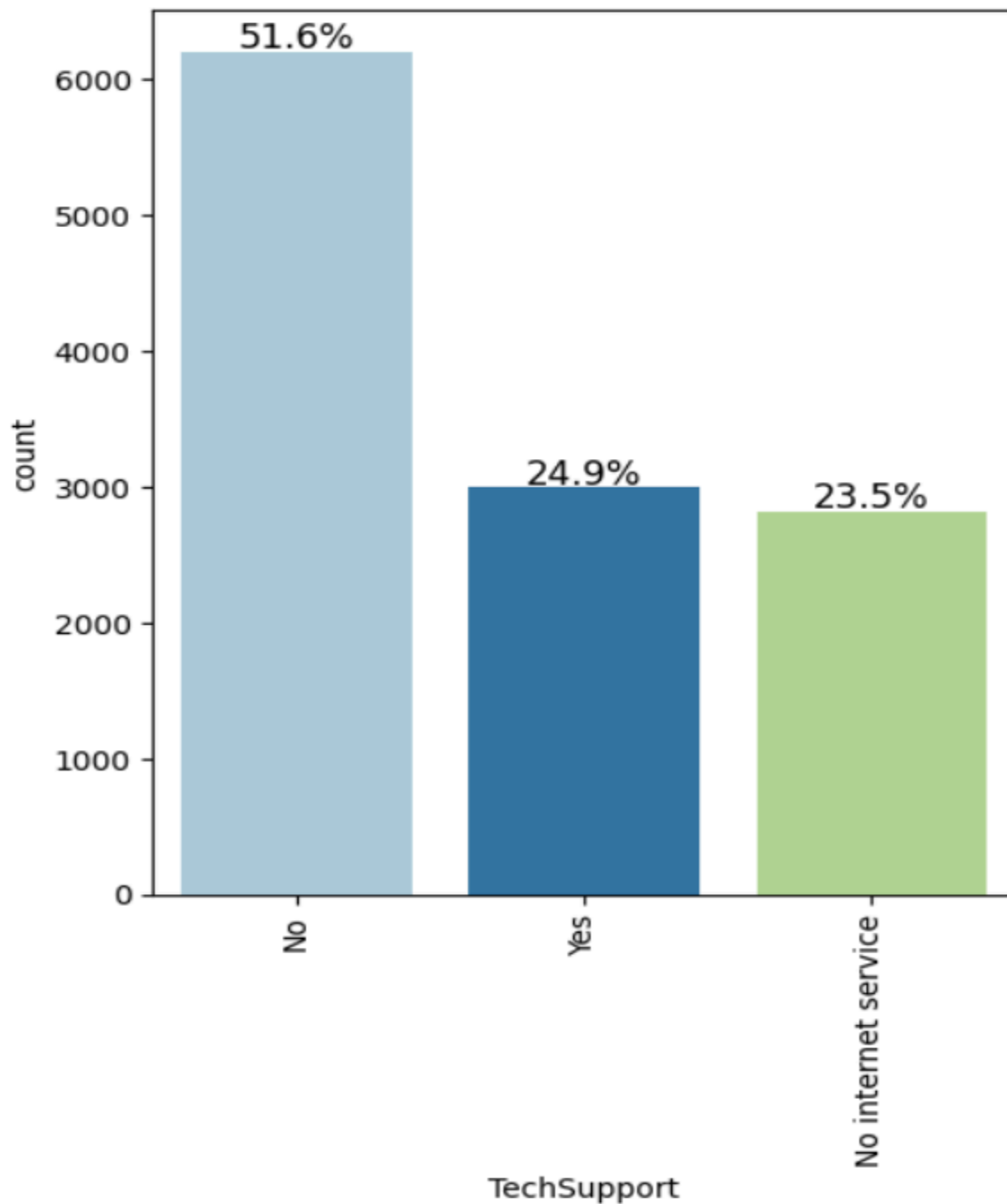
Observation on Device Protection :



Observation on DeviceProtection

- ❖ The vast number of customers(approximately **76.6%**) falls into two categories, **Device Protection Service: Yes**(38.5%) and **No**(38.1%) which are nearly equal to each other.
- ❖ A significantly smaller segment belongs to **No internet service**(23.4%).

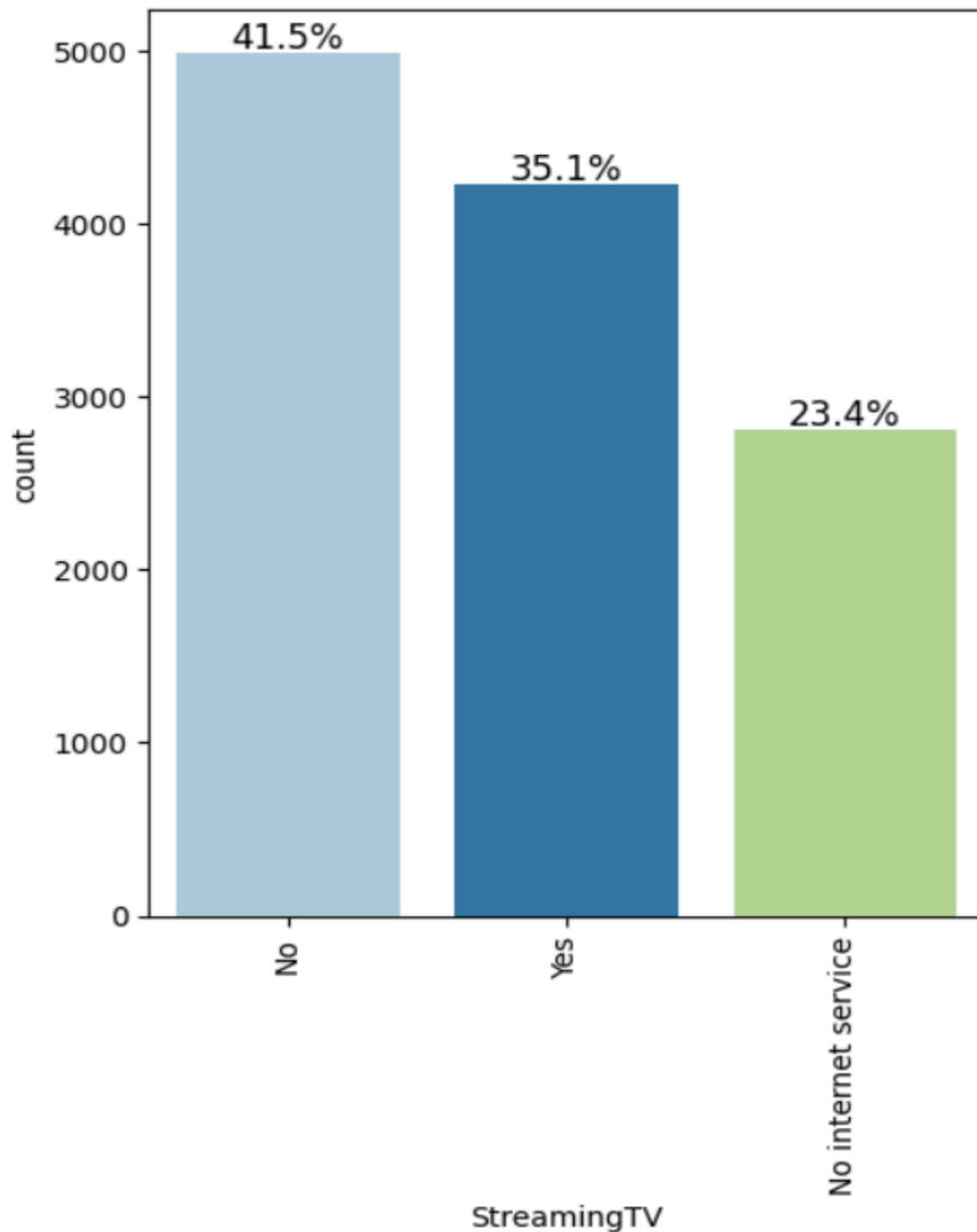
Observation on Tech Support :



Observation on TechSupport

- ❖ The maximum number of customers did not subscribe(51.6%) for technical support services.

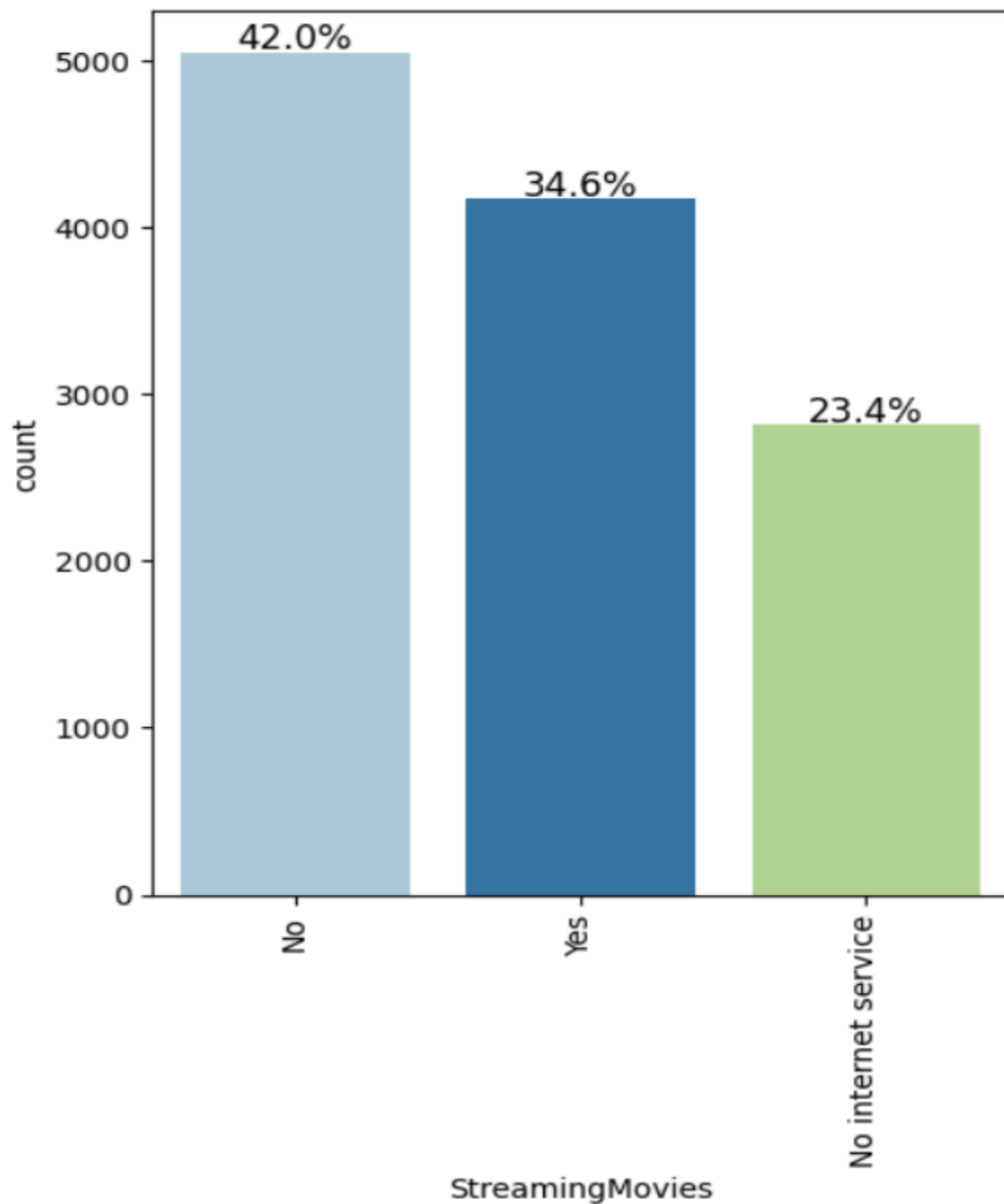
Observation on Streaming TV :



Observation on StreamingTV

- ❖ Most of the customers have not taken **Streaming TV** services(**41.5%**) but a good amount of customers subscribed(**35.1%**) for this service.

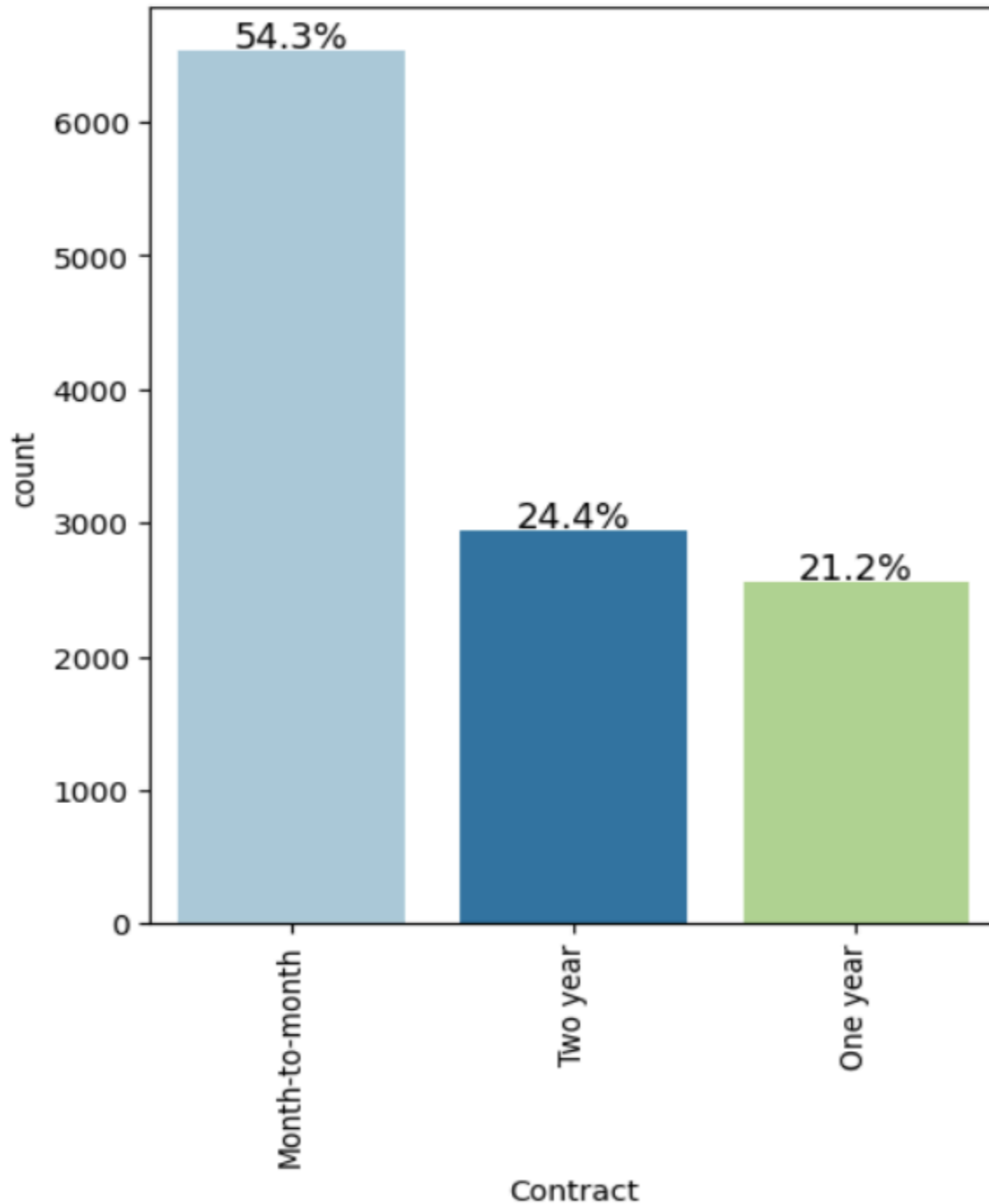
Observation on Streaming Movies :



Observation on StreamingMovies

- ❖ Probably customers are the same as they subscribed/not subscribed for **Streaming Movies** as well as Streaming TV.

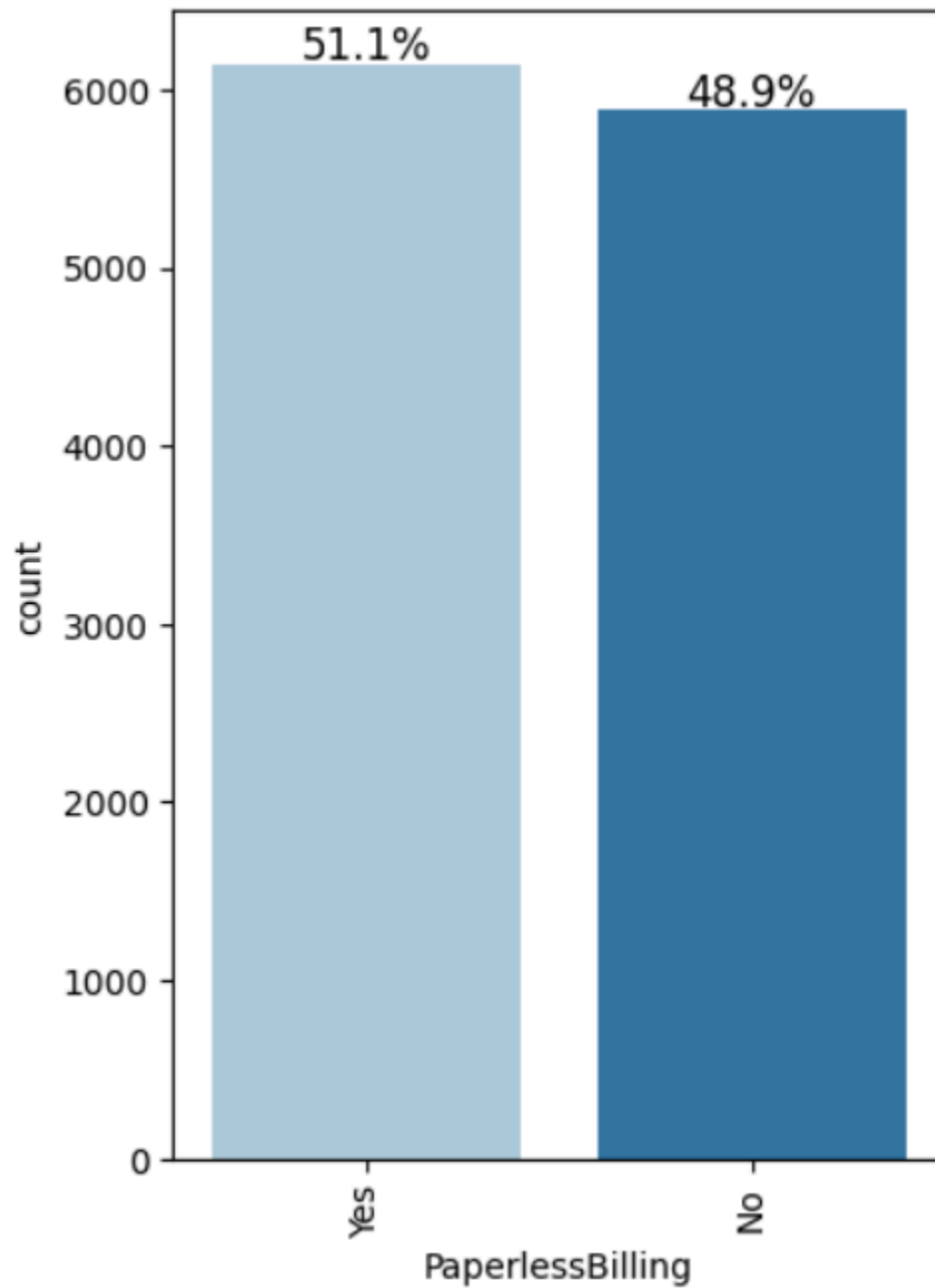
Observation on Contract :



Observation on Contract

- ❖ maximum number of customers(**54.3%**) preferred **Month-to-month** contract. Some customers also go for yearly(one year, two year) contracts but they are less in numbers.

Observation on paperless Billing :



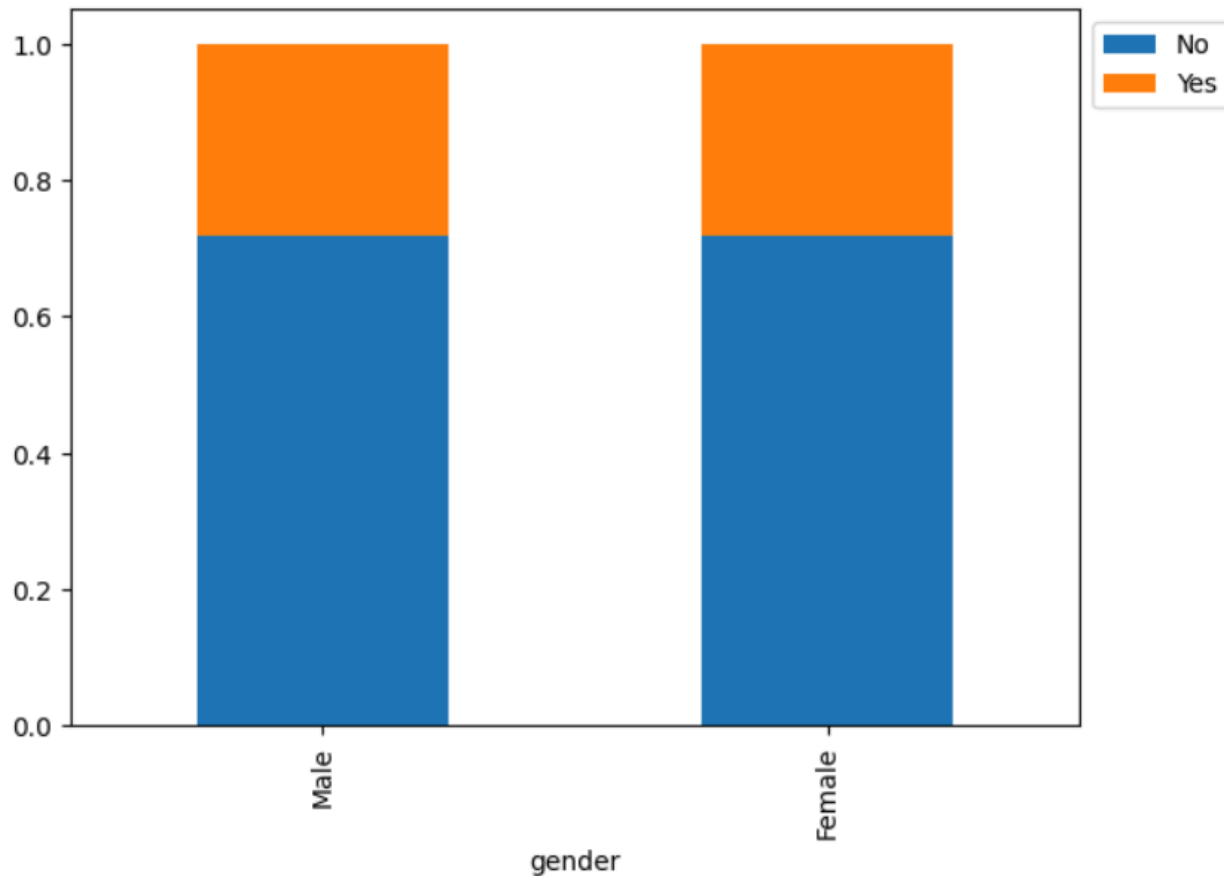
Observation on PaperlessBilling

- ❖ Majority of the customers use **Paperless Billing(51.1%)**.

Bivariate Analysis

Observations on Customer Churn and Gender :

Churn_New	No	Yes	All
gender			
All	8634	3394	12028
Male	4802	1893	6695
Female	3832	1501	5333

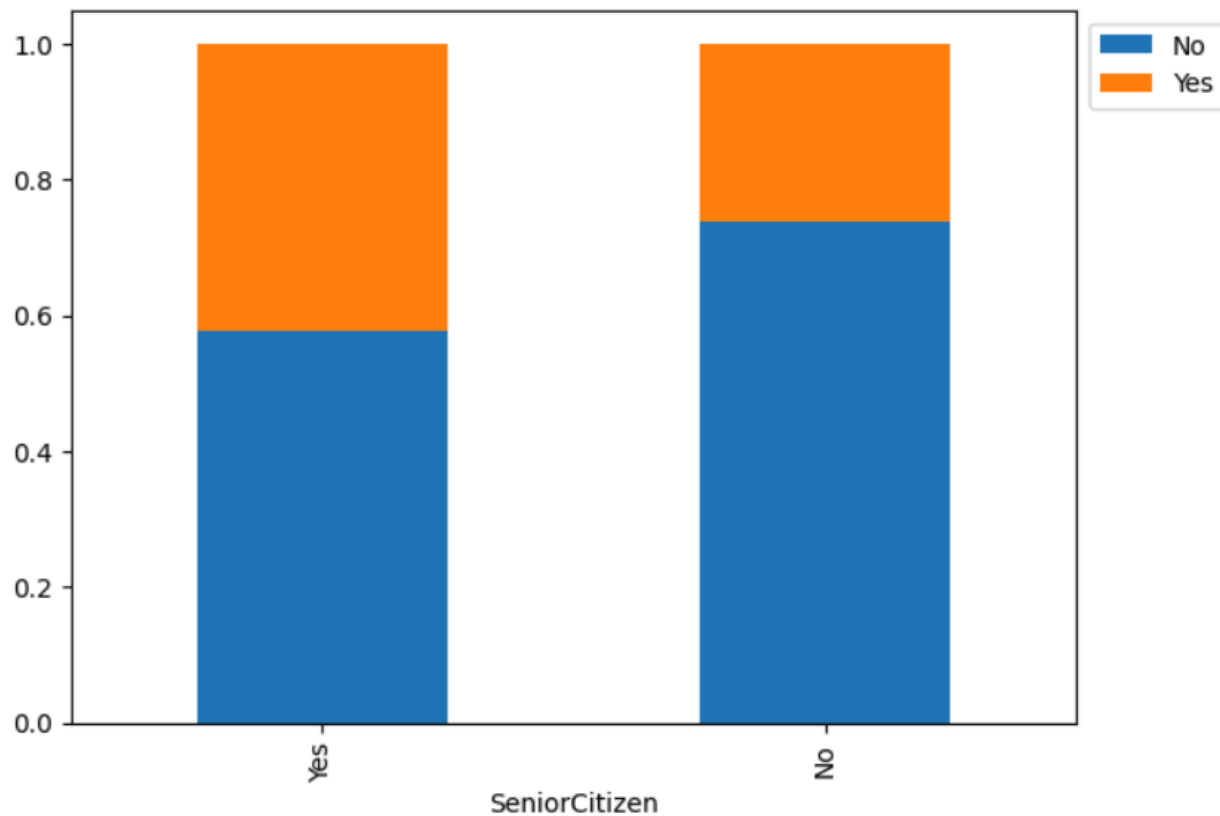


Observation on Customer Churn and gender

- ❖ Both genders have a relatively low churn rate, with the majority of customers in both categories. The non-churning customers make up roughly 70-75% of each gender group.
- ❖ The absolute number of **male** customers who churned(3,394) is nearly double the number of **female** customers who churned(1,893), simply because there are more male customers overall in the dataset.

Observations on Customer Churn and Senior Citizen :

Churn_New	No	Yes	All
SeniorCitizen			
All	8634	3394	12028
No	7816	2792	10608
Yes	818	602	1420

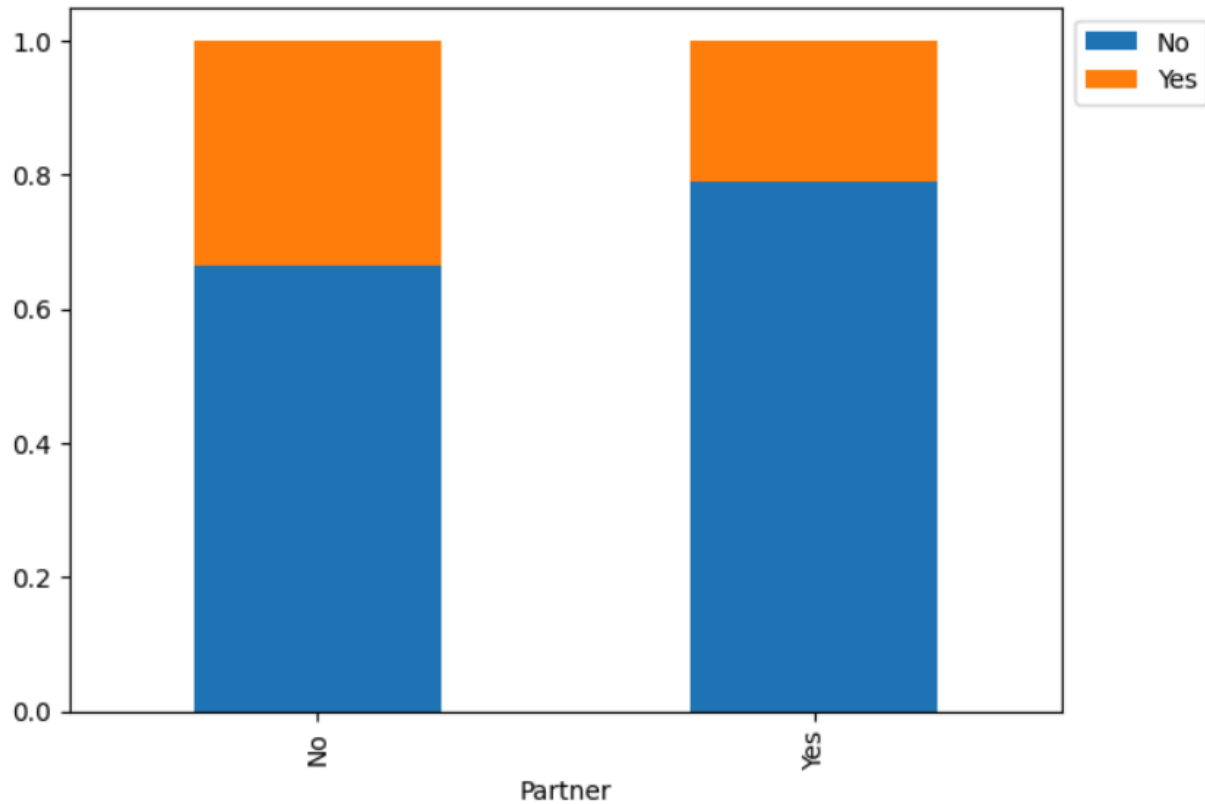


Observation on Customer Churn and SeniorCitizen

- ❖ We observed that the churn rate for **senior citizens** is approximately **42%**(**602** out of **1420**), which is nearly double the rate for **non-senior citizens**, which is around **26%**(**2792** out of **10608**).

Observations on Customer Churn and Partner :

Churn_New Partner	No	Yes	All
All	8634	3394	12028
No	4634	2336	6970
Yes	4000	1058	5058

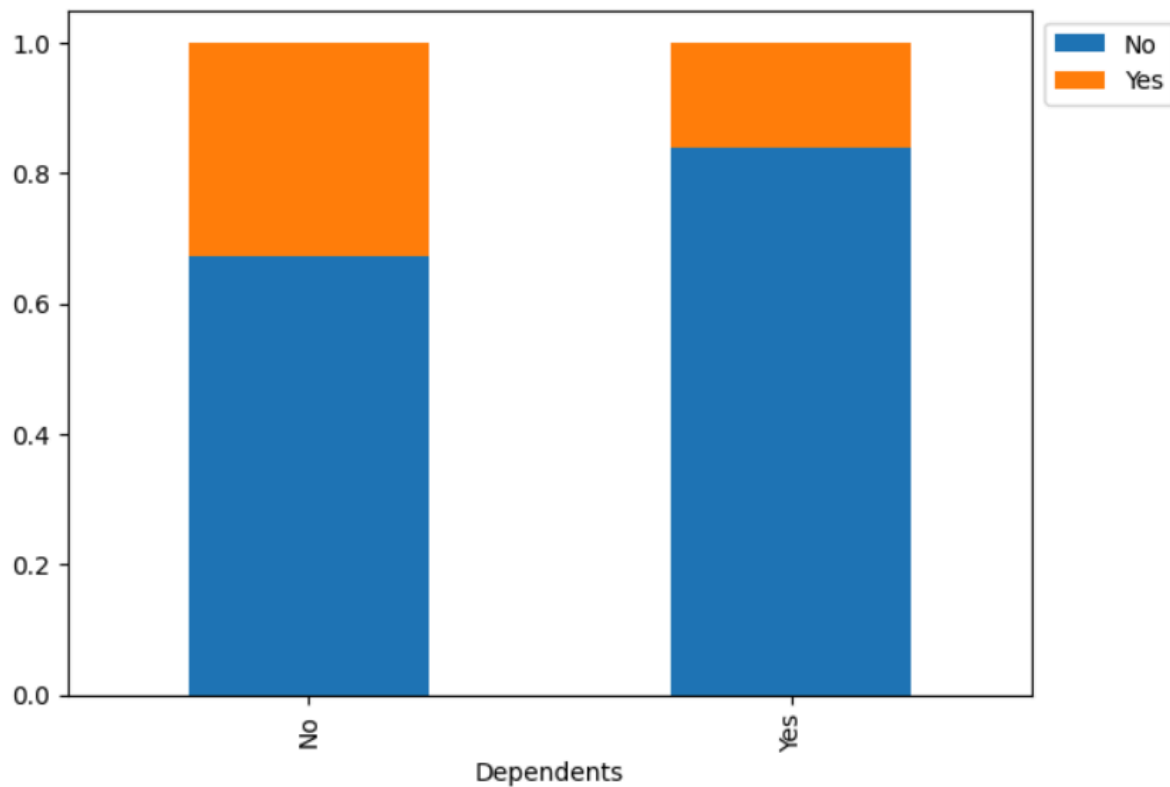


Observation on Customer Churn and Partner

- ❖ Customers who **have a partner are less likely to churn** compared to unpartnered customers. Having a partner appears to be a protective factor against churn.

Observations on Customer Churn and Dependents :

Churn_New	No	Yes	All
Dependents			
All	8634	3394	12028
No	5944	2878	8822
Yes	2690	516	3206

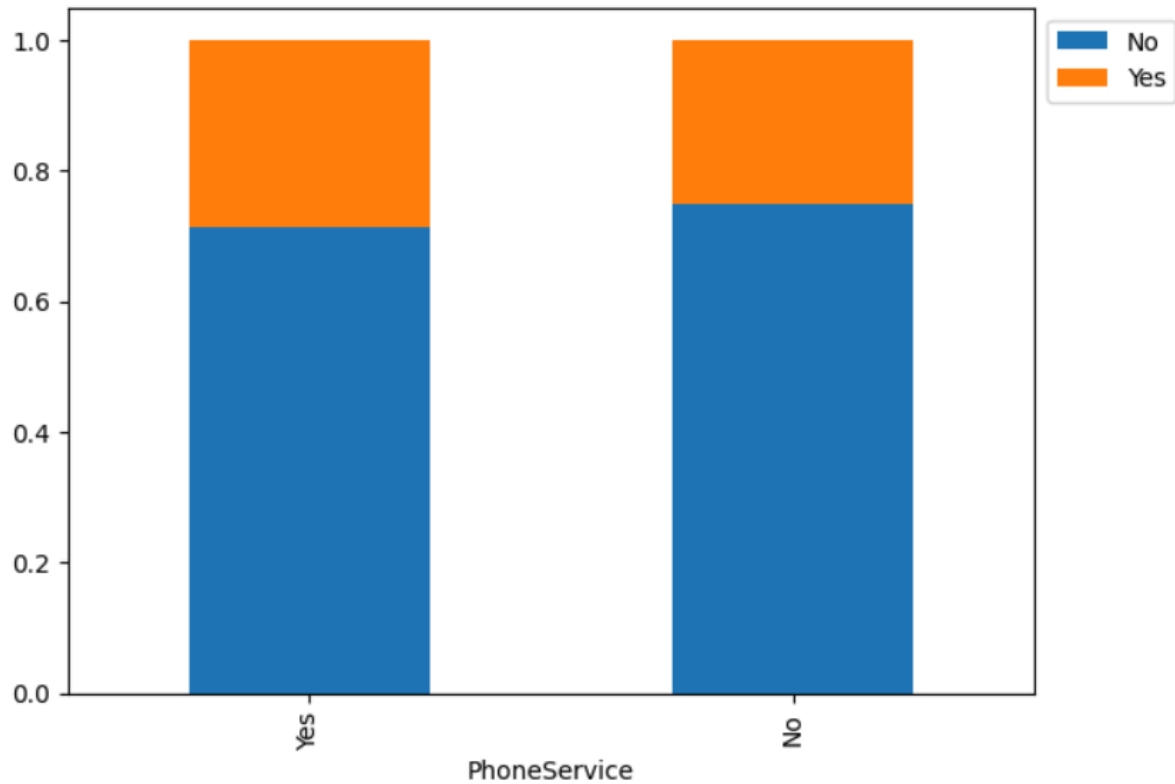


Observation on Customer Churn and Dependents

- ❖ The majority of the customers who do not **have dependents** are more likely to **churn** compared to customers who do have dependents.

Observations on Customer Churn and Phone Service :

Churn_New PhoneService	No	Yes	All
All	8634	3394	12028
Yes	7655	3067	10722
No	979	327	1306

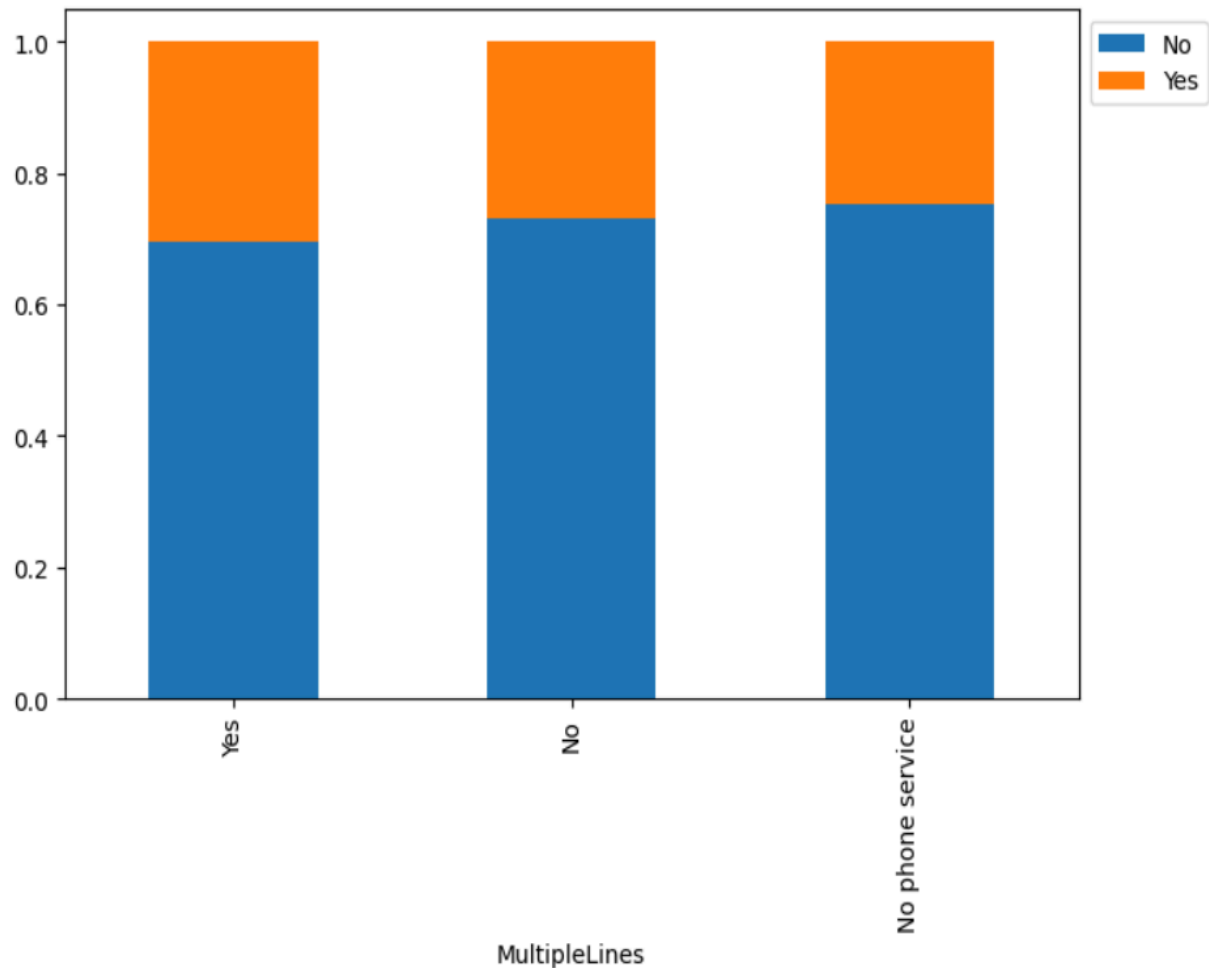


Observation on Customer Churn and PhoneService

- ❖ It is clear that customers who **have been using Phone service** are **more likely to churn**(approximately **28.6%**) than the customers who do not have using Phone service(approximately 25%).

Observations on Customer Churn and Multiple Lines :

Churn_New	No	Yes	All
MultipleLines			
All	8634	3394	12028
Yes	3906	1698	5604
No	3759	1378	5137
No phone service	969	318	1287

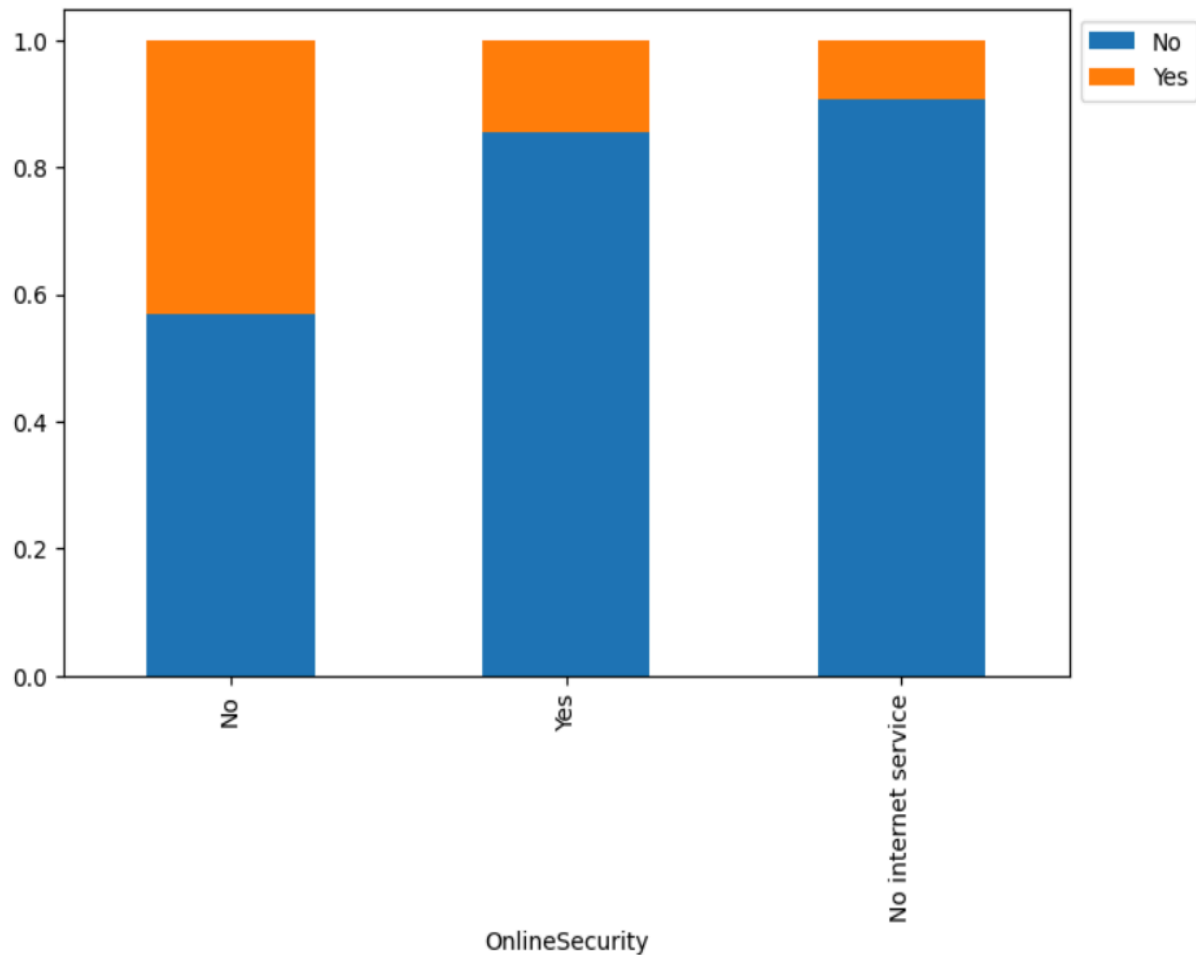


Observation on Customer Churn and MultipleLines

- ❖ The customers who **have used multiple lines are more likely to churn**(approximately **30%**) compared to customers who do not have using multiple lines(approximately **26.9%**) and the customers who do not have any phone service(approximately **24.7%**).

Observations on Customer Churn and Online Security :

Churn_New	No	Yes	All
OnlineSecurity			
All	8634	3394	12028
No	3587	2712	6299
Yes	2499	421	2920
No internet service	2548	261	2809

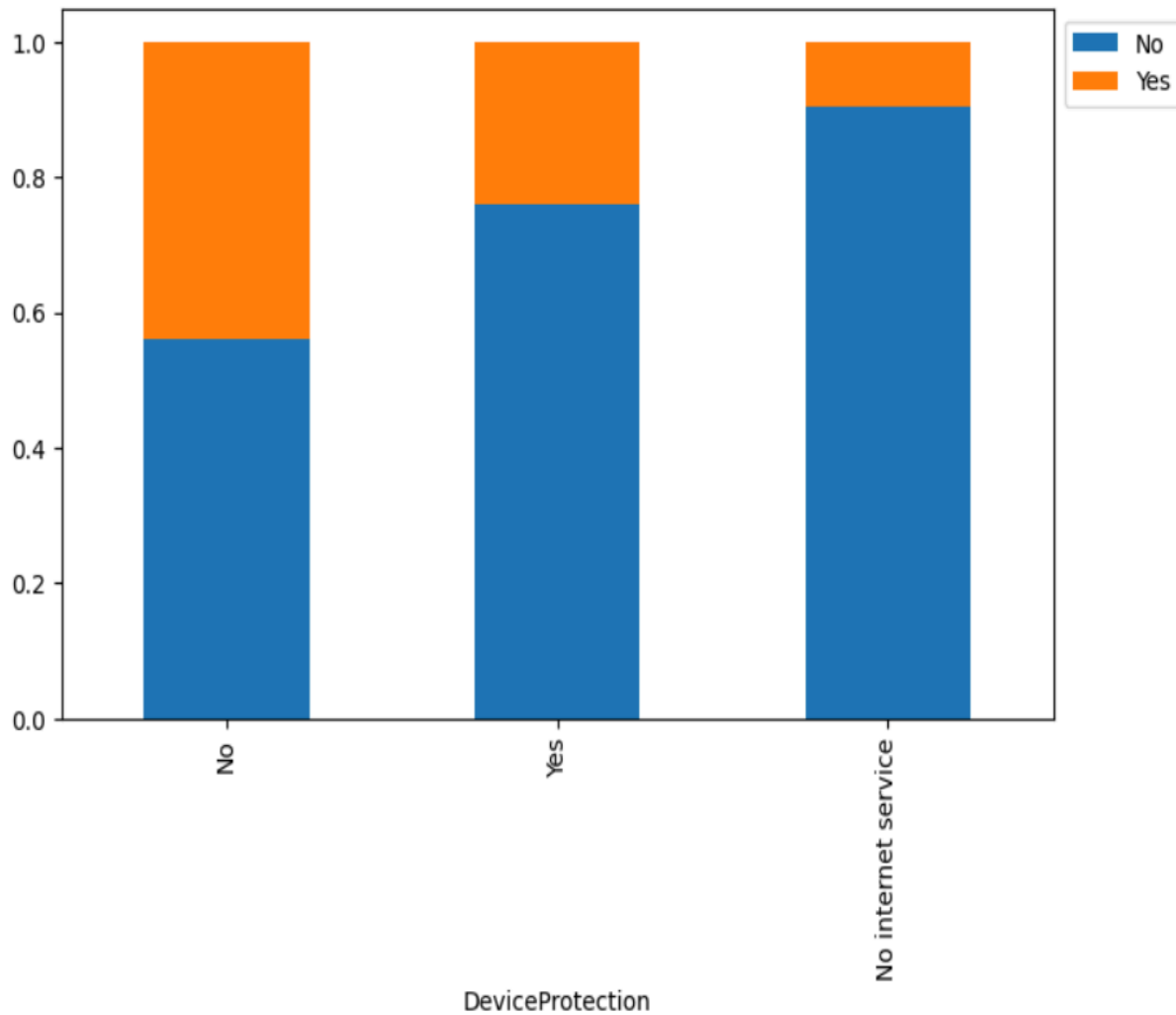


Observation on Customer Churn and OnlineSecurity

- ❖ Customers who subscribe to **online security** are **less likely to leave** the service. And, the customers **without online security** services are substantially **more likely to churn**.

Observations on Customer Churn and Device Protection :

Churn_New	No	Yes	All
DeviceProtection			
All	8634	3394	12028
No	2569	2013	4582
Yes	3512	1115	4627
No internet service	2553	266	2819

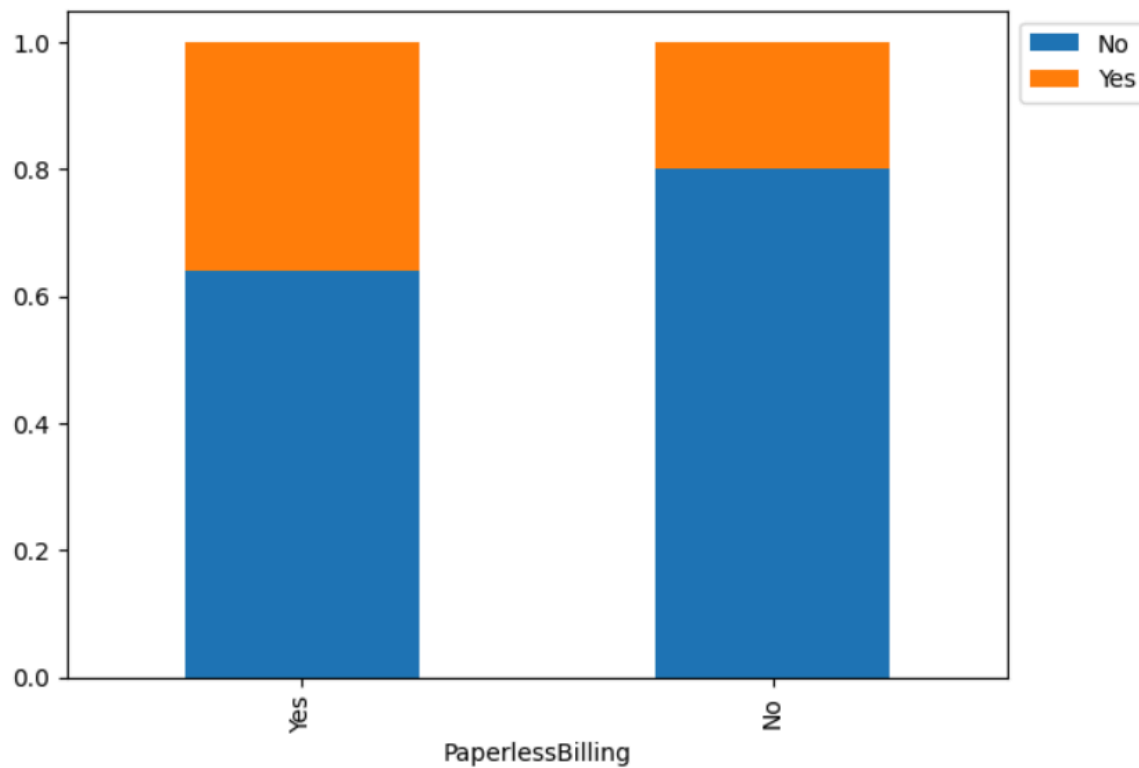


Observation on Customer Churn and DeviceProtection

- ❖ Customers who **have Device protection** are significantly **less likely to churn** compared to those without Device protection service.

Observations on Customer Churn and Paperless Billing :

Churn_New	No	Yes	All
PaperlessBilling			
All	8634	3394	12028
Yes	3929	2215	6144
No	4705	1179	5884



Observation on Customer Churn and PaperlessBilling

- ❖ Customers who use **Paperless billing** show a notably **higher rate of churn(~36%)** compared to those(customers) who use traditional Paper billing(~20%).

Data Preprocessing for Modelling :

Creating Training and Test set

The Training set has 8419 rows and 33 columns, while the Test set has 3609 rows and 33 columns.

Model Building

First, let's create functions to calculate different metrics and confusion matrices so that we don't have to use the same code repeatedly for each model.

- The `model_performance_classification_statsmodels` function will be used to check the model performance of models.
- The `confusion_matrix_statsmodels` function will be used to plot the confusion matrix.

Logistic Regression

Logit Regression Results						
=====						
Dep. Variable:	Churn_New	No. Observations:	8419			
Model:	Logit	Df Residuals:	8386			
Method:	MLE	Df Model:	32			
Date:	Sun, 01 Feb 2026	Pseudo R-squ.:	0.2922			
Time:	09:34:04	Log-Likelihood:	-3546.0			
converged:	True	LL-Null:	-5009.6			
Covariance Type:	nonrobust	LLR p-value:	0.000			
=====						
	coef	std err	z	P> z	[0.025	0.975]

const	-0.5628	0.241	-2.339	0.019	-1.034	-0.091
MonthlyCharges	0.0068	0.003	2.594	0.009	0.002	0.012
TotalCharges	-0.0001	2.33e-05	-5.835	0.000	-0.000	-9.03e-05
gender_Male	0.0710	0.060	1.191	0.234	-0.046	0.188
SeniorCitizen_Yes	0.1997	0.088	2.269	0.023	0.027	0.372
Partner_Yes	-0.0947	0.070	-1.358	0.175	-0.231	0.042
Dependents_Yes	-0.2060	0.081	-2.553	0.011	-0.364	-0.048
PhoneService_Yes	-0.0921	0.184	-0.501	0.617	-0.453	0.269
MultipleLines_No phone service	0.0046	0.187	0.025	0.980	-0.362	0.371
MultipleLines_Yes	0.2881	0.072	3.990	0.000	0.147	0.430
InternetService_Fiber optic	0.6802	0.094	7.201	0.000	0.495	0.865
InternetService_No	0.7650	0.585	1.308	0.191	-0.381	1.911
OnlineSecurity_No internet service	0.3288	0.608	0.541	0.589	-0.863	1.521
OnlineSecurity_Yes	-0.4714	0.082	-5.751	0.000	-0.632	-0.311
OnlineBackup_No internet service	-0.5468	0.563	-0.972	0.331	-1.649	0.556
OnlineBackup_Yes	-0.3262	0.073	-4.440	0.000	-0.470	-0.182
DeviceProtection_No internet service	0.3982	0.555	0.717	0.473	-0.690	1.487
DeviceProtection_Yes	-0.0917	0.069	-1.337	0.181	-0.226	0.043
TechSupport_No internet service	0.1183	0.583	0.203	0.839	-1.025	1.261
TechSupport_Yes	-0.5367	0.086	-6.252	0.000	-0.705	-0.368
StreamingTV_No internet service	-0.7608	0.675	-1.127	0.260	-2.083	0.562
StreamingTV_Yes	0.0796	0.078	1.018	0.309	-0.074	0.233
StreamingMovies_No internet service	-1.2389	0.681	-1.818	0.069	-2.575	0.097

TechSupport_Yes	-0.5367	0.086	-6.252	0.000	-0.705	-0.368
StreamingTV_No internet service	-0.7608	0.675	-1.127	0.260	-2.083	0.562
StreamingTV_Yes	0.0796	0.078	1.018	0.309	-0.074	0.233
StreamingMovies_No internet service	-1.2389	0.681	-1.818	0.069	-2.575	0.097
StreamingMovies_Yes	-0.0153	0.077	-0.198	0.843	-0.167	0.136
Contract_One year	-0.8159	0.093	-8.747	0.000	-0.999	-0.633
Contract_Two year	-1.5841	0.148	-10.734	0.000	-1.873	-1.295
PaperlessBilling_Yes	0.3101	0.064	4.820	0.000	0.184	0.436
PaymentMethod_New_Credit Card (Automatic)	-0.0649	0.107	-0.609	0.543	-0.274	0.144
PaymentMethod_New_Electronic Check	0.4535	0.090	5.017	0.000	0.276	0.631
PaymentMethod_New_Mailed Check	0.1168	0.107	1.097	0.273	-0.092	0.326
tenure_group_Medium (13-36)	-0.7635	0.076	-10.018	0.000	-0.913	-0.614
tenure_group_Long (37-60)	-0.8934	0.100	-8.943	0.000	-1.089	-0.698
tenure_group_Very Long (60+)	-1.0975	0.153	-7.173	0.000	-1.397	-0.798

Logistic Regression Baseline Model

Observations :

- Negative values of the coefficient shows that probability of the customer being a defaulter decreases with the increase of corresponding attribute value.
- Positive values of the coefficient show that the probability of the customer being a defaulter increases with the increase of corresponding attribute value.
- p-value of a variable indicates if the variable is significant or not. If we consider the significance level to be 0.05 (5%), then any variable with a p-value less than 0.05 would be considered significant.
- But these variables might contain multicollinearity, which will affect the p-values.
- We will have to remove multicollinearity from the data to get reliable coefficients and p-values.
- There are different ways of detecting (or testing) multi-collinearity, one such way is the Variation Inflation Factor.

Additional Information on VIF

- **Variance Inflation factor:** Variance inflation factors measure the inflation in the variances of the regression coefficients estimates due to collinearity that exist among the predictors. It is a measure of how much the variance of the estimated regression coefficient β_k is "inflated" by the existence of correlation among the predictor variables in the model.
- **General Rule of thumb:** If VIF is 1 then there is no correlation among the kth predictor and the remaining predictor variables, and hence the variance of β_k is not inflated at all. Whereas if VIF exceeds 5, we say there is moderate VIF and if it is

10 or exceeding 10, it shows signs of high multi-collinearity. But the purpose of the analysis should dictate which threshold to use.

Series before feature selection:

const	72.831
MonthlyCharges	7.597
TotalCharges	2.705
gender_Male	1.009
SeniorCitizen_Yes	1.091
Partner_Yes	1.316
Dependents_Yes	1.240
PhoneService_Yes	4.175
MultipleLines_No phone service	4.240
MultipleLines_Yes	1.442
InternetService_Fiber optic	2.674
InternetService_No	72.439
OnlineSecurity_No internet service	60.634
OnlineSecurity_Yes	1.366
OnlineBackup_No internet service	58.768
OnlineBackup_Yes	1.300
DeviceProtection_No internet service	61.762
DeviceProtection_Yes	1.532
TechSupport_No internet service	59.516
TechSupport_Yes	1.507
StreamingTV_No internet service	70.006
StreamingTV_Yes	1.783
StreamingMovies_No internet service	72.369
StreamingMovies_Yes	1.743
Contract_One year	1.556
Contract_Two year	2.248
PaperlessBilling_Yes	1.197
PaymentMethod_New_Credit Card (Automatic)	1.681
PaymentMethod_New_Electronic Check	2.145
PaymentMethod_New_Mailed Check	1.875
tenure_group_Medium (13-36)	1.361
tenure_group_Long (37-60)	1.742
tenure_group_Very Long (60+)	2.476

Variance Inflation Factor

- None of the variables exhibit high multicollinearity, so the values in the summary are reliable.
- It is important to keep the service-related variables such as no internet service across multiple features. With **high VIF** for business purposes because these are essential for **model interpretability** and identify key **business drivers** of churn.
- Removing them would mean losing crucial insights into what causes customers to leave, even if they are statistically correlated.
- Let's remove the insignificant features (p-value>0.05).

```
['const', 'MonthlyCharges', 'TotalCharges', 'Dependents_Yes',
'PhoneService_Yes', 'MultipleLines_Yes', 'InternetService_Fiber optic',
'OnlineSecurity_Yes', 'OnlineBackup_Yes', 'DeviceProtection_Yes',
'TechSupport_Yes', 'StreamingMovies_No internet service', 'Contract_One
year', 'Contract_Two year', 'PaperlessBilling_Yes', 'tenure_group_Medium
(13-36)', 'tenure_group_Long (37-60)', 'tenure_group_Very Long (60+)',
'PaymentMethod_New_Electronic Check']
```

- The above columns are the significant columns.

Final Model

Logit Regression Results							
=====							
Dep. Variable:	Churn_New	No. Observations:	8419				
Model:	Logit	Df Residuals:	8401				
Method:	MLE	Df Model:	17				
Date:	Sun, 01 Feb 2026	Pseudo R-squ.:	0.2907				
Time:	09:34:06	Log-Likelihood:	-3553.2				
converged:	True	LL-Null:	-5009.6				
Covariance Type:	nonrobust	LLR p-value:	0.000				
=====							
		coef	std err	z	P> z	[0.025	0.975]

const		-0.5365	0.133	-4.040	0.000	-0.797	-0.276
MonthlyCharges		0.0063	0.002	2.808	0.005	0.002	0.011
TotalCharges		-0.0001	2.26e-05	-6.197	0.000	-0.000	-9.59e-05
SeniorCitizen_Yes		0.1911	0.087	2.190	0.029	0.020	0.362
Dependents_Yes		-0.2454	0.076	-3.241	0.001	-0.394	-0.097
MultipleLines_Yes		0.2738	0.071	3.870	0.000	0.135	0.412
InternetService_Fiber optic		0.6583	0.091	7.203	0.000	0.479	0.837
OnlineSecurity_Yes		-0.4793	0.082	-5.873	0.000	-0.639	-0.319
OnlineBackup_Yes		-0.3242	0.073	-4.433	0.000	-0.468	-0.181
TechSupport_Yes		-0.5403	0.085	-6.335	0.000	-0.708	-0.373
StreamingMovies_No internet service		-0.9418	0.127	-7.426	0.000	-1.190	-0.693
Contract_One year		-0.8269	0.092	-8.980	0.000	-1.007	-0.646
Contract_Two year		-1.6062	0.146	-10.982	0.000	-1.893	-1.320
PaperlessBilling_Yes		0.3056	0.064	4.792	0.000	0.181	0.431
PaymentMethod_New_Electronic Check		0.4369	0.065	6.748	0.000	0.310	0.564
tenure_group_Medium (13-36)		-0.7806	0.075	-10.350	0.000	-0.928	-0.633
tenure_group_Long (37-60)		-0.9157	0.099	-9.285	0.000	-1.109	-0.722
tenure_group_Very Long (60+)		-1.1322	0.152	-7.471	0.000	-1.429	-0.835
=====							

Final Regression Model

Now no feature has p-value greater than 0.05, so we'll consider the features in X_train1 as the final ones and lg2 as the final model.

Observations :

- **MonthlyCharges** : The coefficient is positive(0.0063). This suggests that higher monthly charges increase the odds of a customer churning.
- **TotalCharges** : The coefficient is negative(-0.0001). This indicates that as TotalCharges increases, the odds of churning decreases. It might suggest that customers with a high total historical spend are less likely to leave.
- **InternetService_Fibre optic** : The coefficient is positive(0.6583). This indicates customers, those who have used Fibre optic service are more likely to churn. This might point to dissatisfaction with fibre service quality, speed or cost.
- **Contract One year & Two year** : Coefficients for both are negative(-0.8269 & -1.6062). Which indicates customers on longer term contracts are much less likely to churn than those on a month-to-month basis.
- **PaymentMethod_Electronic check** : The coefficient is positive(0.4369). That indicates customers using Electronic checks are more likely to churn compared to other payment methods.

Converting coefficients to odds

- The coefficients of the logistic regression model are in terms of $\log(\text{odds})$, to find the odds we have to take the exponential of the coefficients.
- Therefore, $\text{odds} = \exp(b)$
- The percentage change in odds is given as $\text{odds} = (\exp(b) - 1) * 100$

	const	MonthlyCharges	TotalCharges	SeniorCitizen_Yes	Dependents_Yes	MultipleLines_Yes	InternetService_Fiber optic	OnlineSecurity_Yes	OnlineBackup_Yes	TechSupport_Yes
Odds	0.585	1.006	1.000	1.211	0.782	1.315	1.931	0.619	0.723	0.583
Change_odd%	-41.523	0.630	-0.014	21.054	-21.762	31.490	93.149	-38.081	-27.688	-41.745

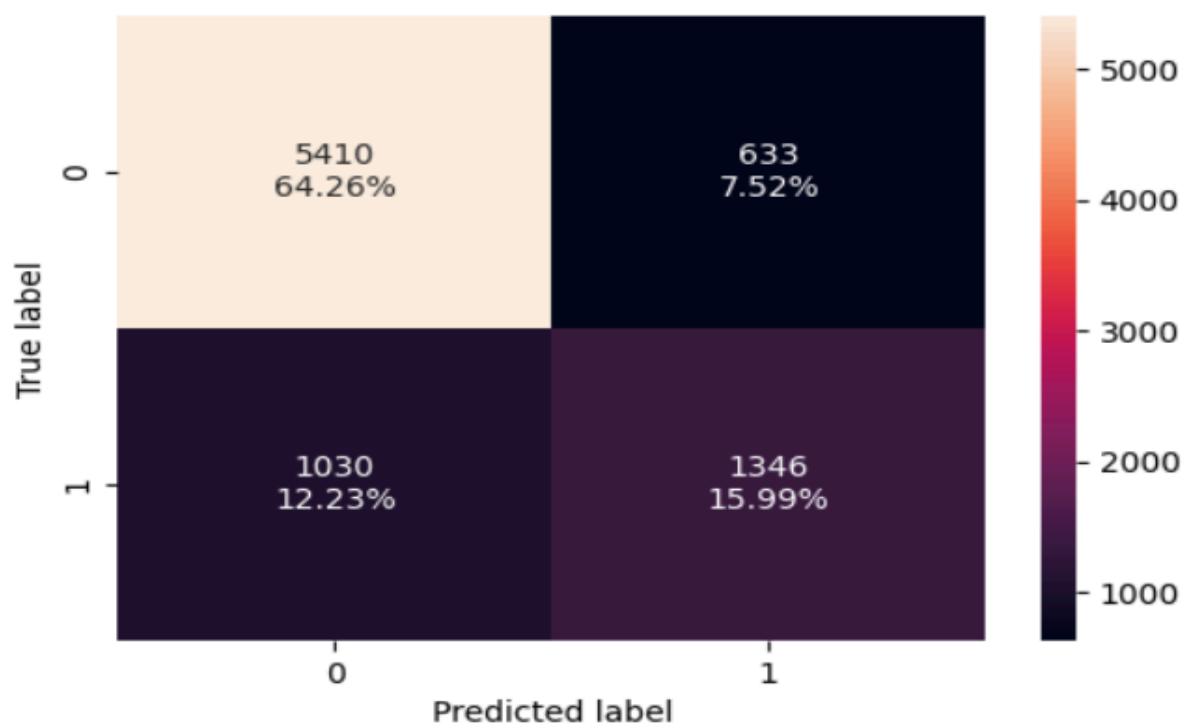
Coefficients to Odds

Coefficient interpretations

- **MonthlyCharges** : Holding all other features constant a unit change in **MonthlyCharges** will increase the odds of a customer being churned by 1.00 times or a 0.63% increase in odds.
- **TotalCharges** : Holding all other features constant a unit change in **TotalCharges** will decrease the odds of a customer being churned by 1.00 times or a 0.01% decrease in odds.

- **SeniorCitizen_Yes**: Holding all other features constant a unit change in **SeniorCitizen_Yes** will increase the odds of a customer being churned by 1.21 times or a 21% increase in odds.
 - **Dependents_Yes** : Holding all other features constant a unit change in **Dependents_Yes** will decrease the odds of a customer being churned by 0.78 times or a 21.76% decrease in odds.
- Interpretation for other attributes can be done similarly.

Checking model performance on the training set



Confusion Matrix on Training set(at default Threshold)

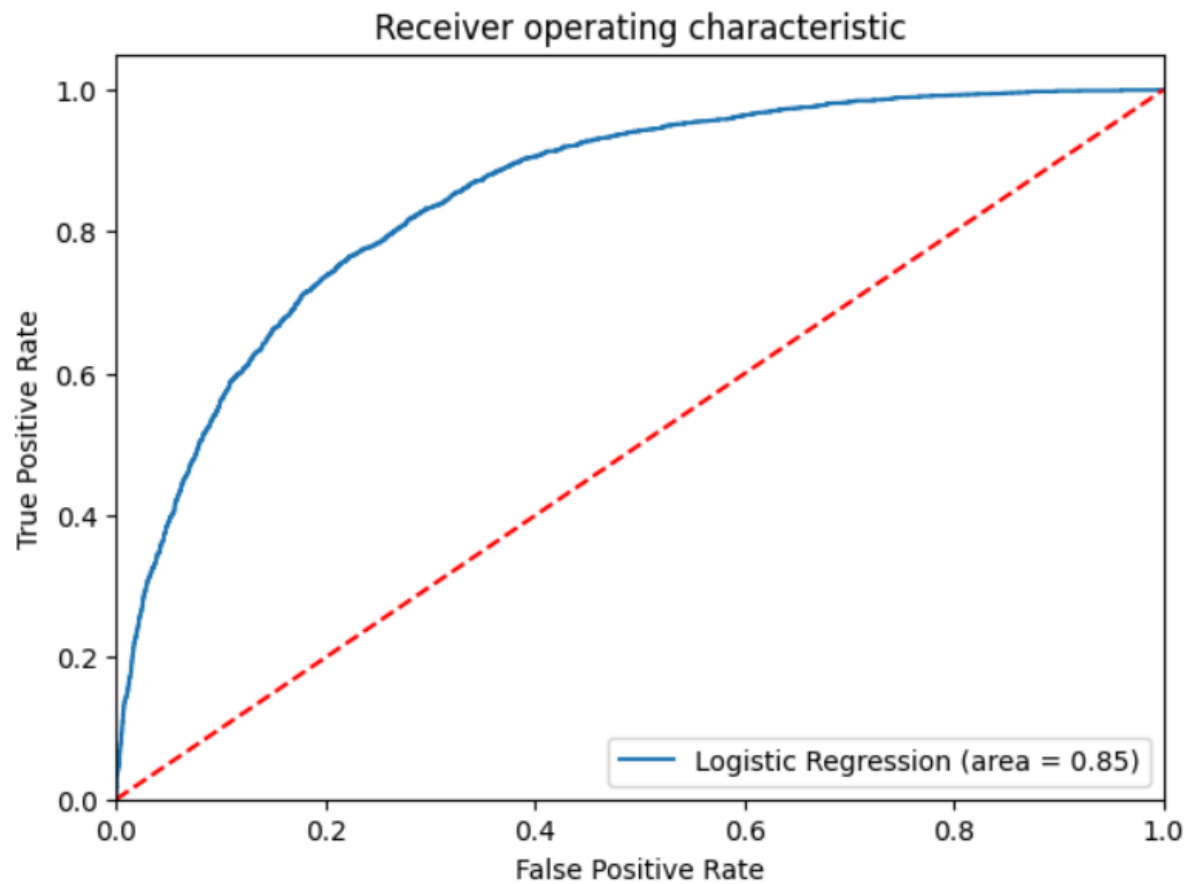
Training Performance

	Accuracy	Recall	Precision	F1
0	0.802	0.566	0.680	0.618

Performance Matrix(at default Threshold)

ROC-AUC

- ROC-AUC on training set

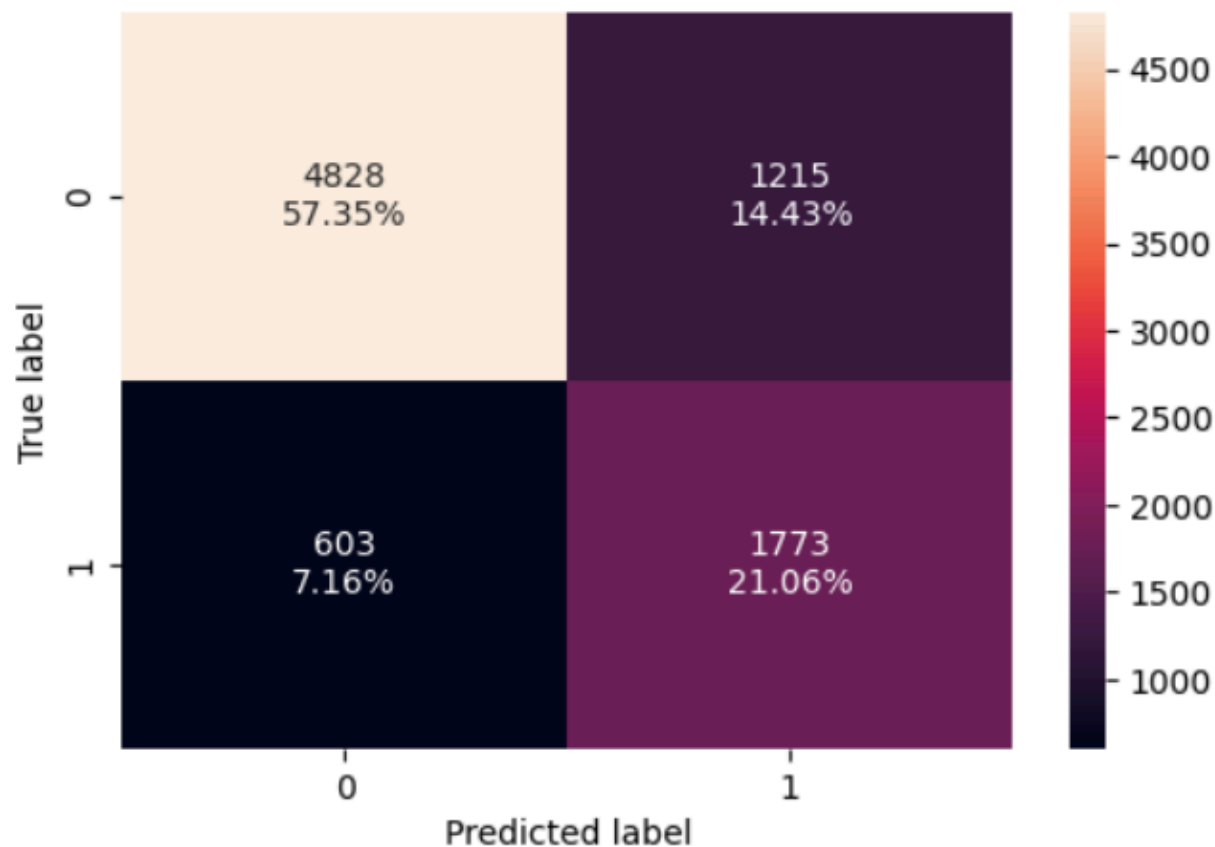


ROC-AUC Curve

Logistic Regression has good recall and ROC-AUC score on the training set.

Optimal threshold using AUC-ROC curve

Confusion Matrix



Confusion Matrix on Training set(at optimal Threshold)

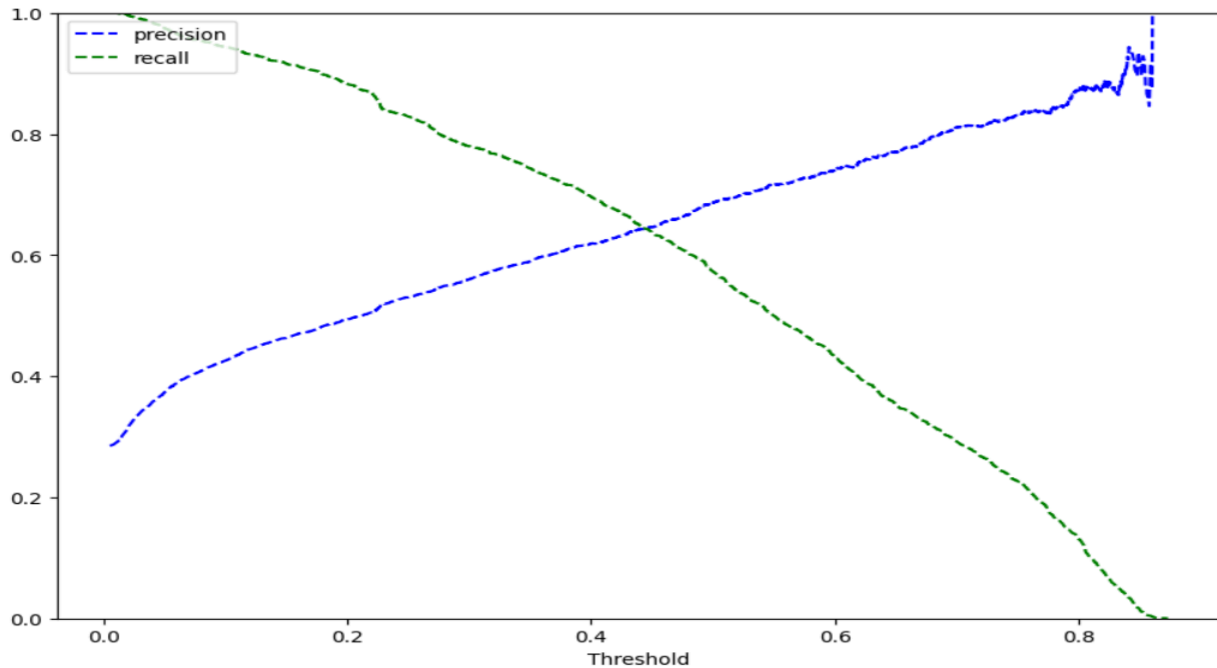
Training Performance

	Accuracy	Recall	Precision	F1
0	0.784	0.746	0.593	0.661

Performance Matrix(at optimal Threshold)

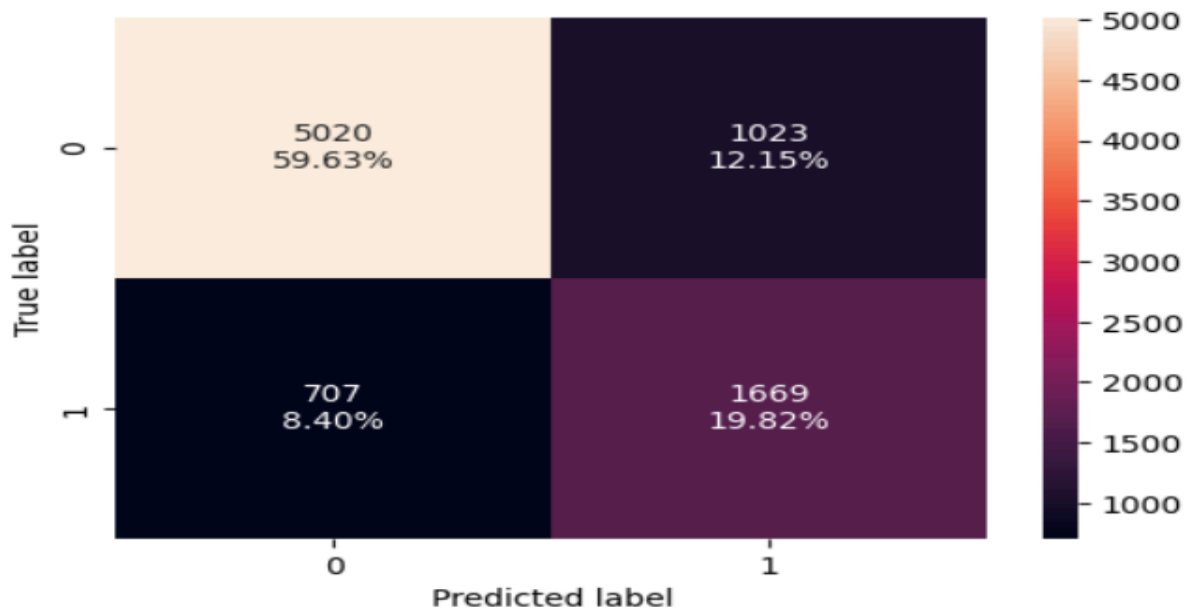
- Model performance has improved.
- The model is giving a recall of 0.74 as compared to the initial model which was giving a recall of 0.56.
- Accuracy and precision has slightly decreased from the initial model.

Let's use Precision-Recall curve and see if we can find a better threshold



- At threshold around **0.45** we will get equal precision and recall but taking a few steps back and selecting value around **0.40** will provide a higher recall.

Checking model performance on training set(with 0.40 threshold)



Confusion Matrix on Training set(at 0.40 Threshold)

Training Performance

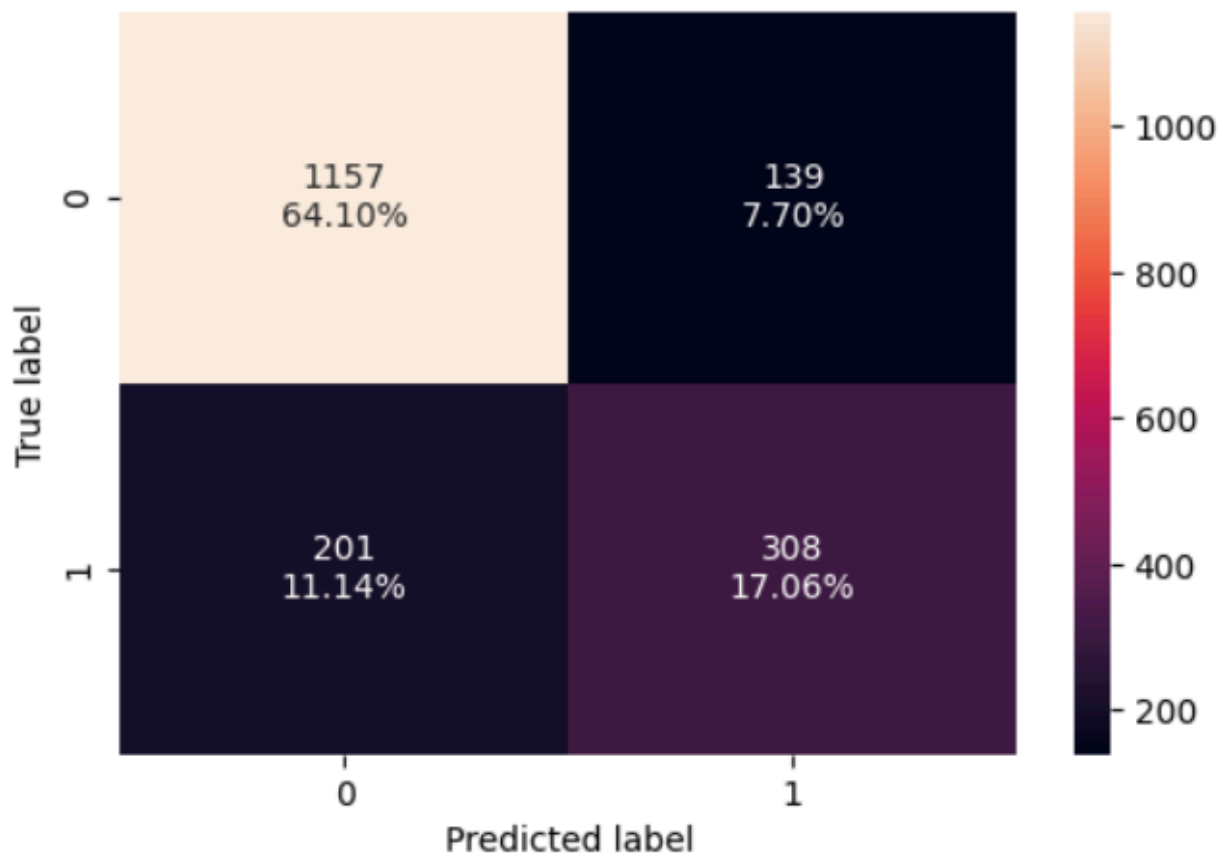
	Accuracy	Recall	Precision	F1
0	0.795	0.702	0.620	0.659

Performance Matrix(at 0.40 Threshold)

- Accuracy and Precision has improved as compared to the initial model.
- The model is giving a better performance with **0.34** threshold found using AUC-ROC curve.

Let's check the performance on the Validation set

- Using model with default threshold



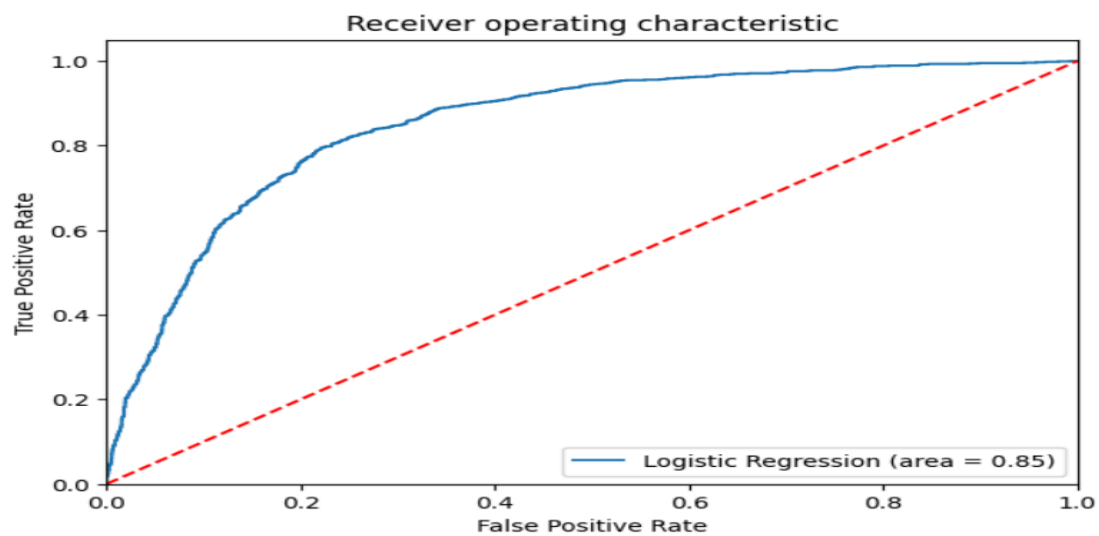
Confusion Matrix on Validation set(at default Threshold)

Validation performance:

	Accuracy	Recall	Precision	F1
0	0.812	0.605	0.689	0.644

Performance Matrix(at default Threshold)

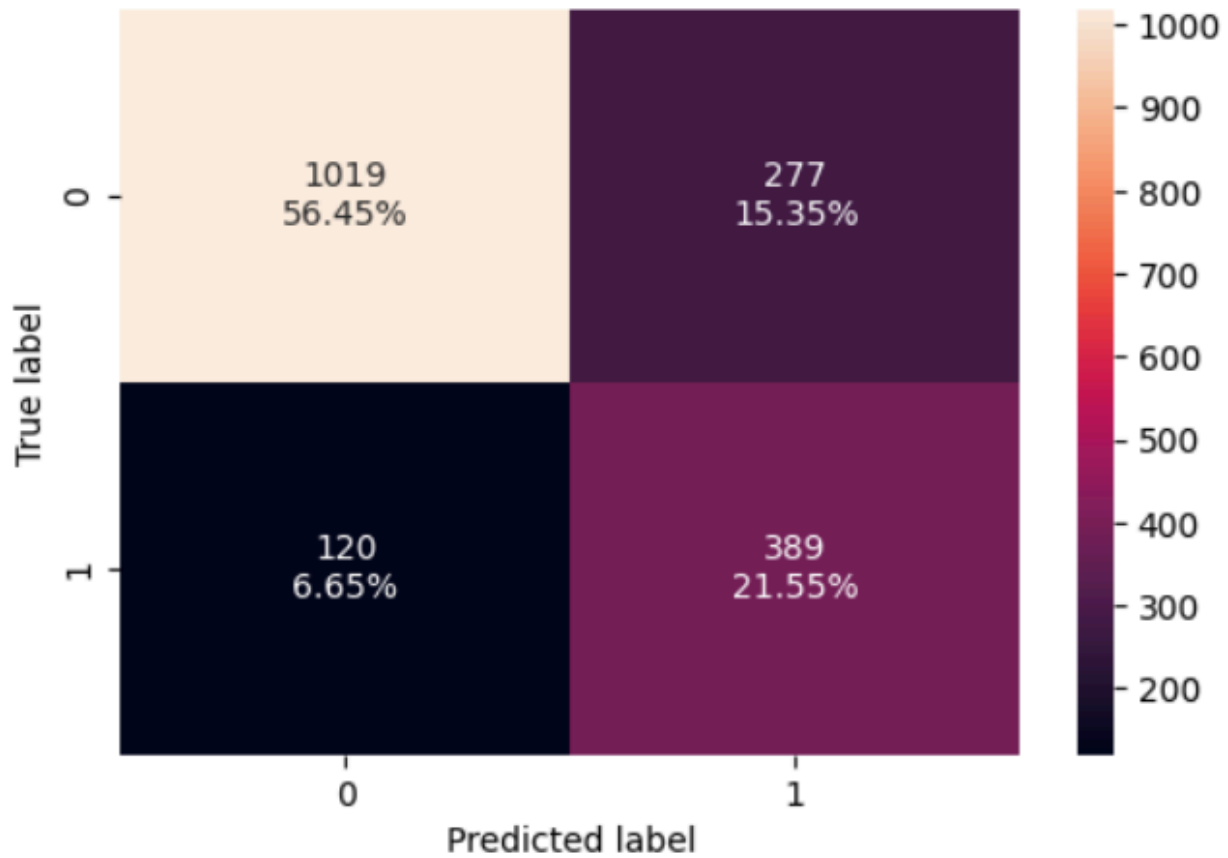
ROC-AUC curve on test set



ROC-AUC curve on test set

- The Logistic Regression model has a good ROC-AUC score on the test set.

Optimal threshold using AUC-ROC curve(with 0.34 threshold)



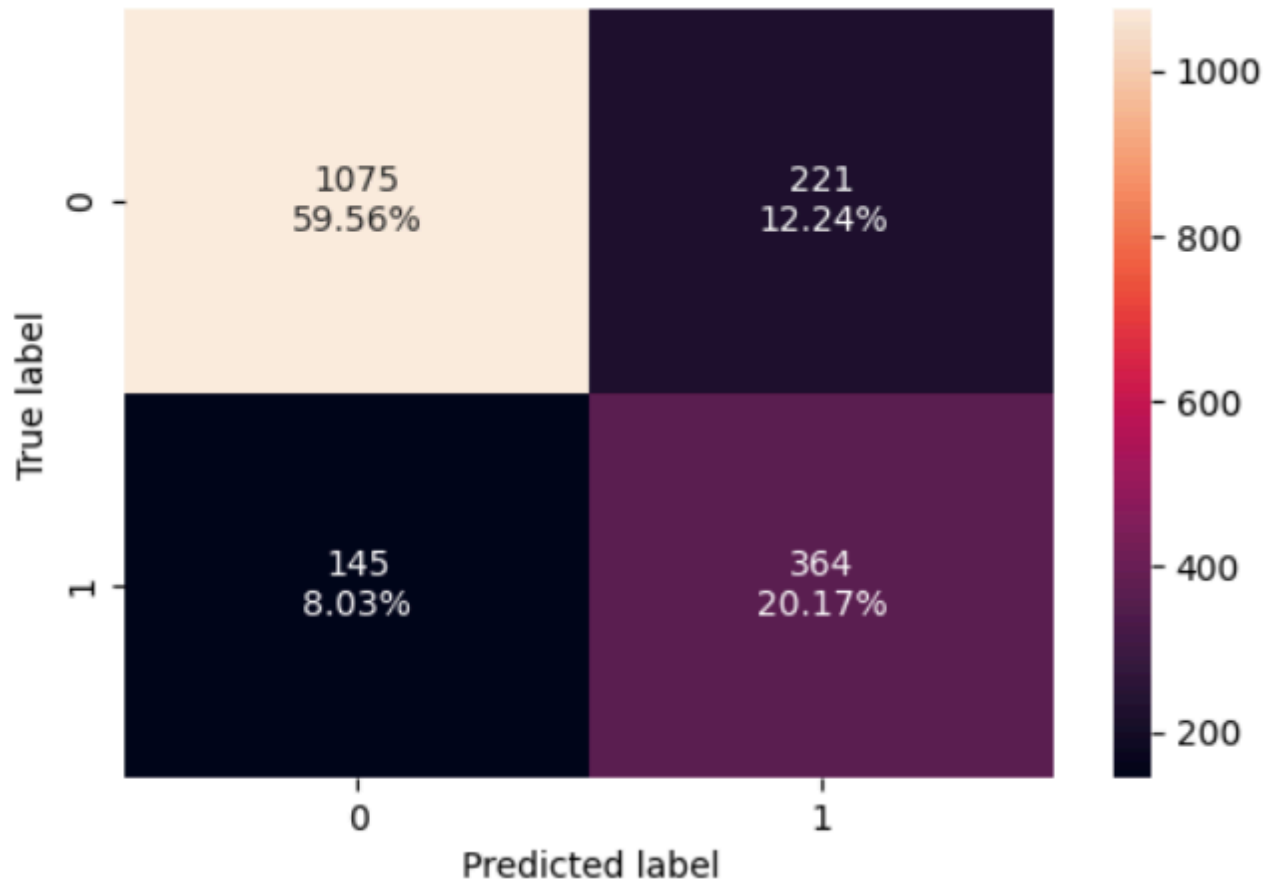
Confusion Matrix at 0.34 Threshold

Validation performance:

	Accuracy	Recall	Precision	F1
0	0.780	0.764	0.584	0.662

Performance matrix at 0.34 Threshold

Checking model performance on test set(with 0.40 threshold)



Confusion Matrix at 0.40 Threshold

Validation performance:

	Accuracy	Recall	Precision	F1
0	0.797	0.715	0.622	0.665

Performance Matrix at 0.40 Threshold

After comparing all models with various Threshold values we observed that the model with Optimal Threshold(0.34) gave us the best result so we consider this model to be the best model.

